

PROPELLER THEORY



Hurricane Mk.2B, Stephen Cobb / Alamy

Propeller Theory

with viscous optimization

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Preface

This text is a basic introduction to classical propeller theory. The intention is to offer the reader an intuitive grasp of the most important concepts in propeller theory. This should help in performing sanity checks on design proposals, parameter studies, and preliminary designs for new propellers. No prior knowledge is assumed except for high school physics, including a basic understanding of NEWTON's law and the concept of kinetic energy.

Appendices give more detailed derivations, and side views on the history of the subject, but should not be needed on a first reading.

There seems to be a fresh attempt at writing an introductory text to propeller theory every few decades. The list of references gives the best known examples from the 1930's onward, through the 1950's and into the 1980's. These references all cover the same basic ground as GLAUERT's 1935 classic synthesis, and the present text is no exception. However, there are some modernizations.

GOLDSTEIN's exact vortex solution is treated in some detail, and taken as the ground truth for all numerical results. Even so, PRANDTL's approximate solution is given prominence for an intuitive understanding of the tip loss.

GLAUERT's optimization of the viscous drag is taken to its logical conclusion, which leads to interesting new results.

The text contains a running example for a lightplane of typical dimensions, showing the effect of thrust loading and pitch angle on the shape and performance of the propeller.

P. Lammertse, 2025.

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1 Introduction

1.1 Scope and reading guide

This text is an introduction to classical propeller theory. It aims to give the reader an intuition of how a propeller works at its design point. The equations can also be used as a starting point for off-design calculations, but this is outside the scope of the present text.

The main text is kept as straightforward as possible. Details of the derivations, side views on alternatives, and historical notes are relegated to a number of appendices.

On a first reading, it should be possible to follow the main text without having to refer to the notes in the appendices.

Readers interested only in new material may choose to read only the chapters on the viscous loss. This contains a simple method of viscous optimization which is original to the current text.

The chapters on vortex theory also contain material which is not new, but which is not often presented in this way.

1.2 Vortex theory

Classical propeller theory has a reputation for being "difficult". This may be partly due to the use of the concept of "circulation" to express the blade lift.

Circulation is closely linked to vortex and potential theory, which are very convenient in describing the propeller. But the concept of circulation tends to be less intuitive to practicing engineers, who are more familiar with the lift coefficient.

But ultimately, the justification of the flow pattern around the tip of a propeller (and of a wing, for that matter) can only be found from a vortex argument. The chapter on tip loss gives a brief introduction to the idea. A separate appendix provides more details if needed.

1.3 Notation

The naming and symbolic notation mostly follows the literature, with the exception of the speed ratio. We use the small x for the local speed ratio $\Omega r/V$, and the capital X for the tip speed ratio $\Omega R/V$.

The literature calls the tip speed ratio either λ or $1/\lambda$, depending on the application area. Using X avoids this confusion, and it is used very extensively in the text.

2 Propeller geometry

2.1 Blade elements

A propeller is a rotating wing, or rather several of them. The wings are called "blades". Like in a wing, the width of the blade is called the "chord". The chord is measured in the nominal streamwise direction.

The propeller has a diameter D , and a maximum radius R which is $D/2$. Locations along the blade are measured by the local radius r . A small length-wise strip dr of the blade is called a "blade element". Figure 2.1 shows these definitions.

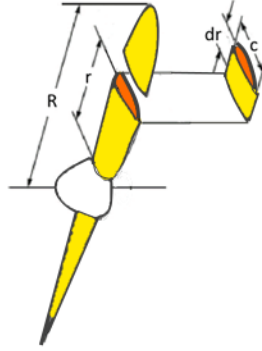


Figure 2.1 : Blade element .

2.2 Geometrical pitch

A propeller's *pitch* is its forward movement during a full revolution of the prop shaft. The result of the forward and rotational motion of a blade element is a spiral path. Figure 2.2 shows the path of a blade element at radius r . The forward motion for one revolution is P and the circumferential motion is $2\pi r$, and so the slope of the local pitch angle ϕ equals P over $2\pi r$:

$$\tan \phi \equiv \frac{P}{2\pi r} \quad (2.1)$$

Most propellers are designed for a certain flying velocity V and shaft RPM. Expressing V in meters per second, and RPM in revs per second, immediately gives the forward travel in meters per prop turn, and this gives the pitch of the propeller. A fixed pitch propeller will often be manufactured with this

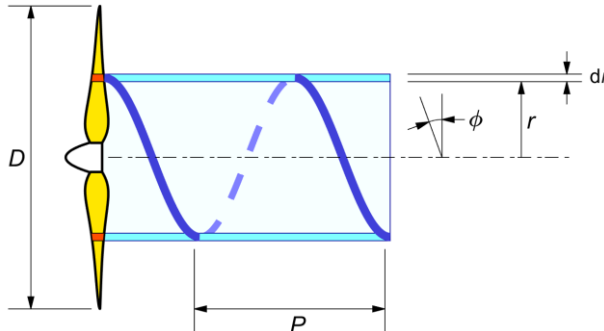


Figure 2.2 : Pitch and pitch angle.

"geometrical pitch" all along the blade. The geometrical pitch *angle* at any radius follows from (2.1), giving the blade its twisted appearance.

Figure 2.2 shows the blade element "living" inside a thin-walled tube of air between the radii r and $r + dr$. Much of classical propeller theory assumes that the blade elements influence only the air between the walls of this imaginary tube, and *not* the air inside r or outside $r + dr$. We will revisit this rather bold assumption later.

2.3 Aerodynamic pitch

In flight, the forward speed and the rate of rotation will generally not match the geometrical pitch of the propeller. The actual conditions will be in line with the blade element velocity diagram of figure 2.3.

The blade circumferential speed Ωr is the product of the angular velocity Ω of the propeller (in radians per second), and the local radius r (in meters).

The velocity V is the undisturbed flying velocity (in meters per second). The effective local velocities V and Ωr will get modified later by the blade's own lift inducing a downwash at the blade.

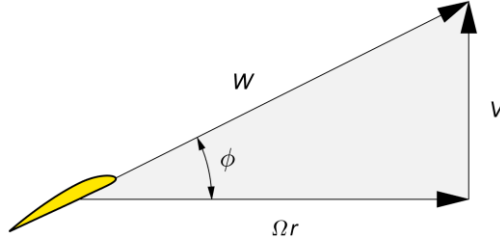


Figure 2.3: Blade element velocity diagram.

2.4 Advance ratio and speed ratio

The propeller literature supports a variety of symbols. Some can even have opposite meanings. Airplane texts generally use the letter λ for the "tip advance ratio" $\lambda \equiv V/\Omega R$. The wind turbine world uses λ for the "tip speed ratio" $\lambda \equiv \Omega R/V$. The difference does make sense. A wind turbine starts from zero RPM, and an airplane starts from zero forward velocity. In both cases we wish to avoid a division by zero in the starting condition, so we do not want the zero in the denominator.

But it is unfortunate that the lowercase letter λ is used for a blade tip condition, since another convention is that local values (like the radius r) are in lowercase, with tip values (like R) in capitals. The capital λ would have been more appropriate. We will avoid the letter λ altogether. Following GLAUERT's convention (in part of his work), we denote the local speed ratio $\Omega r/V$ by the small letter x . By the capital letter convention, we call its tip value X , even though this is a bit unusual. This makes X equal to the λ of the wind turbine world, and to the $1/\lambda$ of the airplane world:

$$\begin{aligned} x &\equiv \frac{\Omega r}{V} = \frac{1}{\tan \phi} && \text{local speed ratio} \\ X &\equiv \frac{\Omega R}{V} = \frac{1}{\tan \phi_{\text{tip}}} && \text{tip speed ratio} \end{aligned} \tag{2.2}$$

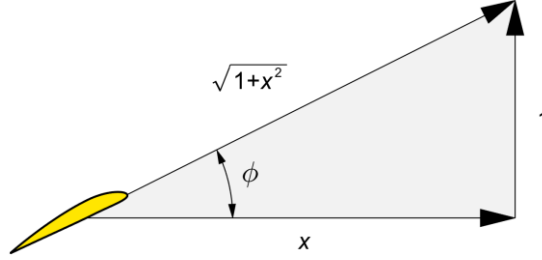


Figure 2.4: The local speed ratio x .

2.5 The local pitch angle

Figure 2.4 gives the velocity diagram in terms of the local speed ratio x . At the tip, we use the capital letter. The figure shows by inspection:

$$\tan \phi = \frac{1}{x} \quad (2.3)$$

$$\sin \phi = \frac{1}{\sqrt{1+x^2}} \quad (2.4)$$

$$\cos \phi = \frac{x}{\sqrt{1+x^2}} \quad (2.5)$$

Another relation which comes up frequently in the theory is :

$$\cos^2 \phi = \frac{x^2}{1+x^2} \quad (2.6)$$

This also means :

$$\frac{1}{\cos^2 \phi} = \frac{1+x^2}{x^2} = 1 + \frac{1}{x^2} \quad (2.7)$$

We will freely use a mixture of function arguments ϕ and x , on the understanding that they are easily converted into each other. In particular the $\cos^2 \phi$ term comes up very frequently, and is more easily recognized when it is expressed in ϕ instead of x .

2.6 Non-dimensional length

The ratio V/Ω is the forward progression of the propeller for one radian of rotation. It has the dimension of a length. We will call this "typical length" U :

$$U \equiv V/\Omega \quad (2.8)$$

The non-dimensional speed ratio x equals the local radius r , divided by this standard length. We can rewrite (2.2) as:

$$x \equiv \frac{r}{U} = r / \left(\frac{V}{\Omega} \right) = r \cdot \left(\frac{\Omega}{V} \right) = \Omega r / V \quad (2.9)$$

$$X \equiv \frac{R}{U} = R / \left(\frac{V}{\Omega} \right) = R \cdot \left(\frac{\Omega}{V} \right) = \Omega R / V$$

At $x = 1$, the pitch angle is 45° and $\tan \phi = 1$. If two propellers have the same tip speed ratio X , then they have the same tip pitch angle Φ . This makes x and X convenient non-dimensional radius values. As a corollary, we have :

$$\frac{r}{R} = \frac{x}{X} \quad (2.10)$$

3 Thrust, power and efficiency

3.1 The thrust coefficient

This text will use the thrust from the propeller as the starting point of the design. We will use two definitions of the non-dimensional thrust coefficient out of the many used in the literature. In analogy with the lift coefficient C_L of a wing, we define the non-dimensional thrust coefficient C_T for use outside propeller theory:

$$C_T \equiv \frac{T}{\frac{1}{2}\rho V^2 \cdot \pi R^2} \quad (3.1)$$

This definition is based on V^2 instead of the more usual $(\Omega R)^2$. This reflects an emphasis in this text on the requirements of the aircraft, rather than on the torque or the RPM of the powerplant.

In detailed derivations it will be more convenient to divide by ρV^2 instead of by $\frac{1}{2}\rho V^2$. We will denote this division by ρV^2 by a prime. This version of the non-dimensional thrust coefficient is then called T' :

$$T' \equiv \frac{T}{\rho V^2 \cdot \pi R^2} \quad (3.2)$$

It follows that :

$$T' = \frac{1}{2} C_T \quad (3.3)$$

Power has the dimension of thrust times velocity. We will therefore make power non-dimensional by dividing by V^3 instead of by V^2 :

$$P' \equiv \frac{P}{\rho V^3 \cdot \pi R^2} \quad (3.4)$$

3.2 The net power

Power equals force times velocity. The net output power that a propeller exerts on an airplane is the thrust force T multiplied by the flying speed V :

$$P = T \cdot V \quad (3.5)$$

It is interesting to note that we can write (3.2) as a power as well as a force :

$$T' = \frac{T \cdot V}{\rho V^2 \cdot \pi R^2} \quad (3.6)$$

We can interpret T' as the non-dimensional thrust, but we can also interpret it as the net power $T \cdot V$ made non-dimensional by the definition of (3.4).

3.3 Power, power loss and efficiency

Power equals force times velocity, but it also equals torque times rate of rotation (expressed in radians per second), and energy per second.

Efficiency is defined as net power *output* divided by gross power *input*. Intuitively, loss of efficiency is associated with a loss of output for a given input. However, losses in a propeller are more easily calculated as an increase of the power *input* needed for a given power *output*. The efficiency is still the same ratio: output over input, but the losses are now written as fractional increases on the input, instead of fractional decreases on the output.

The extra input still goes into energy lost to the wake of the propeller. This is why we keep calling these extra inputs "losses".

3.4 Loss ratios

The momentum loss and the viscous loss are increases on the input power ideally needed for a given output power $T.V$, and we will call these increases P_m and P_v . The propeller's efficiency then becomes :

$$\eta = \frac{T.V}{T.V + P_m + P_v} \quad (3.7)$$

We can divide top and bottom by the net power $T.V$:

$$\eta = \frac{1}{1 + \frac{P_m}{T.V} + \frac{P_v}{T.V}} \quad (3.8)$$

We will call $P_m/(T.V)$ the momentum loss *ratio*, and $P_v/(T.V)$ the viscous loss *ratio*, to indicate that they are relative to the net power.

These power loss ratios are non-dimensional, and they can be expressed in the non-dimensional thrust and power too. By (3.4) and (3.6) we have :

$$\frac{P}{T.V} = \frac{P'}{T'} \quad (3.9)$$

$$\eta = \frac{1}{1 + \frac{P'_m}{T'} + \frac{P'_v}{T'}} \quad (3.10)$$

3.5 Types of loss in a propeller

There are four types of loss in a propeller :

- (axial) momentum loss.
- a rotational overhead on the (axial) momentum loss.
- tip loss, which is a different type of overhead on the momentum loss.
- viscous (frictional) loss.

We will discuss each of these losses in due course. Each have their own distinct influence on the efficiency.

All types of momentum loss rise in direct proportion to the loading. The rotational overhead on the momentum loss also increases with the pitch angle. Tip loss is not a direct loss of power, but by creating gaps in the effective frontal area of the propeller it increases the loading on the active part of the blades, and by that indirect route the momentum loss. It increases with pitch, and with low number of blades.

Viscous loss increases with the C_d/C_l of the blade section used, but also strongly with the tip speed ratio, i.e. roughly with the *inverse* of the pitch angle. This creates a conflict between the best pitch for momentum loss and that for tip loss.

We will discuss the basic (axial) momentum loss first.

4 Axial momentum loss

4.1 The ideal actuator disk

Much of propeller theory is based on the idea that a propeller influences only the air passing through the prop disk, and none of the air outside it. This model is surprisingly accurate, but far from obvious. The real justification comes from a vortex view of the flow. This view is given at the end of this text. For the purposes of this chapter it is best to accept it for now.

The ideal propeller acts like an imaginary disk of frontal area πR^2 , exerting a uniform pressure on the air, and accelerating the flow tube to a uniform velocity. This image is called the ideal "actuator disk". Later chapters will fill in the disk with more detail.

Figure 4.1 shows an actuator disk halfway the length of a solid streamline tube. In the stationary frame of reference, the disk moves forward through the air at the flying speed, exerting a small push force to the rear on the air when passing through it. The acceleration of the air (at almost constant density) makes the tube contract a little. The contraction of the tube moves forward with the disk, in a motion like swallowing.

The disadvantage of using this frame of reference is that the movement of the air at a given viewpoint is time-varying as the disk passes by the observer.

The equations are much simpler if the observer moves alongside with the disk. Relative to us, the disk will then be stationary, and the whole atmosphere will be flowing by. We might call this the "wind tunnel view", or the view from the cockpit. The relative motion is the same as for a propeller moving through a stationary atmosphere, but now the air at a given point is flowing by continuously at the same speed. The air far ahead of the propeller is the undisturbed atmosphere "flowing towards" the actuator disk.

Far behind the actuator disk, the air flows at a slightly increased velocity inside the slightly contracted flow tube. Since the walls of the flow tube far downstream of the disk are straight, there can be no sideways pressure gradient. Any sideways pressure on the air would curve the flow locally. Therefore the pressure in the "far wake" must be identical to the atmospheric pressure. The air in the far wake flows straight to the rear at a constant velocity, "induced" by the propeller.

The actuator disk (like a wing) creates *only* a pressure step. The volume flow through the frontal area is the same just ahead of and just aft of the disk, and so the velocity cannot instantaneously change there. The pressure makes no difference to the volume, because air is practically "incompressible", in the sense that the aerodynamic pressures are so small relative to the massive static pressure from the weight of the atmosphere above us, that the air hardly changes in density due to (subsonic) flow conditions.

The pressure field ahead of, and aft of the disk induces a gradual acceleration of the air to the rear. Because of the constant volume, the downwind tube has a slightly smaller cross-section than the upwind tube. All of the acceleration, and therefore all of the contraction of the tube takes place over a short distance ahead and aft of the actuator disk. The tube walls are curved in an S-curve only in this middle part.

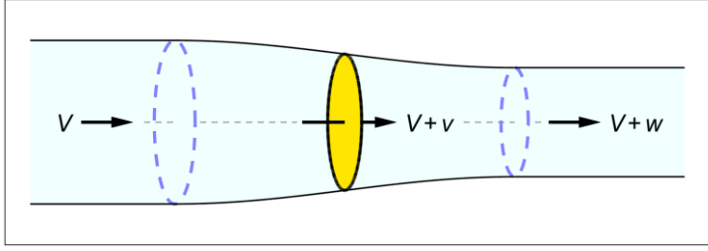


Figure 4.1 : Actuator disk and flow tube.

We will call the axial velocity increase reached *at* the disk v , and the final velocity increase far *behind* the propeller w . We will use the letters a and b for the non-dimensional versions of these so-called "induction ratios" : the first one locally at the disk, and the other one in the "far wake" :

$$a = \frac{v}{V} \quad (4.1)$$

$$b = \frac{w}{V} \quad (4.2)$$

4.2 Newton's second law

One of NEWTON's laws asserts that action equals reaction. An object can only push itself forward by pushing something else back. If that something else is not infinitely heavy, it will start moving in the opposite direction, as anyone can testify who has ever tried stepping ashore from a small boat.

The mechanism applies equally to air. A propeller pushing an airplane forward will have to push some quantity of air to the rear. This will give that air a velocity increase. NEWTON's second law states that the force needed to accelerate a mass is proportional to the mass times its acceleration :

$$F = m \cdot a \quad (4.3)$$

This is not the a of (4.1), but the momentary acceleration. Acceleration is the change of velocity per second. We can also write (4.3) as :

$$F = m \cdot v^\bullet \quad (4.4)$$

The raised dot " $\bullet$ " is NEWTON's symbol for "per second". A lesser known version of the same law is :

$$F = (m \cdot v)^\bullet \quad (4.5)$$

Here the force equals the change in the "quantity of motion" ($m \cdot v$), which is also known as the "momentum". The momentum can change by a change of *mass* per second, just as well as by a change of *velocity* per second. Instead of accelerating the same mass to an ever higher velocity, we can take a new mass $m^\bullet$ every second, and give each new mass the velocity increase v . For example, we can exert a forward force on ourselves by throwing a snowball of mass m to the rear every second, or by squirting a continuous mass flow of water $m^\bullet$ out of a fire hose at a constant velocity v . We can then write (4.5) as :

$$F = m^\bullet \cdot v \quad (4.6)$$

4.3 Newton's momentum law for the propeller.

It is version (4.6) of NEWTON's law which applies to the propeller. Let us say that each second an air mass $m^\bullet$ flows through the prop disk. From figure 4.1, the end result is that this mass gets a velocity increase from V to $V+w$, a net velocity increase of w . The propeller exerts a constant rearward force F on the flow, and by NEWTON's first law receives an equal but opposite forward reaction force which we will call the thrust T . Then NEWTON's second law for a propeller is:

$$T = m^\bullet \cdot w \quad (4.7)$$

This is the basis of what is called the "momentum theory" of the propeller. With the definition for b in (4.2) we can also write :

$$T = m^\bullet \cdot b \cdot V \quad (4.8)$$

4.4 The efficiency at the actuator disk

We now look at the power. First we calculate the power at the prop disk. With (3.5) for the net power and (4.7) or (4.8) for the thrust we have :

$$P_{\text{net}} = T \cdot V = m^\bullet \cdot w \cdot V = m^\bullet \cdot b \cdot V^2 \quad (4.9)$$

From figure 4.1, the propeller works in a local flow of increased velocity of its own making, $(V+v)$. On this flow, the thrust of the propeller needs to exert the following power, since power is force times velocity :

$$P_{\text{gross}} = T \cdot (V + v) = m^\bullet \cdot b \cdot V^2 \cdot (1 + a) \quad (4.10)$$

The local momentum loss (actually, the gross power increase) is:

$$P_m = P_{\text{gross}} - P_{\text{net}} = m^\bullet \cdot b \cdot V^2 \cdot a \quad (4.11)$$

Note that P_m is not the total momentum power, but only the extra power due to the induction velocity v , on top of the power $T \cdot V$ needed to drive the airplane forward at its flying velocity V .

The momentum *loss ratio* is the ratio between (4.11) and (4.9) :

$$\frac{P_m}{T \cdot V} = \frac{m^\bullet \cdot b \cdot V^2}{m^\bullet \cdot b \cdot V^2} \cdot a = a \quad (4.12)$$

By (3.9), the non-dimensional form of this ratio is :

$$\frac{P'_m}{T'} = a \quad (4.13)$$

The propulsive efficiency is the ratio between net output power and the gross input power. It does not depend on the mass flow, only on the induction ratio a . From (3.10), the momentum efficiency at the prop disk is :

$$\eta_m = \frac{1}{1 + a} \quad (4.14)$$

4.5 The efficiency in the far wake

The wake velocity $w = b \cdot V$ dropped out of the momentum loss ratio (4.12) when dividing (4.9) by (4.10). But the wake velocity is interesting in its own right, so we derive it from an energy argument.

The propeller leaves behind a velocity w , and therefore also a kinetic energy, in the atmosphere. The equation for kinetic energy is $\frac{1}{2} m v^2$, but in this case the velocity is w , and the mass is m^* per second. Energy per second is the definition of a power, and so the kinetic energy lost to the wake represents the following power :

$$P_m = \frac{1}{2} m^* \cdot w^2 = \frac{1}{2} m^* \cdot b^2 \cdot V^2 \quad (4.15)$$

With the net power (4.9), we have an alternative expression for the loss ratio :

$$\frac{P_m}{T \cdot V} = \frac{\frac{1}{2} m^* \cdot b^2 \cdot V^2}{m^* \cdot b \cdot V^2} = \frac{\frac{1}{2} b^2}{b} = \frac{1}{2} b \quad (4.16)$$

The mass flow m^* drops out of this ratio like it did in (4.12). Once again, the efficiency does not depend on the mass flow. This time, it only depends on the far wake induction ratio b . By (3.9), the non-dimensional form yields :

$$\frac{P'_m}{T'} = \frac{1}{2} b \quad (4.17)$$

From (3.10), the far wake efficiency is:

$$\eta_m = \frac{1}{1 + \frac{1}{2} b} \quad (4.18)$$

4.6 Equating the two efficiencies

We now have two ways of calculating the momentum loss ratio, and they must be equal. Equating (4.17) to (4.13) we have :

$$a = \frac{1}{2} b \quad (4.19)$$

The flow towards the disk and the flow away from it are anti-symmetrical. Air accelerates towards the oncoming propeller, and then accelerates by the same amount aft of it. The propeller disk itself is exactly halfway.

There is a direct parallel with the downwash of a wing, which we will discuss in later chapters on vortex theory.

4.7 The nature of the momentum loss

The momentum loss is a loss of kinetic energy to the atmosphere, and the kinetic energy $\frac{1}{2} m v^2$ is quadratic in the velocity. The thrust on the other hand, is proportional to the momentum "lost" to the atmosphere, i.e. the velocity increase of the mass flow.

As a result, the ... parabolic plot ??

,

5 Thrust and the mass flow

5.1 Non-dimensional thrust and mass flow

We wish to express the thrust equation (4.7) in non-dimensional terms.

Applying (3.2) to (4.7) we have :

$$T' \equiv \frac{m^{\bullet} \cdot w}{\rho V^2 \cdot \pi R^2} \quad (5.1)$$

We already have the non-dimensional expression (4.2) for w and so :

$$T' = \frac{m^{\bullet} \cdot V}{\rho V^2 \cdot \pi R^2} \cdot b \quad (5.2)$$

It is then convenient to define a non-dimensional momentum flow :

$$m' \equiv \frac{m^{\bullet} \cdot V}{\rho V^2 \cdot \pi R^2} \quad (5.3)$$

This changes (5.2) to :

$$T' = m' \cdot b \quad (5.4)$$

To find m' , we turn to figure 4.1. In this figure, the velocity of the air flowing through the actuator disk is $(V+v)$. Its frontal area is πR^2 . For an air density of ρ kg/m³, the mass flow per second is :

$$m^{\bullet} = \rho \cdot \pi R^2 \cdot (V+v) \quad (5.5)$$

Using (4.1) to make v non-dimensional, we have :

$$m^{\bullet} = \rho V \cdot \pi R^2 \cdot (1+a) \quad (5.6)$$

Substituting this into (5.3) gives:

$$m' = \frac{\rho V^2 \cdot \pi R^2 \cdot (1+a)}{\rho V^2 \cdot \pi R^2} = 1+a \quad (5.7)$$

First off, we note that $m^{\bullet} V$ is **... not ...** the momentum that $m^{\bullet}$ had at the velocity V far ahead of the propeller, not after it has been (halfway) accelerated into the prop disk. It is *not* the momentum flow *at* the propeller, but the momentum *inflow* into the propeller from far ahead.

We then look at the situation far ahead of the propeller. Here the same mass $m^{\bullet}$ flows with velocity V . With (5.6) and with incompressible air of constant density ρ , this means that the propeller takes in air from a frontal area of $\pi R^2 \cdot (1+a)$. In other words, m' is the ratio between the "effective" frontal area of the propeller, and its physical disk area.

The contraction on the inflow makes the propeller take in more mass and momentum than would normally flow through its physical diameter. By the contraction, the propeller delivers more thrust for the same induction and hence the same efficiency. Put differently, the propeller acts partly as if it were located at the start of the contraction, instead of at the physical disk. With (5.7) in (5.4), the non-dimensional thrust becomes :

$$T' = b \cdot (1+a) \quad (5.8)$$

It is sometimes more intuitive to call it the non-dimensional mass flow, but it is really the momentum.

5.2 The light loading assumption

Airplane propellers have relatively small induction ratios in cruising flight, on the order of $b \cong 0.1$ like in our running numerical example.

Equation (5.8) and many equations to follow will stay much simpler if we assume that $\frac{1}{2} b \ll 1$, and therefore $(1 + a) \cong 1$. It is common to develop the theory mostly under this so-called "light loading assumption". This turns (5.8) into :

$$T' = b \quad (5.9)$$

The induction ratio $b(x)$ may vary with the radius x , but for the actuator disk it is constant ("uniform") over the whole disk, so we can write :

$$b(x) = T' \quad (5.10)$$

By (4.18) and (3.3), the efficiency of the ideal, lightly loaded actuator disk is :

$$\eta_m = \frac{1}{1 + \frac{1}{2} T'} = \frac{1}{1 + \frac{1}{4} C_T} \quad (5.11)$$

With (5.9), the momentum loss ratio (4.17) is:

$$\frac{P'_m}{T'} = \frac{1}{2} T' = \frac{1}{4} \cdot \frac{T}{\frac{1}{2} \rho V^2 \cdot \pi R^2} \quad (5.12)$$

5.3 Heavy loading

We briefly touch on the general case without the light loading assumption. The mass loading term $(1 + \frac{1}{2} b)$ in (5.8) increases the thrust without affecting the efficiency (4.18). This slightly mitigates the linear increase in the momentum loss ratio with the thrust. The contraction makes the propeller act as if it has a slightly larger diameter than it actually has. We can solve (5.8) for b by writing it as a quadratic:

$$b^2 + 2b - 2T' = 0 \quad (5.13)$$

Solving the quadratic gives one positive solution:

$$b = \sqrt{1 + 2T'} - 1 \quad (5.14)$$

Figure 5.1 shows this relation. For a given induction ratio and efficiency, the contraction of the flow tube draws more air through the prop disk, and this increases the thrust at the same efficiency, which gives a slight advantage.

In the main text we will stay with the light loading assumption, because it keeps the main argument easier to follow. In the math appendix we follow the $(1 + \frac{1}{2} b)$ term through for some of the detailed calculations.

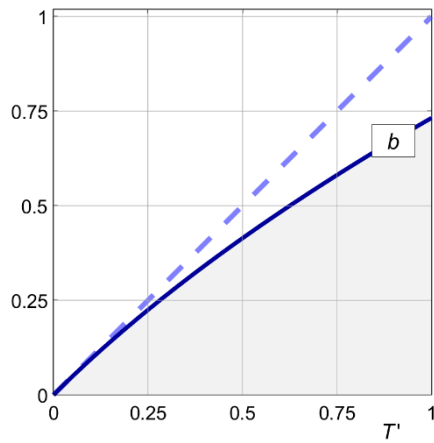


Figure 5.1 : Axial induction ratio b versus T' .

TODO Consider a loading T' which is decreased for contraction.

6 Concentric tubes

6.1 The independence of the rings

Most of propeller theory rests on the idea that the momentum tube of figure 4.1 can be subdivided into set of thin-walled concentric tubes, sliding (and rotating) inside each other like a pirate's telescope. These tubes will cut the propeller disk into nested concentric rings. Figure 2.2 showed such a tube.

The tubes are supposed not to influence each other very much, but this mutual independence is by no means obvious. The justification can be found in vortex theory, which will be discussed much later. In practice, the independence of the tubes holds up much better than might reasonably be expected.

6.2 The effective area per ring

If the rings and tubes of a propeller do not interact, then each ring is like a separate propeller. It has its own local induction ratio, its own pitch angle and its own blade elements. The propeller as a whole is just a superposition of independent, nested ring propellers, flying in close formation.

It is usual to derive the conditions for a single ring, and then integrate the ring contributions into an overall value for the whole propeller.

We will use the prefix "d" for variables belonging to a ring of "infinitesimal" width dr . For the thrust, the contribution of a ring follows NEWTON's momentum law (4.8) just the same as for a complete disk :

$$dT = dm^* \cdot b(x) \cdot V \quad (6.1)$$

The difference is that dm^* is the air flowing through a ring instead of all of the air flowing through the whole disk. The induction $b(x)$ may vary among the rings, and so we often use the argument x on it. We could do the same for the local mass flow, since this will also vary with the ring area. We can make the thrust on a ring non-dimensional by the same definition (3.2) we used for the overall thrust :

$$dT' \equiv \frac{dm^* \cdot V \cdot b(x)}{\rho V^2 \cdot \pi R^2} \quad (6.2)$$

The local equivalent of the non-dimensional momentum flow (5.6) is :

$$dm' \equiv \frac{dm^* \cdot V}{\rho V^2 \cdot \pi R^2} \quad (6.3)$$

In (5.5) the mass flow m^* was for the whole disk area. For a ring of width dr at radius r , the equivalent area is $2\pi r \cdot dr$. This changes the overall mass flow (5.5) into the local mass flow for a ring :

$$dm^* = \rho \cdot V \cdot 2\pi R \cdot dr \cdot (1 + a) \quad (6.4)$$

The local non-dimensional momentum flow dm' becomes :

$$dm' = \frac{\rho V^2 \cdot 2\pi r \cdot dr \cdot (1 + a)}{\rho V^2 \cdot \pi R^2} \quad (6.5)$$

Most of the terms in (6.5) cancel between the numerator and the denominator, but r and R remain and they are still dimensional. However, by (2.10) we can replace r/R by x/X and dr/R by dx/X , leaving only :

$$dm' = \frac{2x \cdot dx}{X^2} \cdot (1 + a) \quad (6.6)$$

The value of $b(x)$ may vary from ring to ring, but the relation $a = \frac{1}{2} b$ from (4.19) remains valid, so we can also write :

$$dm' = \frac{2x \cdot dx}{X^2} \cdot (1 + \frac{1}{2} b(x)) \quad (6.7)$$

The “effective area” interpretation of dm' is the same as that for the m' of the whole disk : dm' is the ratio between the effective ring area far ahead of the propeller, before contraction and the *overall* geometrical disk area πR^2 . The total effective mass flow m' is the sum of the individual dm' contributions integrated over the prop disk.

This result will be a weighted average of $(1 + a)$ over the prop disk.

6.3 The effective area per ring at light loading

In the light loading approximation, we have $(1+a) \cong 1$. This simplifies the dm' of (6.6) to :

$$dm' \cong \frac{2x \cdot dx}{X^2} \quad (6.8)$$

This is simply the geometrical frontal area of the ring as a fraction of the frontal area of the whole disk.

The integral of (6.8) from the hub to the radius x/X is the area of a disk of radius x/X :

$$m'(x) = \int_0^x dm' = \frac{1}{X^2} \int_0^x 2x \cdot dx = \left(\frac{x}{X}\right)^2 \quad (6.9)$$

Figure 6.1 shows the result. For light loading, the effective area $m'(X)$ for the whole prop disk is simply 1.

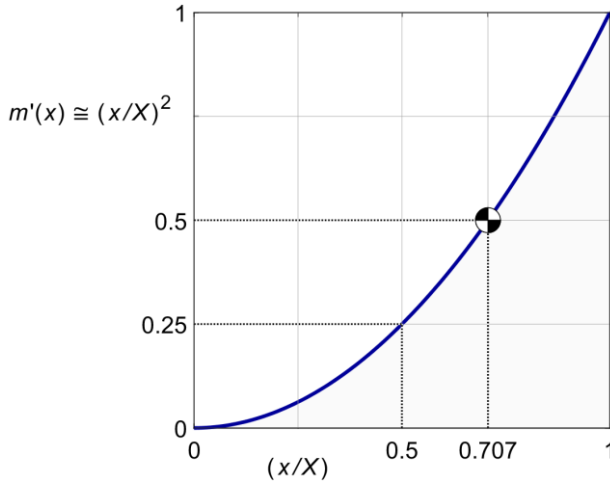


Figure 6.1 : Relative disk area within a radius (6.9), for light loading.

Figure 6.1 conveys an important message : the median of the disk area is located at the radius $r/R = 0.7$ (actually, $r/R = \frac{1}{2} \sqrt{2}$). Conventional wisdom therefore has it that a propeller is characterized by its radius $0.7 R$.

The area within the radius $r/R = 0.5$ is one quarter of the total disk area, and the area within $r/R = 0.25$ is only $1/16$ of the total disk area.

Also, the shape of the blade shank near the hub is dominated by structural considerations, and the central part will be covered by a spinner. All in all, the inner 25% of the blade radius is mostly along for the ride. Integrating the thrust over the rings

By analogy with (***) and by (6.6), the local non-dimensional ring thrust is :

$$dT' = b(x) \cdot dm' = b(x) \cdot (1+a) \cdot \frac{2x \cdot dx}{X^2} \quad (6.10)$$

Integrating dT' over the rings gives the overall thrust:

$$T' = \int dT' = \frac{1}{X^2} \int_0^X b(x) \cdot (1+a(x)) \cdot 2x \cdot dx \quad (6.11)$$

For light loading, we can use the dm' of (6.8), which yields :

$$dT' = b(x) \cdot dm' \cong b(x) \cdot \frac{2x \cdot dx}{X^2} \quad (6.12)$$

The light loading thrust integral is :

$$T' = \int_0^X b(x) \cdot dm' = \frac{1}{X^2} \int_0^X b(x) \cdot 2x \cdot dx \quad (6.13)$$

If $b(x)$ is constant over the whole blade, then so is a , and we have :

$$T' = b \cdot (1+a) \frac{1}{X^2} \int_0^X 2x \cdot dx \quad (6.14)$$

Since the primitive of x is $\frac{1}{2} x^2$, this reproduces (5.8) for the actuator disk :

$$T' = b \cdot (1+a) \quad (6.15)$$

With the light loading assumption, this retrieves (5.9).

Figure 6.2 shows the ring thrust contributions (6.10). The area under the curve is the integral (6.14). For uniform thrust, the graph is a right triangle of area $(1+a)$. Its center of mass is at $\frac{2}{3}X$ from the left. This is by the way not the same as the median of the area at $\frac{1}{2} X\sqrt{2}$, but it is close. We will meet this center of mass in the calculation of the viscous drag.

The curved line is a more typical (normalized) thrust distribution. It retains the nominally triangular shape, but it is modified by local variations of $b(x)$ due to rotational and viscous optimization, and by local variations on $dm'(x)$ due to tip loss. The value of $(1+a)$ is an average over the rings in this general case.

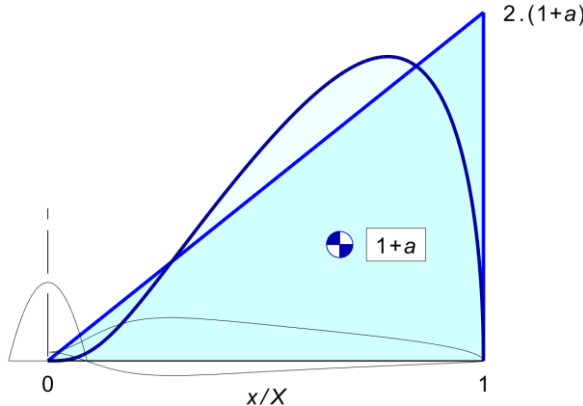


Figure 6.2 : Integrating (6.10) for uniform thrust, and for a

(normalized) more typical thrust distribution.

6.4 Numerical integration

The typical functions in propeller theory are of the form $h(x)$, where $h(x)$ only depends on x , and typically has a teardrop shape which starts with $h(0) = 0$ at the hub and ends with $h(X) = 0$ at the tip. They are often shaped are much like the "typical" thrust line in figure 6.2.

We will often wish to integrate them over the prop disk, like this :

$$H(X) \equiv \int_0^X h(x) \cdot \frac{2x \cdot dx}{X^2} \quad (6.16)$$

The effective local disk area dm' always contains a term $2x \cdot dx/X^2$, so we show it separately here, but that is not even necessary to the argument.

Many of these functions are not as easily integrated as (6.14), and in these cases we will use numerical integration. Once $h(x)$ is available (6.16) is easily integrated numerically on a laptop. The integral over the infinitesimal width dx becomes a sum over finite widths $\Delta x = X/N$:

$$H(X) \cong \frac{1}{X \cdot N} \sum_0^N h(x) \cdot 2x \quad (6.17)$$

This expression sums over $N+1$ points, and divides by N rings. With the end values of $h(x)$ both zero, this is equivalent to the trapezoidal rule.

A typical value for N used to be $N = 10$ in the days of the slide rule, and the result was often improved by SIMPSON's weighting rule with end correction. Today, $N = 10,000$ is more common and (6.17) is adequate.

6.5 A note on graph axes

In graphs however, the inner half of the blade tends to take up as much attention as the outer half. This is a bit misleading.

There is a remedy for this which we will not be using, but which is of enough interest to mention. It is to plot all variables (including thrust and losses) not against x , but against x^2 . This scale makes the width of the "bars" for each ring proportional to the ring area, while the height of the bars remains unchanged. The area under the graph will now be a visually correct representation of the integral.

Another way to understand this is by noting that $2x \cdot dx$ is the derivative of x^2 , so an integral like $\int F \cdot 2x \cdot (dx)$ can also be interpreted as $\int F \cdot (dx^2)$.

7 Rotational loss

7.1 Introduction

So far we discussed a purely axial momentum theory. But the blades of a propeller also cause a rotation of the wake, and this causes an increase in the momentum loss. This increase is a factor on the axial loss. We could call it the rotational overhead on the momentum loss. The rotational overhead depends only on the local pitch angle ϕ .

This chapter discusses the overhead without regard to tip loss. We will see later that the analysis also holds true in the case of a finite number of blades. We will first calculate the rotational loss, then optimize it.

7.2 The momentum loss ratio with rotation

The previous chapter subdivided the prop disk into a nested assembly of independent hollow flow tubes, intersecting the prop disk in concentric rings. We consider the conditions in one of these tubes.

Figure 2.3 already showed how the blades meet the air at a local aerodynamic pitch angle ϕ . By the very definition of lift, the blade lift is orthogonal to the local velocity vector W . The local lift dL is tilted relative to the direction of flight by the same pitch angle. Figure 7.1 shows the basic force diagram. The thrust is the forward component of the lift:

$$dT = dL \cdot \cos \phi \quad (7.1)$$

In the actuator disk, the pressure force pointed straight to the rear, causing a purely axial induced velocity v at the prop disk. With the lift tilted by the angle ϕ , by NEWTON's law of action and reaction the velocity increment is tilted by the angle ϕ .

We will call the induced velocity at the prop disk v' . Figure 7.2 shows the velocity diagram. The induced velocity is orthogonal to the local blade velocity W' . Therefore, v' is tilted relative to the flying velocity V . In lightly loaded propellers, v' is much smaller than W' . This makes W and W' almost the same length, and v' nearly orthogonal to both.

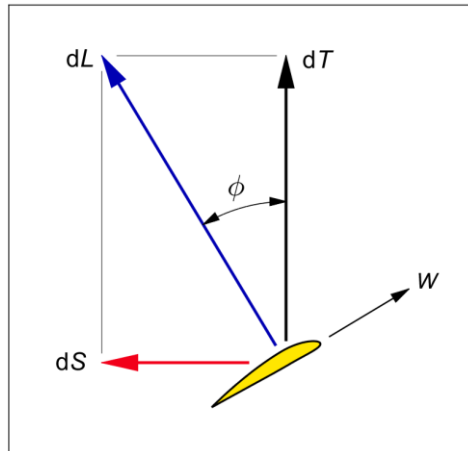


Figure 7.1 : Force diagram for a ring.

All of the values in the diagram are vectors and should formally be printed in boldface. We will use regular font to indicate the length of the vector. The length of the axial component of the induced velocity at the prop disk is v . We will call the rotational component u . We approximate ϕ' by ϕ , which is in keeping with the light loading assumption. From the diagram we have:

$$v = v' \cdot \cos \phi \quad (7.2)$$

and so :

$$v' = \frac{v}{\cos \phi} \quad (7.3)$$

The axial induction ratio w in the far wake is twice that at the prop disk. We use the same notation with the prime as in v' at the disk, for the tilted induced velocity w' in the far wake :

$$w' = \frac{w}{\cos \phi} \quad (7.4)$$

With the definition $w \equiv b \cdot V$ from (4.1), the kinetic energy dP_m left behind in the far wake of a ring becomes :

$$dP_m = \frac{1}{2} \cdot dm^* \cdot (w')^2 = \frac{1}{2} \cdot dm^* \cdot V^2 \cdot \frac{b^2}{\cos^2 \phi} \quad (7.5)$$

Dividing by the **local thrust** (6.10) gives us the local momentum loss ratio for any induction ratio $b(x)$ for any value of dm^* :

$$\frac{dP_m'}{dT'} = \frac{dP_m}{dT \cdot V} = \frac{1}{2} \cdot \frac{dm^* \cdot V^2 \cdot \frac{b^2}{\cos^2 \phi}}{dm^* \cdot V^2 \cdot b} = \frac{1}{2} \cdot \frac{b(x)}{\cos^2 \phi} \quad (7.6)$$

This local momentum loss ratio depends on the local induction ratio, but not on the mass flow through the ring, since the factor dm^* drops out of (7.6).

Compared to the axial loss ratio (4.16), the rotation increases the momentum loss by a local factor of $1/\cos^2 \phi$, for any local axial induction ratio $b(x)$. The ring efficiency becomes:

$$d\eta_m = \frac{1}{1 + \frac{dP_m'}{dT'}} = \frac{1}{1 + \frac{1}{2} \cdot \frac{b(x)}{\cos^2 \phi}} \quad (7.7)$$

We loosely use the symbol $d\eta_m$ for the local momentum efficiency, but we note that such local ratios and efficiencies cannot be integrated over the rings like the overall thrust and power can.

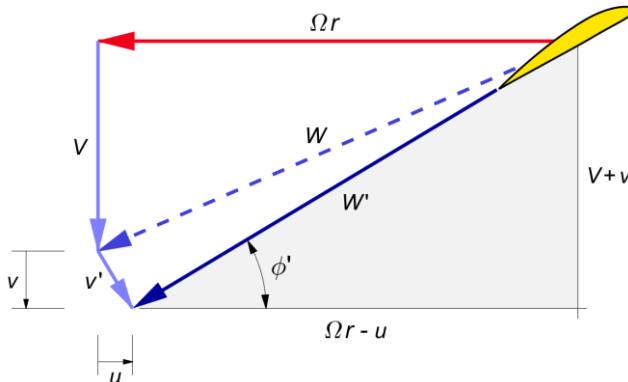


Figure 7.2 : Velocity diagram for a ring.

7.3 Integrating the rotational power loss

For the overall efficiency, we cannot integrate the power loss *ratio* (7.6) over the propeller. As a counterexample, this will not work if half the propeller has a nice power loss ratio of maybe zero, but there is no thrust on that half.

We will have to separately integrate the power and the thrust over the propeller, and find the ratio of those two overall values. We do the integration in non-dimensional form. We can make (7.5) non-dimensional by the definition of a non-dimensional power (3.3) :

$$dP_m' = \frac{1}{2} \cdot \frac{dm^* \cdot V^2}{\rho V^3 \cdot \pi R^2} \cdot \frac{b^2}{\cos^2 \phi} \quad (7.8)$$

Following the steps from (6.5) to (6.8), for lightly loaded propellers we have :

$$dA' = \frac{dm^* \cdot V^2}{\rho V^3 \cdot \pi R^2} \cong \frac{2x \cdot dx}{X^2} \quad (7.9)$$

This turns (7.8) into :

$$dP_m' = \frac{1}{2} \cdot \frac{b(x)^2}{\cos^2 \phi} \cdot \frac{2x \cdot dx}{X^2} \quad (7.10)$$

The integral is :

$$P_m' = \int dP_m' = \frac{1}{X^2} \int_0^X \frac{1}{2} \cdot \frac{b(x)^2}{\cos^2 \phi} \cdot 2x \cdot dx \quad (7.11)$$

This is the generic, non-dimensional, overall momentum loss for a lightly loaded propeller including rotation, at an arbitrary induction ratio distribution $b(x)$.

The matching thrust integral was given by (6.13). The ratio of the two integrals yields the overall loss ratio, which is P_m'/T' by (3.9) :

$$\frac{P_m'}{T'} = \frac{\int dP_m'}{\int dT'} = \frac{\int_0^X \frac{1}{2} \cdot \frac{b(x)^2}{\cos^2 \phi} \cdot 2x \cdot dx}{\int_0^X b(x) \cdot 2x \cdot dx} \quad (7.12)$$

This equation looks a bit daunting, at first, but the next chapter will derive the optimum distribution $b(x)$ which minimizes the rotational loss. As a side benefit, it will also give a very simple solution for (7.11).

8 Optimizing the rotational loss

8.1 Marginal loss

We will now find the thrust distribution which minimizes the rotational loss for a given overall thrust, and its local and overall efficiency by the method of "marginal loss".

The rotational momentum loss ratio (7.6) and the efficiency (7.7) vary along the blade with the local pitch angle. Intuitively, we can move thrust from a radius with low efficiency to one with a high efficiency, making the whole propeller more efficient.

We see by inspection of (7.7) that we can get the same efficiency everywhere by scheduling $b(x)$ with $\cos^2 \phi$, and we will find below that this is in fact the optimal solution. But having the same efficiency everywhere is not formal proof of optimality. The proof requires that we make the *marginal* efficiency the same everywhere. This is not the same thing.

A small change in the local thrust dT , or better in the local net power contribution $dT.V$, will cause a small change in the local power loss. The ratio between the extra power loss and the extra net power increase is called the *marginal* loss. If the derivative $d(dP)/d(dT.V)$ is not the same everywhere, we gain efficiency by taking increments of thrust away from locations with high marginal loss, and moving them to locations with low marginal loss. We will stop moving thrust only when the derivative has reached the same value everywhere, i.e. when the *marginal* loss ratio $d(dP)/d(dT)$ has the same value everywhere. We will call this constant marginal loss ratio C :

$$\frac{d(dP_m)}{d(dT.V)} = C \quad (8.1)$$

The ring thrust dT is the variable driving the change. We could express dP_m as a function of dT and take the derivative with respect to dT directly in (8.1). But it is easier to use the local value of the induction ratio $b(x)$ as the variable driving both dT and dP_m . For ease of notation we leave off the (x) argument for a moment, but we note that each ring has its own value of b .

The marginal loss becomes : **Dit kan ook met P' , T' en m' .**

$$\frac{d(dP_m)/db}{d(dT.V)/db} = C \quad (8.2)$$

Substituting (7.5) for P_m and (4.9) for $T.V$ yields:

$$\frac{\frac{d}{db} \left(dm^* . V^2 . \frac{1/2 b^2}{\cos^2 \phi} \right)}{\frac{d}{db} (dm^* . V^2 . b)} = C \quad (8.3)$$

For heavy loading, dm^* contains a term $(1 + 1/2 b)$, via the term $(V + v)$ in **(Error! Reference source not found.** which is equal to $V . (1 + a) = V . (1 + 1/2 b)$. This gives terms in b^3 in the power and b^2 in the thrust, which will appear in the derivatives. That is not a problem in itself, but it adds terms to the derivation which somewhat obscure the main argument. For clarity, we will therefore use the light loading assumption.

In that case the dm^* of (Error! Reference source not found. does not depend on b . It can be taken outside the derivatives where it will cancel between the numerator and the denominator. This simplifies the marginal loss for light loading to:

$$C = \frac{d(dP_m)}{d(dT.V)} = \frac{\frac{d}{db} \left(\frac{\frac{1}{2} b^2}{\cos^2 \phi} \right)}{\frac{d}{db} \left(\frac{b}{\cos^2 \phi} \right)} = \frac{b}{\cos^2 \phi} \quad (8.4)$$

For optimality the marginal loss C must be a constant. As a result, the axial induction ratio b will have to vary along the blade :

$$b(x) = C \cdot \cos^2 \phi \quad (8.5)$$

8.2 The shape of the reduction factor

The rotationally optimal induction (8.5) contains a factor which we will meet in many places in the theory of propellers. We will give it a name, but we will usually write it out in full :

$$g(x) = \cos^2 \phi \quad (8.6)$$

By (8.5), $g(x)$ is a reduction factor on the local induction C relative to an ideal actuator disk of uniform induction C . The factor is always smaller than 1, since the cosine is always smaller than 1. With (2.6) we can also express $\cos^2 \phi$ directly in x :

$$g(x) = \frac{x^2}{1 + x^2} \quad (8.7)$$

For $x = 1$, the fraction equals $\frac{1}{2}$. This fits with the $\cos^2 \phi$ formulation, since the local pitch angle at $x = 1$ is $\phi = 45^\circ$, and $\cos(45^\circ) = 1/\sqrt{2}$.

For large x the fraction goes to 1 at a rate of $1/x^2$:

$$\frac{x^2}{1 + x^2} = 1 - \frac{1}{1 + x^2} \cong 1 - \frac{1}{x^2} \quad (8.8)$$

Figure 8.1 shows the shape of (8.5) on the x scale. On a finite propeller blade, x will vary from $x = 0$ at the root to X at the tip.

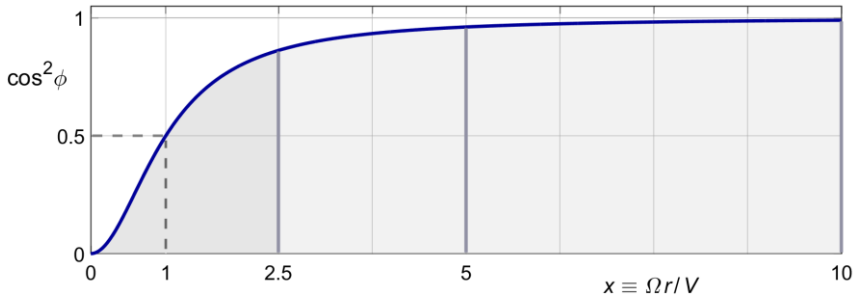


Figure 8.1 : The rotationally optimal induction factor (8.6).

8.3 The efficiency for a given C

Having found the marginal loss C for the lightly loaded propeller, we are also interested in the *actual* loss ratio. Substituting the rotationally optimal distribution of (8.5) into the local loss ratio (7.6), the $\cos^2\phi$ terms cancel, to yield:

$$\frac{dP'_m}{dT'} = \frac{1}{2} C \quad (8.9)$$

The actual local loss ratio differs from the marginal one in (8.4) by a factor of $\frac{1}{2}$. This is not a coincidence. **Same in uniform loss.** It is the natural ratio between a quadratic function and its derivative, and the marginal momentum loss is the derivative of the local momentum loss (7.5), which is quadratic in the thrust.

The local momentum loss ratio being constant at every radius, the overall loss ratio will have the same value:

$$\frac{P'_m}{T'} = \frac{1}{2} C \quad (8.10)$$

By the efficiency equation (3.8), the overall momentum efficiency becomes:

$$\eta_m = \frac{1}{1 + \frac{1}{2} C} \quad (8.11)$$

We now have an expression for the optimal momentum efficiency of the propeller which looks a lot like the efficiency (4.18) of a uniform actuator disk with a constant induction of $b(x) = C$.

But by (8.5), for the same efficiency (i.e. for the same overall loss ratio $\frac{1}{2}C$), the thrust of this propeller will be less than that of the actuator disk.

9 The efficiency in design parameters

9.1 Integrating the optimal thrust

For a given marginal loss C , (8.5) will reduce the local induction everywhere along the blade. We are interested in the average thrust reduction over the whole prop disk. For this we need to integrate the thrust over the rings. With (8.5) as the optimal induction and (6.8) for dm' , both valid for light loading, the local thrust from (6.12) becomes :

$$dT' = b(x) \cdot dm' \cong C \cdot \cos^2 \phi \cdot \frac{2x \cdot dx}{X^2} \quad (9.1)$$

The integral for the overall thrust T' is:

$$T' = C \cdot \int_0^X \cos^2 \phi \cdot \frac{2x \cdot dx}{X^2} \quad (9.2)$$

This integral is the weighted average of $\cos^2 \phi$ over the prop disk. Its value is always smaller than one. From here on we will call such overall thrust reduction factors κ (Greek letter kappa), in a notation due to THEODORSEN. Since the rotational reduction factor depends only on ϕ , we will call this factor κ_ϕ :

$$\kappa_\phi(X) \equiv \frac{1}{X^2} \int_0^X \cos^2 \phi \cdot 2x \cdot dx \quad (9.3)$$

But with (8.7) we can write it directly in x :

$$\kappa_\phi(X) = \frac{1}{X^2} \int_0^X \frac{x^2}{1+x^2} \cdot 2x \cdot dx \quad (9.4)$$

The integral can be solved analytically by a substitution of $u = 1 + x^2$. This gives $x^2 = (u - 1)$ and $du/dx = 2x$, from which $2x \cdot dx = du$:

$$\frac{1}{X^2} \int_0^X \frac{x^2}{1+x^2} \cdot 2x \cdot dx = \frac{1}{X^2} \int_1^{1+X^2} \left(1 - \frac{1}{u}\right) \cdot du \quad (9.5)$$

With $\int 1/u = \ln u$ this yields the classical result for κ_ϕ :

$$\kappa_\phi(X) = 1 - \frac{\ln(1+X^2)}{X^2} \quad (9.6)$$

The value of κ_ϕ depends only on the tip speed ratio X .

Figure 9.1 shows the graph of (9.6). This figure gives the *overall* thrust reduction, plotted against the *tip* speed ratio X . The shape is essentially different from the local reduction shown against the local radius x in figure 8.1.

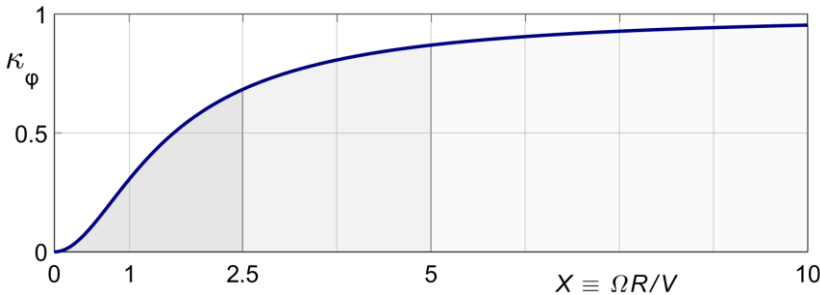


Figure 9.1 : The average thrust reduction factor $\kappa_\phi(X)$.

9.2 The rotational reduction in design parameters

With κ_ϕ now known, we can rewrite (9.2) as:

$$T' = C \cdot \kappa_\phi \quad (9.7)$$

which yields:

$$C = \frac{T'}{\kappa_\phi} \quad (9.8)$$

The local axial induction ratio (8.5) expressed in design parameters is :

$$b(x) = T' \cdot \frac{\cos^2 \phi}{\kappa_\phi} \quad (9.9)$$

9.3 The rotational efficiency in design parameters

Equation (9.8) expresses the marginal loss C in the design parameters T' and X , the latter via $\kappa_\phi(X)$. From (8.9) to (8.11) this also gives us the momentum loss and efficiency in the design parameters :

$$\frac{dP'_m}{dT'} = \frac{1}{2} \frac{T'}{\kappa_\phi} \quad (9.10)$$

$$\frac{P'_m}{T'} = \frac{1}{2} \frac{T'}{\kappa_\phi} \quad (9.11)$$

$$\eta_m = \frac{1}{1 + \frac{1}{2} \frac{T'}{\kappa_\phi}} \quad (9.12)$$

9.4 The approximate inverse of the reduction

Figure 9.1 gave the graph of the thrust reduction factor κ_ϕ as a function of the tip speed ratio X . This factor is always smaller than 1, which gives it the feel of an efficiency. But that is not what it is. It is really a fill factor on the thrust over the prop disk.

The actual efficiency is (9.12). If the thrust loading T' is small, then the efficiency is high even when the fill factor κ_ϕ is small. It is more helpful to say that $1/\kappa_\phi$ is an overhead factor on the thrust loading, and therefore an overhead factor on the momentum loss.

We get a better intuition of the rotational loss by studying $1/\kappa_\phi$ rather than κ_ϕ itself. Comparing (9.12) to (5.11) for the ideal actuator disk, the rotation increases the momentum loss by a factor of $1/\kappa_\phi$.

Figure 9.2 gives the graph of $1/\kappa_\phi$ against X . The graph is easily calculated from (9.6), but that equation is not very intuitive. A simpler formula can give a better feel. One such formula is :

$$\frac{1}{\kappa_\phi(X)} \cong 1 + \left(\frac{\pi}{2 \cdot X} \right)^{1.7} \quad (9.13)$$

This approximation is merely a numerical fit. The factor $\pi/2$ has no physical meaning. It is more of an in-joke: the numerical value of the parameter just happens to be close to 1.57.

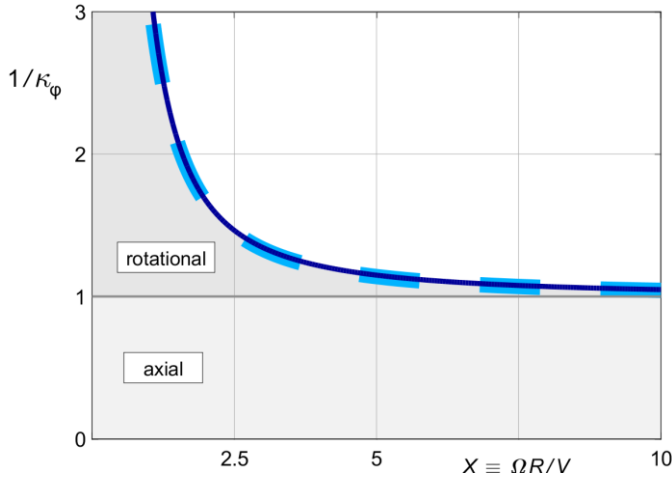


Figure 9.2 : The rotational overhead factor $1/\kappa_\phi$ and its approximation (9.13).

Figure 9.2 shows the exact solution by (9.6) as a solid line, and the approximation (9.13) as a dashed line. The approximation is very close.

By (9.13), the rotational overhead is slightly less than quadratic in $1/X$. For large tip speed ratios the rotational loss is negligible: in the limit for $X \rightarrow \infty$, (9.13) goes to 1, and the propeller approaches the axial efficiency (5.11). In our standard example of $X = 5$, the rotational loss overhead is still limited. Below $X = 2.5$ however, it does become very significant.

10 Inflating the shape to the design thrust

10.1 From thrust reduction to thrust redistribution

By (8.5), the thrust for a rotationally optimal propeller was reduced by the factor $g(x) = \cos^2\phi$, relative to an actuator disk of the same efficiency. The factor reduces the local thrust, and therefore in the end also the overall thrust.

By (9.9), the thrust can be increased back to its original design value T' by increasing the marginal loss C by a factor $1/\kappa_\phi$, and by (8.11) this increases the momentum loss by the same factor.

We can say that (8.5) gives us the thrust reduction for equal efficiency, and that (9.9) gives us the thrust redistribution for equal thrust.

Figure 10.1 repeats figure 8.1, but this time for $\cos^2\phi/\kappa_\phi$. This normalizes the reduction factor to an average of 1 over the whole blade, or in other words, it replaces the reduction by a redistribution around an average of 1. The shapes in figure 10.1 all have the same unit thrust, for their particular tip speed ratio X .

Figures 8.1 and 10.1 both show a memorable fact : all rotationally optimal propellers are identical in shape. The ones with lower tip speed ratio X are just cut off shorter.

As a preparation for the later addition of tip loss in (16.3), we mention that before tip loss, using (9.9) in the light loading thrust (6.12) yields :

$$dT' = b(x) \cdot dm' \cong T' \cdot \frac{\cos^2\phi}{\kappa_\phi} \cdot \frac{2x \cdot dx}{X^2} \quad (10.1)$$

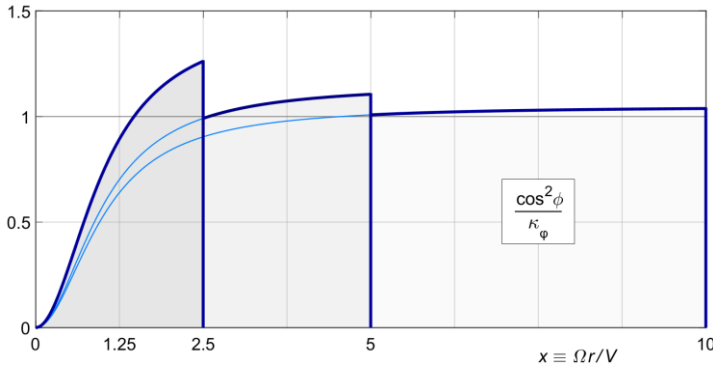


Figure 10.1 : The optimal induction factor inflated to unit thrust.

11 Blade chords

11.1 Blade chord from lift

Equation (9.9) gives the optimum thrust distribution including rotation. We will derive the corresponding blade chord. This chord does not yet include tip loss and viscous optimization. We will add those in later chapters.

Equation (**Error! Reference source not found.**) and (**Error! Reference source not found.**) gave the thrust on a ring. For light loading we have $(V+v) \cong V$, and so:

$$dT = dm \cdot V. b(x) = b(x) \cdot \rho V^2 \cdot 2\pi r \cdot dr \quad (11.1)$$

The aerodynamic lift on the blade elements inside the ring must provide this thrust. The lift coefficient C_L for a wing of area S is defined by:

$$dL = C_L \cdot \frac{1}{2} \rho V^2 \cdot S \quad (11.2)$$

The blade area inside a ring is $dS = B \cdot c \cdot dr$, coming from B strips width dr and chord c . The chord is defined in the direction of the local wind velocity W , as shown in the velocity diagram of figure 7.2. The lift generated by the total blade area dS within the ring is:

$$dL = C_L \cdot \frac{1}{2} \rho W^2 \cdot B \cdot c \cdot dr \quad (11.3)$$

Also from figure 7.2, the local wind velocity W equals $V/\sin\phi$:

$$dL = C_L \cdot \frac{1}{2} \rho \frac{V^2}{\sin^2 \phi} \cdot B \cdot c \cdot dr \quad (11.4)$$

And from the force diagram in figure 7.1, the thrust component of dL is:

$$dT = dL \cdot \cos \phi \quad (11.5)$$

We now have (11.1) for the ring thrust needed to generate an induction ratio $b(x)$, and (11.5) for the ring thrust aerodynamically available from the blade elements to do this. The two must be equal:

$$\rho \cdot 2\pi r \cdot dr \cdot b \cdot V^2 = C_L \cdot \frac{1}{2} \rho \frac{V^2}{\sin^2 \phi} \cdot B \cdot c \cdot dr \cdot \cos \phi \quad (11.6)$$

Dividing out the common factors ρV^2 and dr , we have:

$$\frac{B \cdot c}{2\pi r} = \frac{2}{C_L} \cdot b(x) \cdot \sin \phi \cdot \tan \phi \quad (11.7)$$

The left hand term represents the fraction of the ring that is "covered by chord". We will call it the local solidity, but some care is needed, because "solidity" goes under many other definitions in the literature.

11.2 The non-dimensional solidity

Equation (2.9) defines $x \equiv \Omega r/V$ as the non-dimensional version of the radius r . We similarly define the non-dimensional chord c' :

$$c' \equiv \Omega c/V \quad (11.8)$$

The ratio between c and r stays the same in the non-dimensional form. This will allow us to draw the blade planform (11.7) of c versus r in the same "true" proportions on a scale of c' versus x :

$$\frac{c}{r} = \frac{c'}{x} \quad (11.9)$$

The non-dimensional local solidity stays the same as in (11.7):

$$\frac{B \cdot c'}{2\pi x} = \frac{2}{C_L} \cdot b(x) \cdot \sin \phi \cdot \tan \phi \quad (11.10)$$

With (2.3) and (2.4) we can express this completely in x :

$$\frac{B \cdot c'}{2\pi x} = \frac{2}{C_L} \cdot b(x) \cdot \frac{1}{\sqrt{1+x^2}} \cdot \frac{1}{x} \quad (11.11)$$

We will draw the blade shapes in proper proportion on the non-dimensional scale. With x on the horizontal axis, we have for the chord:

$$B \cdot c' = \frac{4\pi}{C_L} \cdot b(x) \cdot \sin \phi = \frac{4\pi}{C_L} \cdot b(x) \cdot \frac{1}{\sqrt{1+x^2}} \quad (11.12)$$

11.3 The blade shape factor for uniform loading

For uniform loading, (5.10) gives us the constant induction $b(x) = T'$. This turns (11.11) into:

$$B \cdot c' = \frac{4\pi}{C_L} \cdot T' \cdot \sin \phi = \frac{4\pi}{C_L} \cdot T' \cdot \frac{1}{\sqrt{1+x^2}} \quad (11.13)$$

The non-dimensional blade shape is proportional to $\sin \phi$. We will call $\sin \phi$ the chord shape factor $f_{c,1}$ for uniform loading:

$$f_{c,1} = \sin \phi = \frac{1}{\sqrt{1+x^2}} \quad (11.14)$$

Figure 11.1 shows this factor against x , as a red line.

For larger values of x , this line approaches $1/x$, since $\sqrt{1+x^2}$ tends to $1/x$ for larger x . Near the hub, $\phi \cong 90^\circ$ and $\sin \phi \cong 1$. The non-dimensional chord factor becomes 1. Dividing the chord by $2\pi x$ to get the "cover factor" on each ring shows that the local solidity will go to infinity near the hub. The propeller blades become layered like a Da Vinci helicopter, and in the end the chords wrap around the hub infinitely many times, for an infinitely small radius.

Such shapes have actually been attempted in the past, but they are both impractical and unnecessary. In the rotationally optimal load distribution, the $\cos^2 \phi$ term comes to the rescue.

11.4 The blade shape factor including rotation

Rotational optimization will scale the thrust by $\cos^2 \phi$. With the light loading $b(x)$ of (9.9), the non-dimensional chord (11.12) becomes:

$$B \cdot c' = \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_\phi} \cdot \cos^2 \phi \cdot \sin \phi \quad (11.15)$$

With (2.6) for $\cos^2 \phi$, the blade shape factor becomes:

$$f_{c,2} = \cos^2 \phi \cdot \sin \phi = \frac{x^2}{1+x^2} \cdot \frac{1}{\sqrt{1+x^2}} \quad (11.16)$$

Figure 11.1 shows this factor as a solid blue line.

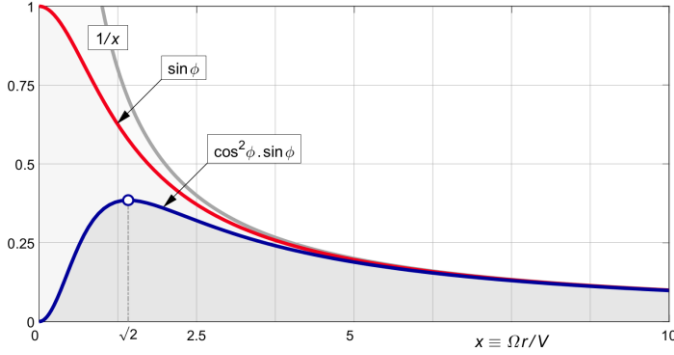


Figure 11.1: Chord shape factors (11.14) and (11.16).

For large x , where $(1+x^2) \cong x^2$, the blade shape factor is approximately $f_{c,2} \cong 1/x$ just like in (11.14) for the uniform propeller.

For small x we have $(1+x^2) \cong 1$ and this yields $f_{c,2} \cong x^2$. The local solidity (11.15) divides this by a factor of $1/x$, but even then the result does not go to infinity. The local solidity in fact goes to zero at the hub, at a rate of $1/x$. Unlike the uniformly loaded propeller, the rotationally optimal propeller has a viable blade shape. In practice, the chord will not go completely to zero near the hub, for mechanical reasons.

The location of the widest chord is found by setting the derivative of (11.16) to zero. This yields $\tan^2 \phi = 1/2$, hence $x = \sqrt{2}$:

$$x_{\text{wide}} = \sqrt{2} \quad (11.17)$$

The chord factor at this point is :

$$f_{c,2 \text{ max}} = \frac{2}{3} \cdot \sqrt{\frac{1}{3}} \cong 0.385 \quad (11.18)$$

Figure 11.1 indicates the widest point by a circle marker.

11.5 Inflating the chord factor to unit thrust

To normalize the thrust, we can make the factor $1/\kappa_\phi$ in (11.15) part of the blade shape factor:

$$f_{c,3} = \frac{1}{\kappa_\phi} \cdot \cos^2 \phi \cdot \sin \phi \quad (11.19)$$

Figure 11.2 shows the result. As for the thrust in figure 10.1, all rotationally optimal propellers are identical in shape, since the shape depends only on x . Blades of smaller X are just "cut off" sooner, at the desired tip speed ratio X .

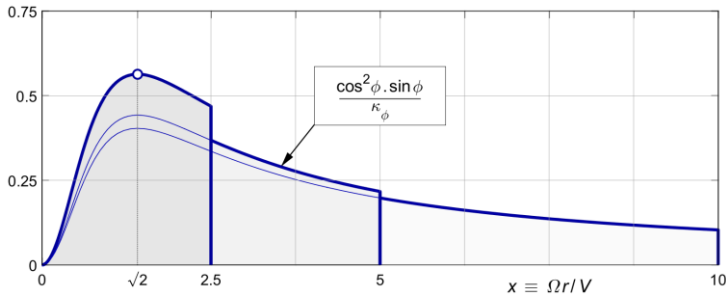


Figure 11.2: Chord shape factor (11.19).

11.6 The dimensional chord

Figures 11.1 and 11.2 showed the non-dimensional chord c' against x . The non-dimensional radius runs from 0 to X . This makes it hard to compare the shapes and the relative widths of the blades, since the blade planforms extend over different tip radius lengths X .

We get a better impression of the shape variation if we plot all blades to a standard radius from 0 to 1. We achieve this by plotting c/R versus r/R . Relative to figure 11.2, we need to reduce both the horizontal and the vertical axis by $1/X$. This changes (11.15) to :

$$B \cdot \frac{c}{R} = B \cdot \frac{c'}{X} = \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_\phi} \cdot \frac{1}{X} \cdot \cos^2 \phi \cdot \sin \phi \quad (11.20)$$

Figure 11.3 shows the result for two-bladed propellers of various X . This figure is basically the same as figure 11.2, but this time with all blades scaled to a single radius. This new plot is dimensioned for our typical lightplane example of $T' = 0.1$ and $C_L = 0.4$.

By (11.17) the widest chord is located at $r/R = \sqrt{2}/X$, and by (11.18) the maximum width is :

$$B \cdot \frac{c_{\max}}{R} = \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_\phi} \cdot \frac{0.385}{X} \quad (11.21)$$

These points are marked by a circle marker. Apart from the secondary effect of $1/\kappa_\phi$, all else being equal, the widest chord varies with $1/X$.

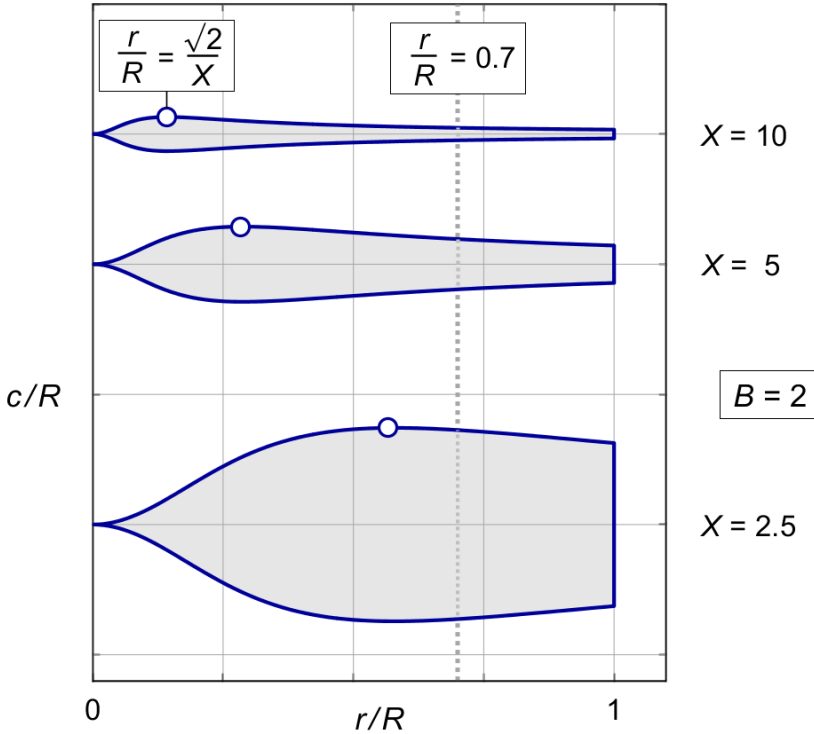


Figure 11.3: Rotationally optimal chords for varying X .
Two blades, thrust $T' = 0.1$, $C_L = 0.4$.

11.7 The chord at the tip

At the tip, we have $x = X$. Expressing $f_{c,2}$ in x like in (11.16), but substituting X for x , equation (11.20) now reads:

$$B \cdot \frac{c_{\text{tip}}}{R} = \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_\phi} \cdot \frac{1}{X} \cdot \frac{X^2}{1+X^2} \cdot \frac{1}{\sqrt{1+X^2}} \quad (11.22)$$

For larger X we have $(1+X^2) \cong X^2$, and we can approximate the tip chord by:

$$B \cdot \frac{c_{\text{tip}}}{R} \cong \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_\phi} \cdot \frac{1}{X^2} \quad (11.23)$$

The tip chord is therefore roughly proportional to $1/X^2$.

11.8 The chord at $0.7R$

The tip is not always the most convenient place to measure the chord. The typical working radius for a propeller blade is $r = 0.7R$, or $x = 0.7X$.

If the value of $0.7X$ is fairly large relative to 1, the same logic applies as for the tip chord, and we have:

$$B \cdot \frac{c_{0.7}}{R} \cong \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_\phi} \cdot \frac{1}{0.7X^2} \quad (11.24)$$

The chord at 70% of the radius is roughly proportional to $1/X^2$. Higher pitch propellers therefore have much wider blades, as was already obvious from figure 11.3.

11.9 Blade taper

With (11.21) the widest chord at $x = \sqrt{2}$ is proportional to $1/X$, and by (11.23) and (11.24) the chords farther out are proportional to $1/X^2$. Figure 11.3 shows how it works out in practice. Propellers of high pitch (hence of low X) have paddle-like shapes, with the tips nearly as wide as the widest point at $x = \sqrt{2}$. Propellers of high X are clearly tapered. Chapter 23 will show that this result holds even after tip loss.

The reason for the chord variation is the variation in the local airspeed on the blade elements. The velocity diagram of figure 2.3 shows that for larger values of $x \equiv \Omega r/V$, the local airspeed W is very similar to Ωr . This makes W proportional to x , with $W \cong x \cdot V$. The lift on a blade element is proportional to the local airspeed squared. All else being equal, this makes the lift per chord width locally proportional to x^2 , and at the tip to $1/X^2$. This ratio is reflected in the tip chord width of (11.23).

For any local radius, the lift is also proportional to the square of the local circumferential speed Ωr . This makes it proportional to r^2 . This lift has to accelerate the air in a local ring of area $2\pi r \cdot dr$, which is proportional only to r . For a given X , the local chord therefore varies with $1/x$ relative to the tip chord (OR RELATIVE TO THE WIDEST CHORD TIMES A FACTOR?).

12 Betz's Archimedes screw

12.1 Introduction

We will mention here an interesting corollary to the rotationally optimal induction distribution. It is known as BETZ'S ARCHIMEDES screw wake. While this shape may seem as not much more than a geometrical coincidence, it is used extensively in the later tip loss calculations.

12.2 The rotating, telescoping wake

The induction velocity v' as shown in figure 7.2 is not purely axial. It has a circumferential (tangential, sideways) component too. The axial and tangential components can both be expressed as functions of twice the pitch angle, the "double angle" 2ϕ .

The axial reduction equals $\cos^2\phi$. From trigonometry, we know that $\cos^2\phi = \frac{1}{2} + \frac{1}{2}\cos 2\phi$.

The tangential component equals the axial component $\cos^2\phi$ multiplied by $\tan\phi$. From trigonometry we know that $\tan\phi \cdot \cos^2\phi = \sin\phi \cdot \cos\phi$, and $\sin\phi \cdot \cos\phi = \frac{1}{2}\sin 2\phi$.

Figure 12.1 shows that by these two equations, the ends of the induced velocity vector v' lie on a circle with radius $\frac{1}{2}C$ and phase 2ϕ . Far from the hub, the local pitch angles ϕ go to zero. The induced velocity points almost straight aft (down in the picture), and its length is nearly C . At the radius $x = 1$, the induced velocity points 45° sideways and its length is $\frac{1}{2}\sqrt{2}C$. Very near to the hub, v' is almost completely tangential and its length is almost zero.

From this picture, the flow tube must be filled with hollow, telescoping tubes of air, which both rotate and slide relative to each other. This inconvenient truth is why the independence of the rings is an essential assumption in the theory of optimal propellers.

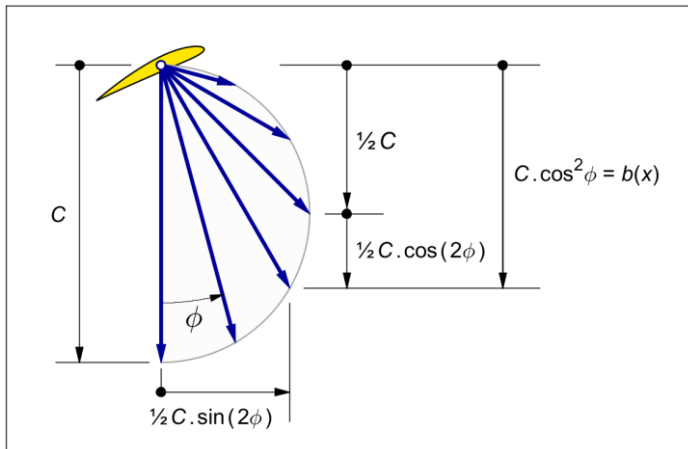


Figure 12.1 : The thrust reduction vector.

12.3 Betz's apparent Archimedes screw

BETZ was the first to point out that his optimal velocity distribution results in a wake shaped like a regular rigid ARCHIMEDES screw, which seems to drift straight to the rear without rotating.

From the previous section we know that in reality, each ring has its own rearward velocity and its own rate of rotation.

The apparent lack of rotation of the ARCHIMEDES screw is due to an optical effect called the "barber pole illusion". Figure 12.2 shows how the apparent rearward velocity v'' comes about. The sheet surface *seems* to move straight to the rear with v'' if we ignore the fact that it is actually moving in a spiral with v' , and not really sliding backward this fast, but rotating as well.

Interestingly, the efficiency (7.7) of the rotationally optimal propeller matches the purely axial efficiency (4.18) of the actuator disk if we replace v by v'' , or b by $b/\cos^2\phi$. Some texts will give this apparent axial induction a central role in the theory, which it most likely does not deserve.

The rigid ARCHIMEDES wake shape however is used extensively in both PRANDTL's approximate theory of the tip loss, and in the exact vortex solution of that loss, as we will see in later chapters.

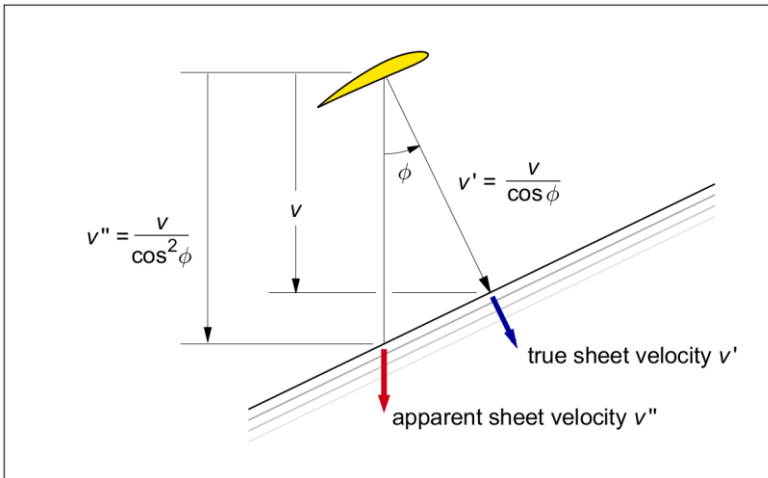


Figure 12.2 : The apparent axial sheet velocity of the spirals, partly due to the barber pole illusion..

13 Viscous loss

13.1 Introduction

Viscous loss is the "leadscrew" part of the power loss. The leadscrew acts in a flow accelerated by the propeller's own thrust, but the screw itself does not know it. It just threads itself forward relative to the surrounding air, even if that air is slipping backward.

At the design stage, the sliding friction force is proportional to the blade lift force. The ratio is called the friction coefficient. It is usually indicated by μ (Greek letter mu). The ratio is the tangent of the angle at which a friction block would begin to slide down a slope.

The aerodynamic equivalent is the inverse of the glide ratio C_l/C_d of the blade sections. At the design stage, we will use the prop blades at their best C_l/C_d , so we can consider the "friction angle" of the blade sections as a constant. In off-design calculations this will no longer hold. In that case the behavior is closer to a constant C_d at varying C_l . There are also effects of REYNOLDS and MACH number which are best solved by numerical methods.

Figure 13.1 shows a horizontal force pushing a block up a slope. On a slope of 1:10, the horizontal force F travels ten times the distance of the vertical (weight) force W . From the power balance, the force F will be ten times lower than the force W , in the ideal case with no friction.

For small angles, the force normal to the surface approximately equals the block weight W , and the friction force almost directly opposes the horizontal driving force F . Force F will have to move the friction force over 10 times the distance of the vertical displacement. This costs 10 times more friction power than if F could have lifted the force W directly. The frictional loss relative to weight is (approximately) inversely proportional to the slope. A more precise analysis in the next section will confirm this.

Most mechanical leadscrews more or less have a single slope angle, for a single outside diameter. Their thread does not cut very deep. In propellers, the "thread" goes right to the core. This causes considerable pitch angle variation with radius. Therefore we will subdivide the prop disk radially into rings, and analyze each radius separately.

Unlike the momentum loss, the frictional efficiency does not depend on the loading. It does depend on the average radius where the load applies, and on the local friction coefficient.

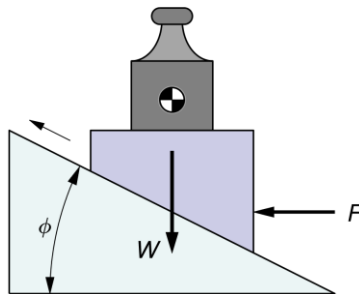


Figure 13.1 : Horizontal force pushing a block up a slope.

13.2 Leadscrew efficiency

We now look at the lift and drag forces on a blade element at radius r . Figure 13.2 shows the force diagram. We assume light loading for the moment, so the actual pitch angle ϕ' is approximately the same as the nominal pitch angle $\phi = \text{atan}(V/\Omega r)$ for the free stream conditions. In more precise calculations, we can substitute the actual ϕ' for ϕ .

The forward velocity of the airplane is V , and the propeller rate of rotation is Ω (in rad/s). The local lift L is orthogonal to the local velocity W . The drag D is that part of the aerodynamic force which is orthogonal to L . The resultant of L and D is called R .

The drag D is an order of magnitude smaller than L . We can interpret the resultant R as a slightly rotated version of L .

The viscous backward angle of tilt ϵ follows from the ratio between D and L . In wing sections this is called the "glide ratio":

$$\tan \epsilon \equiv \frac{D}{L} = \frac{C_d}{C_l} \quad (13.1)$$

We can separate R into an alternative pair of orthogonal components: the thrust and tangential forces dT and dS , with "d" meaning "for a ring", as in **(Error! Reference source not found..** From figure 13.2 we have:

$$dT = R \cdot \cos(\phi + \epsilon) \quad (13.2)$$

$$dS = R \cdot \sin(\phi + \epsilon) \quad (13.3)$$

With (13.2), the net power exerted by the ring on the airplane is:

$$dT \cdot V = R \cdot \cos(\phi + \epsilon) \cdot V \quad (13.4)$$

From figure 13.2, for a blade element dr at radius r the torque Q is:

$$dQ = dS \cdot r \cdot dr \quad (13.5)$$

The power needed to turn the propeller is the shaft torque Q multiplied by the rotation rate Ω in rad/s. With (13.3) for S , the shaft power is:

$$dQ \cdot \Omega = R \cdot \sin(\phi + \epsilon) \cdot \Omega r \cdot dr \quad (13.6)$$

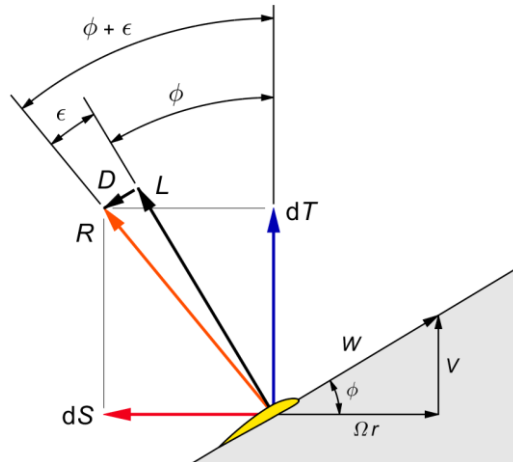


Figure 13.2: Thrust and torque on a blade element.

The viscous efficiency of the blade element is net output power over shaft input power:

$$d\eta_v = \frac{dT \cdot V}{dQ \cdot \Omega} = \frac{\cos(\phi + \epsilon)}{\sin(\phi + \epsilon)} \cdot \frac{V}{\Omega r} \quad (13.7)$$

With $\tan \phi = V/\Omega r$, we have:

$$d\eta_v = \frac{\tan \phi}{\tan(\phi + \epsilon)} \quad (13.8)$$

This formula goes back to FROUDE. It is the standard efficiency equation for mechanical leadscrews with friction angle ϵ .

For zero friction, i.e. for $\epsilon = 0$, the propeller is like an ideal leadscrew. It will convert shaft torque into forward thrust with perfect efficiency. For pitch angles ϕ of $(90^\circ - \epsilon)$ and higher, the efficiency becomes zero. The leadscrew seizes, or "locks up". In propellers this condition occurs close to the hub. We will ignore it, and analyze the efficiency only for moderate angles of ϕ and small angles of ϵ .

13.3 The viscous loss ratio

We find the local power loss ratio from the viscous efficiency. Equating (13.8) to the efficiency definition of (3.8) we have :

$$\frac{1}{1 + \frac{dP_v}{dT \cdot V}} = \frac{\tan \phi}{\tan(\phi + \epsilon)} \quad (13.9)$$

Some reworking gives :

$$\frac{dP_v}{dT \cdot V} = \frac{\tan(\phi + \epsilon)}{\tan \phi} - 1 = \frac{\tan(\phi + \epsilon) - \tan \phi}{\tan \phi} \quad (13.10)$$

We will call the viscous power ratio μ (Greek letter mu), deliberately giving it the feel of a friction coefficient. The analogy is not complete, but we will see later that μ is close to a projected version of the surface friction coefficient ϵ :

$$\mu(\phi) \equiv \frac{dP_v}{dT \cdot V} \quad (13.11)$$

Figure 13.3 shows μ as a function of ϕ . Textbooks on mechanical drive components give a full analysis of (13.8) which we will not repeat here. In the region beyond $\phi = 90^\circ - \epsilon$ the leadscrew is "self locking". The figure shows this region as a shaded vertical bar, for the case of $\epsilon = 0.02$.

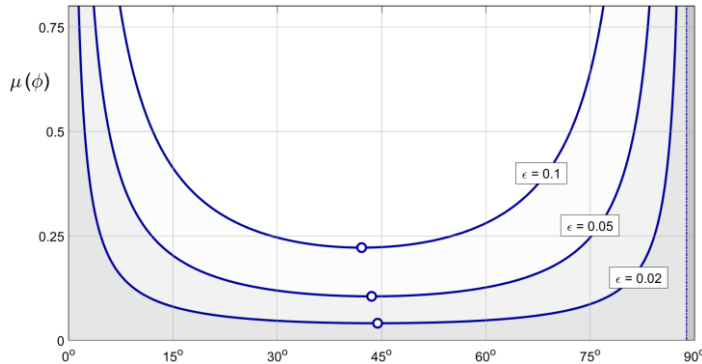


Figure 13.3 : The local leadscrew loss ratio $\mu(\phi)$.

The viscous efficiency is optimal for a pitch angle of $(45^\circ - \frac{1}{2}\epsilon)$. It is left-right symmetrical around this vertical line. At the optimum point, the loss ratio is nearly 2ϵ . If ϵ is small, the optimum angle is close to 45° .

Most of the thrust in a propeller is delivered on the outboard part of the blade at relatively flat pitch angles. Low pitch angles are favorable with respect to rotational momentum loss, but they are the *least* favorable condition with respect to profile drag.

From a viscous point of view, all thrust should really be concentrated in a short paddle near the 45° pitch angle radius. Simple wind turbines with sheet metal blades of poor lift to drag ratio $\epsilon = C_d/C_l$ in fact use exactly this arrangement.

13.4 A near-perfect approximation

Expression (13.10) for the frictional loss ratio can be greatly simplified with almost no loss of precision, by noting that the numerator is a finite difference between two tangents. Since the friction angle ϵ is usually small, we can approximate this difference by ϵ times the derivative of the tangent. That derivative is $1/\cos^2\phi$:

$$\tan(\phi + \epsilon) - \tan\phi \cong \epsilon \cdot \frac{1}{\cos^2\phi} \quad (13.12)$$

This changes (13.10) into :

$$\mu(\phi) = \epsilon \cdot \frac{1}{\cos^2\phi} \cdot \frac{1}{\tan\phi} \quad (13.13)$$

Using $\tan\phi = 1/x$ from (2.3), we have a simple, yet almost exact expression for μ , which is easiest remembered in mixed coordinates ϕ and x :

$$\mu(x) = \frac{\epsilon \cdot x}{\cos^2\phi} \quad (13.14)$$

Eliminating ϕ completely by (2.7), this is the same as :

$$\mu(x) = \epsilon \cdot \left(x + \frac{1}{x}\right) \quad (13.15)$$

Figure 13.4 shows gives the exact value (13.10) of $\mu(x)$ as a solid dark blue line. It is the same as figure 13.3, only skewed because it is against x instead of against ϕ . The figure also shows approximation (13.14) as a dashed line.

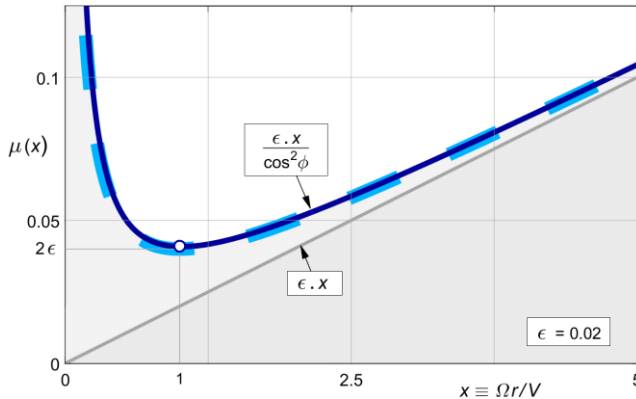


Figure 13.4 : The local leadscrew loss ratio $\mu(x)$.

For all but the smallest values of x , i.e. for all pitch angles except $\phi \cong 90^\circ$, the approximation is nearly perfect. The factor μ reaches a minimum of 2ϵ near $x = 1$, the point where ϕ equals 45° .

Some of the literature, like GLAUERT will set $\cos^2 \phi \approx 1$:

$$\mu(x) \cong \epsilon \cdot x \quad (13.16)$$

Figure 13.4 shows this further simplification as a straight line. It comes closest to an interpretation of μ as the "geared" friction coefficient ϵ .

In the vector diagram of figure 13.2, for small angles ϕ a small driving force S is equivalent to a large thrust force T by a factor of $x = 1/\tan \phi$. In the same way, a small friction force becomes a large drag force.

13.5 Vector derivation of the same approximation

The nearly exact $\mu(x)$ of (13.14) can also be loosely derived directly from the force diagram. We decompose the viscous drag vector D in two orthogonal components, which we shall call ΔdT and ΔdS .

Via the steps of $\Delta dT \cong D \cdot \sin \phi$, $D \cong \epsilon \cdot L$, and $L \cong dT/\cos \phi$, we have :

$$\Delta dT \cong dT \cdot \epsilon \cdot \frac{\sin \phi}{\cos \phi} = dT \cdot \epsilon \cdot \tan \phi \quad (13.17)$$

Similarly, via $\Delta dS \cong D \cdot \cos \phi$, we have :

$$\Delta dS \cong dT \cdot \epsilon \cdot \frac{\cos \phi}{\cos \phi} = dT \cdot \epsilon \quad (13.18)$$

With some hand waving, we can say that if we multiply these two orthogonal drag forces each by the local velocity in their own direction, we will obtain the local power loss due to drag.

The drag component ΔdT directly opposes the flying velocity V , and the drag component ΔdS directly opposes the circumferential blade element velocity Ωr . The total local drag power loss is :

$$dP_v \cong \Delta dT \cdot V + \Delta dS \cdot \Omega r \quad (13.19)$$

Substituting (13.17) and (13.18) and dividing by $dT \cdot V$, the local loss ratio is :

$$\frac{dP_v}{dT \cdot V} = \frac{dT \cdot \epsilon \cdot (\tan \phi \cdot V + \Omega r)}{dT \cdot V} = \epsilon \cdot \left(\tan \phi + \frac{\Omega r}{V} \right) \quad (13.20)$$

With (2.2) we have $\tan \phi = 1/x$ and $\Omega r/V = x$, and so (with (3.9)) :

$$\frac{dP_v'}{dT'} = \frac{dP_v}{dT \cdot V} = \epsilon \cdot \left(\frac{1}{x} + x \right) = \frac{\epsilon \cdot x}{\cos^2 \phi} \quad (13.21)$$

This recovers (13.14). The derivation shows that the linearization (13.16) amounts to looking only at the effect of the viscous drag on the torque, and neglecting its effect on the thrust.

13.6 Neglecting a small momentum term

The power calculation of (13.19) misses a subtle point. The viscous drag component ΔdT of (13.17) detracts from the propeller's thrust, and to maintain the original thrust we need to increase the momentum thrust by that small amount. This will cause a small additional momentum loss. We will discuss this point further in the notes, but following the rest of the literature, in the main text we will neglect this extra loss.

14 Rotationally optimized viscous loss

14.1 Viscous loss in the rotationally optimal propeller

For the rotationally optimal propeller, the momentum loss (**Error! Reference source not found.**) is constant along the blade due to the special loading distribution (9.9). The viscous loss however varies along the blade, and its local loss ratio (13.21) is independent of the load. It only depends on the radius. The combined local loss ratio for the rotationally optimal propeller is :

$$\frac{dP'_m}{dT'} + \frac{dP'_v}{dT'} = \frac{1}{2} \cdot \frac{T'}{\kappa_\phi} + \frac{\epsilon \cdot x}{\cos^2 \phi} \quad (14.1)$$

Figure 14.1 shows the graph. It is basically figure 13.4 shifted up by the constant momentum loss. Clearly, the *combined* local loss ratio is not constant along the blade. Neither is the marginal loss, as we shall see in a later chapter.

The rotationally optimized propeller is *not* optimal in the presence of viscous loss. We will return to this in the next chapter.

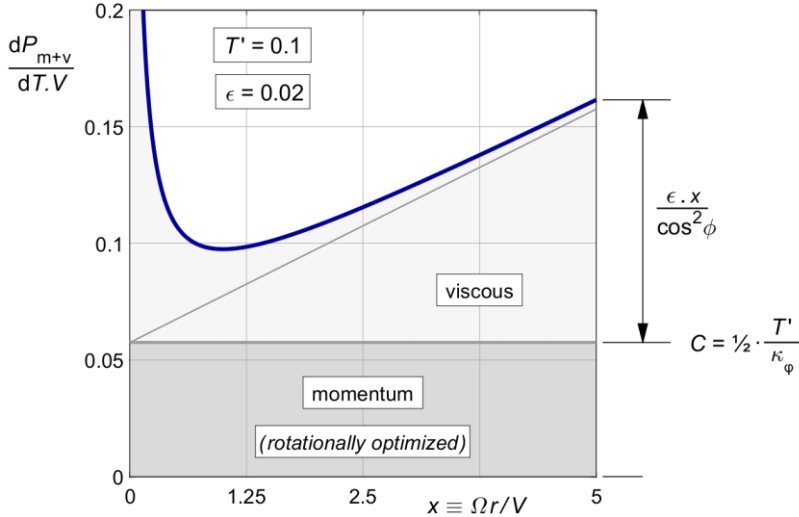


Figure 14.1 : Local combined loss ratio (14.1), rotationally optimal thrust.

14.2 Integrating the rotationally optimized viscous loss

We wish to find an average viscous loss over the propeller. The average will depend on the thrust distribution, because the local viscous loss ratio depends on the radius. We need to make a choice, and we will use the rotationally optimized propeller as a baseline. We noted above that this is not the optimal choice, but will use it because it is common in the literature, and it gives a simple result.

Loss ratios and efficiencies cannot be integrated over the propeller disk. As a counter-example, it will not help if half the disk area has a perfect zero percent loss ratio, if all the thrust is on the non-perfect half. We will have to integrate the local loss itself, not the ratio.

The actual local loss equals the loss ratio times the local thrust. From (13.21) we have:

$$dP_v' = dT' \cdot \frac{\epsilon \cdot x}{\cos^2 \phi} \quad (14.2)$$

For lightly loaded propellers without tip loss, dm' follows from (6.8), dT' follows from (6.10), and the viscous loss on a ring is:

$$dP_v' = b(x) \cdot \frac{\epsilon \cdot x}{\cos^2 \phi} \cdot \frac{2x \cdot dx}{X^2} \quad (14.3)$$

In the rotationally optimal distribution, $b(x)$ follows from (9.9). Substituting this into (14.3) the $\cos^2 \phi$ terms cancel, giving:

$$dP_v' = \epsilon \cdot \frac{T'}{\kappa_\phi} \cdot x \cdot \frac{2x \cdot dx}{X^2} \quad (14.4)$$

The integral is particularly simple for this rotational case. We can the integral by T' to get the overall loss ratio':

$$\frac{P_v'}{T'} = \epsilon \cdot \frac{1}{\kappa_\phi} \cdot \frac{1}{X^2} \int_0^X 2x^2 \cdot dx \quad (14.5)$$

The primitive of $2x^2$ is $\frac{2}{3}x^3$, so the overall viscous loss ratio for the rotationally optimal propeller becomes:

$$\frac{P_v'}{T'} = \frac{1}{\kappa_\phi} \cdot \frac{2}{3} \cdot \epsilon \cdot X \quad (14.6)$$

The averaging factor $1/\kappa_\phi$ is not very different from the local factor $1/\cos^2 \phi$ at $0.7R$, and therefore (14.6) supports the classical rule of thumb that the *overall* viscous efficiency of a propeller can be approximated by the *local* viscous efficiency at the radius $0.7R$. This was given by (13.21), with $x = 0.7X$ and the appropriate ϕ at $0.7X$.

Figure 14.2 show this average viscous loss over the whole propeller disk, using $\epsilon = 0.02$ as an example. Unlike figure 13.4 and 14.1, this figure has the *tip* speed ratio X on the horizontal axis, not the local x .

The shape is similar to figure 13.4, but the minimum occurs near $X = 1.35$ instead of $x = 1$, and the slope of the asymptote is $\frac{2}{3}\epsilon$ instead of ϵ .

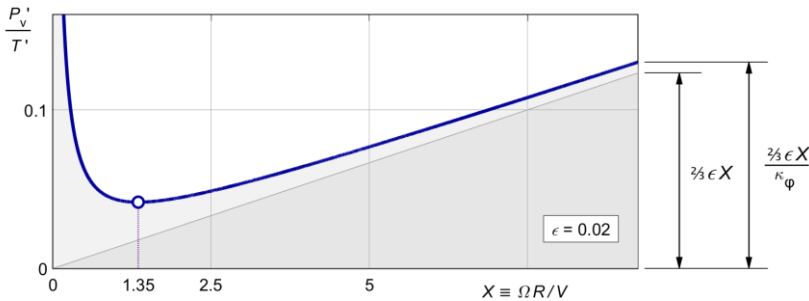


Figure 14.2 : The overall viscous loss ratio (14.6) as a function of X , in the rotationally optimal propeller.

14.3 The combined overall loss for the rotational case

We can now sum the averaged values of momentum and viscous losses (**Error! Reference source not found.**) and (14.6), like we did for the local losses in (14.1). Note that these are values for the rotationally optimal propeller, for light loading, before tip loss, and before viscous optimization :

$$\frac{P'_{m+v}}{T'} \equiv \frac{P'_m}{T'} + \frac{P'_v}{T'} = \frac{\frac{1}{2} T'}{\kappa_\phi} + \frac{\frac{2}{3} \epsilon \cdot X}{\kappa_\phi} \quad (14.7)$$

Figure 14.3 shows the result. The basis is a constant loss due to the optimized momentum loss, and a straight slope due to the corresponding viscous loss.

The viscous loss term gives a steady rise of the combined loss ratio with increasing tip speed ratio X .

The factor $1/\kappa_\phi(X)$ which restores the design thrust after the rotational optimization adds a sharply curved sweep-up for the lower X values on top of this straight line.

The loss increase at both ends of the X range means that there is a minimum halfway. The figure has a circle marker at this minimum for our standard example of $T' = 0.1$, $\epsilon = 0.02$. It has its optimum at $X = 2.9$.

The location of the optimum depends on the balance between momentum (rotational) and viscous loss. This balance will not change if we scale up both T' and ϵ by the same factor. The whole graph will then just scale up. The shape of the curve and the location of the minimum do shift if we change the ratio ϵ/T' . The figure shows the effect of varying ϵ for a given thrust.

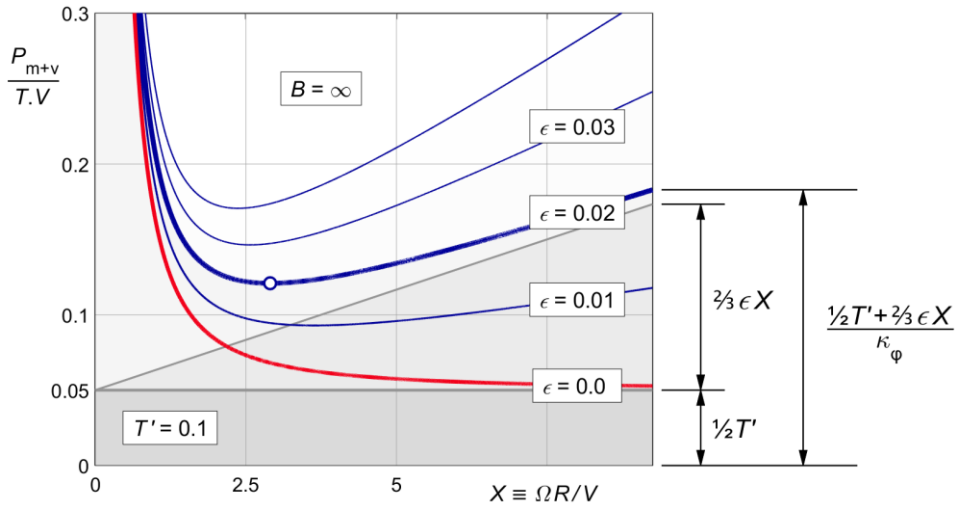


Figure 14.3 : Overall combined loss ratio (14.7),
for rotationally optimal thrust.

14.4 The optimal tip speed ratio in a ducted propeller

Figure 14.3 shows a minimum for the combined momentum and viscous loss of the rotationally optimal propeller before tip loss. The location of the optimum can be found by taking the derivative of (14.7) with respect to X , and setting it to zero. Indicating the X derivative by a subscript X , the standard rules for the derivative of a quotient yield :

$$\left(\frac{P_{m+v}}{T'} \right)_X = -\frac{1}{2} T' \cdot \frac{(\kappa_\phi)_X}{(\kappa_\phi)^2} + \frac{\frac{2}{3} \cdot \epsilon}{\kappa_\phi} - \frac{2}{3} \cdot \epsilon \cdot X \cdot \frac{(\kappa_\phi)_X}{(\kappa_\phi)^2} \quad (14.8)$$

Equating this to zero and collecting terms we have :

$$\frac{\kappa_\phi}{(\kappa_\phi)_X} = \frac{3}{4} \cdot \frac{T'}{\epsilon} + X \quad (14.9)$$

This equation has not been found in the literature studied so far, but it seems useful. We have κ_ϕ from (9.6). Its derivative with respect to X is :

$$(\kappa_\phi)_X = \frac{2 \cdot \ln(1 + X^2)}{X^3} - \frac{2}{X^3} \quad (14.10)$$

Between this derivative and (9.6) for κ_ϕ itself, solving (14.9) analytically is not trivial; but fortunately it is easily solved graphically or by iteration.

Figure 14.4 gives the graphs of the left and right hand terms of (14.9). The optimum values of X are at the crossings of these lines.

The optimum X depends only on the value of T'/ϵ . For large ϵ , the optimum shifts to the purely viscous optimum at $X = 1.35$ of figure 14.2. For very low ϵ , the optimum shifts to very high X , since the momentum loss continuously decreases for larger X . A typical intermediate value is $X = 2.9$, for our standard example $T' = 0.1$, $\epsilon = 0.02$, as shown before in figure 14.3.

These are optimum tip speed ratios before tip loss. Strictly speaking, they only apply in a duct. Chapter 24 will add the tip loss to this equation. This will shift the optimum X to higher values.

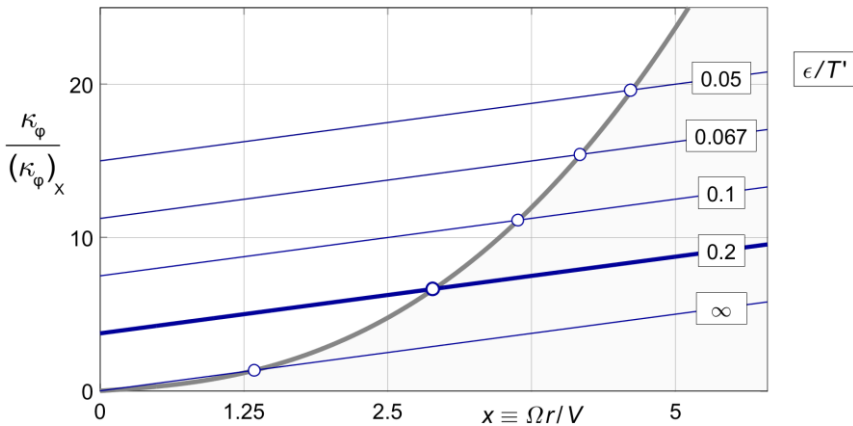


Figure 14.4: Finding the optimum X for $B = \infty$ with (14.9).
See figure 24.3 for arbitrary number of blades.

15 Optimizing the viscous loss

15.1 Relevance, your honor

The classical way to handle viscous loss is the one shown in the previous chapter: the propeller is designed for optimal rotation (and later, tip loss). The viscous loss is then calculated almost as an afterthought, and the outcome is accepted as a fact of life.

But this runs against the spirit of propeller optimization, which is to minimize the overall loss by scheduling the thrust for constant marginal loss.

Historically, GLAUERT already gave the proper solution, but his treatment is not complete, which may be the reason why later texts fail to include it. This chapter picks up GLAUERT's argument and takes it to its logical conclusion.

Viscous optimization has been a wrinkle in the classical theory of propellers for a long time, and this chapter puts it to rest.

The conclusion of this chapter is that in one way the literature is right: for normal propellers, not much efficiency is gained by doing a proper viscous optimization. But it is the proper thing to do, and it is not difficult after all.

Perhaps more importantly, there is a side benefit in the shape of the blade. Readers who use this text as their first introduction to propeller theory may wish to skip this chapter on a first reading. Those looking for new material however, or who like to tie up loose ends, are encouraged to read this chapter for their personal enjoyment.

15.2 A basic trade-off

Figure 13.4 showed that for minimal viscous loss, all thrust should be concentrated on a paddle close to the hub at $x = 1$, for a 45° pitch angle. By contrast, figure 9.1 showed that the optimal momentum efficiency moves the thrust outboard towards the flattest pitch angles. This difference creates a basic trade-off in blade shape design. We attack this trade-off by our trusted method of constant marginal loss.

Momentum goes with thrust. Viscous ratio is independent of thrust.

15.3 The viscous marginal loss

We will be needing the viscous marginal loss. We may hope to optimize it, but somewhat surprisingly, this is not how it works.

By (13.21), the viscous loss is directly proportional to the thrust. An increase of thrust will give an increase of loss in the same proportion. This makes the marginal loss ratio the same as the actual loss ratio. We will not be able to change the local viscous loss ratio by any optimization strategy.

Formally, we can write ((13.21) as:

$$dP_v = \frac{\epsilon \cdot x}{\cos^2 \phi} \cdot d(T \cdot V) \quad (15.1)$$

Taking the derivative with respect to $dT \cdot V$ like in (8.1) gives:

$$\frac{d(dP_v)}{d(dT \cdot V)} = \frac{\epsilon \cdot x}{\cos^2 \phi} \quad (15.2)$$

This marginal loss is exactly the same as the actual local loss ratio (13.21). GLAUERT used the simplified viscous drag ratio (13.16) here, but we will use the (almost) exact equation for simpler and more accurate results.

15.4 The rotational marginal momentum loss

The previous chapter discussed the viscous loss case of the rotationally optimized propeller, without any attempt at optimization of the momentum and viscous loss together. Almost all of the literature after GLAUERT either tacitly accepts this solution, or claims that viscous optimization is not worth the trouble. We will examine this claim by taking the trouble and seeing where it leads us.

We apply the method of constant marginal loss which we applied to the rotational momentum loss alone in section 8.1, this time to the case including viscous loss.

We combine (15.1) with the marginal momentum loss for an arbitrary (but light loading) induction $b(x)$ in (8.4) :

$$\frac{d(dP_m)}{d(dT.V)} + \frac{d(dP_v)}{d(dT.V)} = \frac{b(x)}{\cos^2 \phi} + \frac{\epsilon}{\cos^2 \phi} \cdot x \quad (15.3)$$

The combined marginal loss will now depend on $b(x)$ through the momentum loss. We will first look at the combined marginal loss for the rotationally optimized propeller, without optimization for the viscous loss. This is an interesting reference case, since most of the literature accepts this as the standard solution.

The $b(x)$ for the rotationally optimal propeller was given by (9.9), and the resulting constant marginal momentum loss by (9.8). Substitution into (15.3) yields :

$$\frac{d(dP_m)}{d(dT.V)} + \frac{d(dP_v)}{d(dT.V)} = \frac{T'}{\kappa_\phi} + \frac{\epsilon \cdot x}{\cos^2 \phi} \quad (15.4)$$

This combined marginal loss is very similar to (14.1) which gave the actual local loss for the rotationally optimal propeller. The only difference is that the marginal momentum loss is exactly twice the local momentum loss, while the marginal viscous loss remains the same.

Figure 15.1 shows the result, with numerical values taken from our standard lightplane example. The plot is almost identical to figure 14.1, again with only the difference in the doubling of the momentum loss.

The combined marginal loss is not constant, and so this propeller is not optimal. Moving thrust from the tips to the region around $x = 1$ should improve the efficiency.

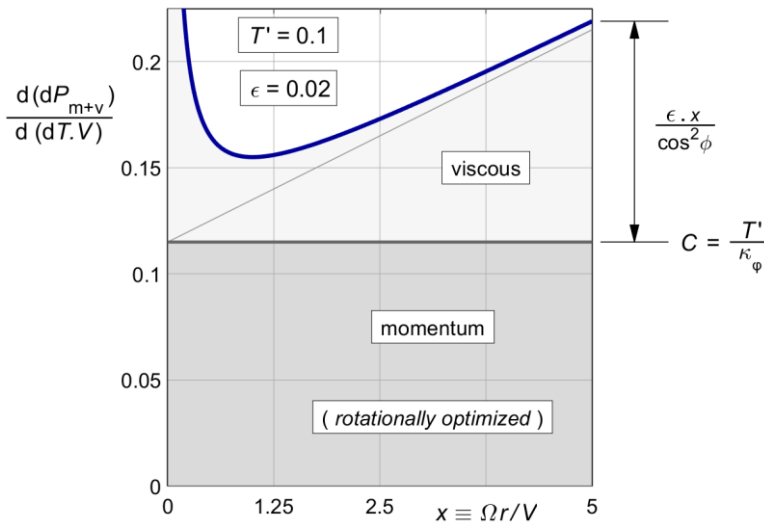


Figure 15.1 : The marginal loss in the rotationally optimal propeller, (15.4).

15.5 Constant combined marginal loss

We will now turn the combined marginal loss into a constant. Since the new constant will have a different value than before, we will use a new letter A instead of the C of (8.3). We will require:

$$\frac{d(dP_m)}{d(dT.V)} + \frac{d(dP_v)}{d(dT.V)} = A \quad (15.5)$$

With (15.3) this gives:

$$A = \frac{b(x)}{\cos^2 \phi} + \frac{\epsilon}{\cos^2 \phi} \cdot x \quad (15.6)$$

Since A is a constant, the combined optimal distribution becomes:

$$b(x) = A \cdot \cos^2 \phi - \epsilon \cdot x \quad (15.7)$$

Relative to (8.5), the right hand term in x is new. The math appendix gives a sketch of this local reduction based on A , which is a replay of GLAUERT's figure. The thrust is reduced to zero near $X = A/\epsilon$. However, Glauert's figure is subtly misleading if we do not realize that A is much larger than the C of (9.8). Therefore we will first express A in the design parameters.

GLAUERT himself used the approximate $\mu(x)$ of (13.16), which yields:

$$b(x) = (A - \epsilon \cdot x) \cdot \cos^2 \phi \quad (15.8)$$

We will use (15.7), but **we will also follow through (15.8) in the notes, to show that it does not make the derivations simpler, but in fact more complex.**

15.6 The marginal loss in the design parameters

We express A in terms of T' , ϵ and X . Using the induction $b(x)$ of (15.7) in the light loading integral $()$, the overall thrust is:

$$T' = \frac{1}{X^2} \int_0^X (A \cdot \cos^2 \phi - \epsilon \cdot x) \cdot 2x \cdot dx \quad (15.9)$$

The left hand integral was called κ_ϕ in (9.6), and the primitive of $2x^2$ is $\frac{2}{3}x^3$. Between the limits of $x = 0$ and $x = X$ this gives $\frac{2}{3}X^3$, and so:

$$T' = A \cdot \kappa_\phi - \frac{2}{3} \cdot \epsilon \cdot X \quad (15.10)$$

The marginal loss expressed in design parameters is therefore:

$$A = \frac{T' + \frac{2}{3} \cdot \epsilon \cdot X}{\kappa_\phi} \quad (15.11)$$

Substituting this into (15.7) gives the optimal $b(x)$ in design parameters:

$$b(x) = \frac{T' + \frac{2}{3} \epsilon \cdot X}{\kappa_\phi} \cdot \cos^2 \phi - \epsilon \cdot x \quad (15.12)$$

This is the equivalent of the induction (9.9) for optimal rotational loss, only this time including the viscous optimization. The original "amplitude" T'/κ_ϕ is increased by $\frac{2}{3}\epsilon \cdot X/\kappa_\phi$. This fixed term makes up for the loss of thrust from the radius dependent $\epsilon \cdot x$ term.

We will take a closer look at the shape of (15.12) in figure 15.4, but first we look at the two components of the marginal loss.

15.7 Components of the marginal loss

The marginal viscous loss is independent of $b(x)$ but the marginal momentum loss is not. Substituting the $b(x)$ of (15.7) into the marginal momentum loss (8.4) we have :

$$\frac{d(dP_m)}{d(dT.V)} = \frac{T' + \frac{2}{3} \epsilon \cdot X}{\kappa_\phi} - \frac{\epsilon \cdot x}{\cos^2 \phi} \quad (15.13)$$

The left hand term is the constant marginal loss (15.11). The right hand term effectively subtracts the viscous marginal loss (15.2) from this, leaving a marginal momentum loss which is no longer a constant. The momentum loss by itself will no longer be optimal. Interestingly, the marginal loss is made constant by modifying the *momentum* term, not the viscous term.

Figure 15.2 shows the result. The optimization moves thrust towards the region of lower viscous loss near $x = 1$, as shown in figure 13.4. Increasing the thrust there means that we have to accept an uneven momentum loss.

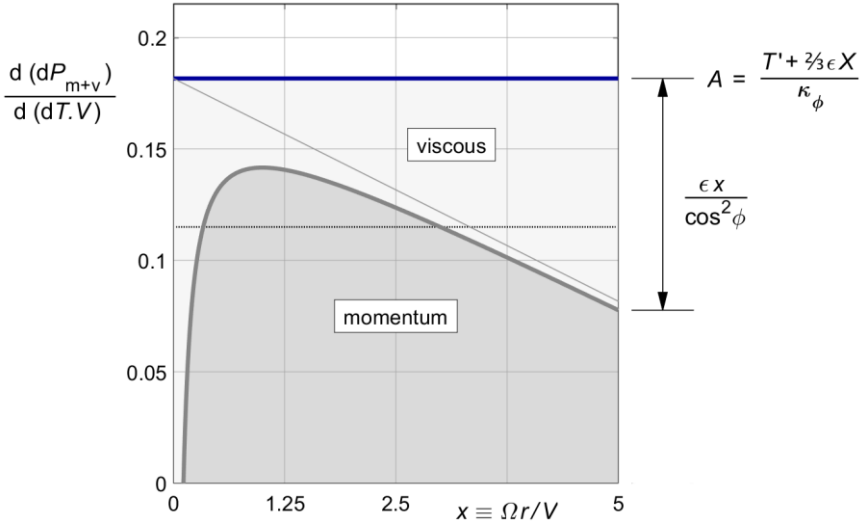


Figure 15.2 : Combined marginal loss, optimized by (15.3).

15.8 The optimized local loss ratio is not a constant

The combined marginal loss is constant along the blade. But unlike for the purely rotational optimum, the actual loss ratio is not.

We know from (13.21) that the local viscous loss ratio does not depend on the local loading $b(x)$, whether optimal or not. The local momentum loss does depend on the loading, and due to the viscous optimization it now also varies with the radius. Therefore, the momentum loss ratio will also vary with the radius.

Substituting the combined optimum $b(x)$ of (15.12) into the generic momentum loss (7.6) gives :

$$\frac{dP_m}{dT.V} = \left(\frac{1}{2} \cdot \frac{T'}{\kappa_\phi} + \frac{1}{2} \cdot \frac{\frac{2}{3} \cdot \epsilon \cdot X}{\kappa_\phi} \right) - \frac{1}{2} \cdot \frac{\epsilon \cdot x}{\cos^2 \phi} \quad (15.14)$$

The bracketed term is a constant. It equals $\frac{1}{2}A$. The first term within the brackets is the usual rotationally optimal momentum loss ratio (***). The second term within the brackets is exactly half the *average* rotationally

optimal viscous loss ratio of (14.6). Unlike the terms within the brackets, the only term outside them is *not* a constant. It varies with x both directly, and via $\cos^2\phi$. This term is exactly half the local viscous loss ratio of (13.21). The viscous loss has been split into two halves : one a constant term depending only on X , and the other a varying term depending only on x .

Figure 15.3 shows the result for our standard lightplane example. The actual loss is not constant along the blade. But compared to figure 14.1, the difference with a constant local loss ratio has approximately been halved.

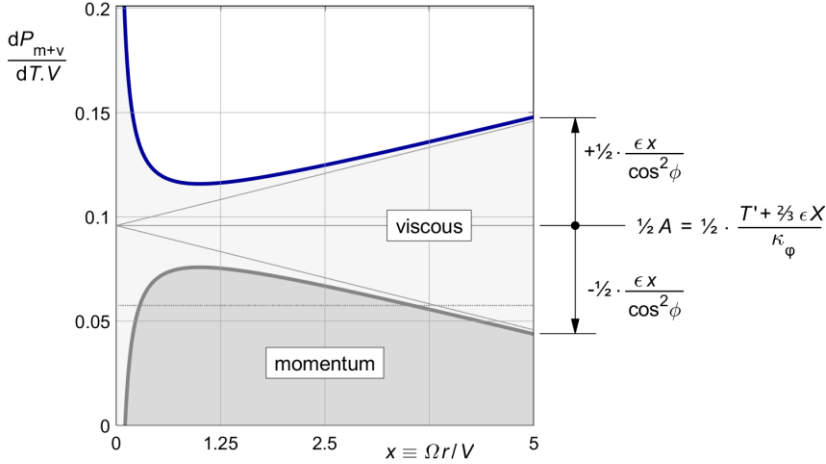


Figure 15.3 : Combined local loss, optimized.

15.9 Blade shape variation with the section drag

Figure 15.4 gives the shape of the optimal induction distribution according to (15.12), for our standard lightplane example of $T' = 0.1$, for viscous friction values of $\epsilon = 0$ and $\epsilon = 0.02$. The zero-friction case of figure 10.1 is shown in red. There is a cross-over at $0.7R$, and this is easier understood by collecting terms in T' and in ϵ (15.12) :

$$b(x) = \frac{T'}{\kappa_\phi} \cdot \cos^2\phi + \epsilon \cdot \left(\frac{2}{3} \cdot X \cdot \frac{\cos^2\phi}{\kappa_\phi} - x \right) \quad (15.15)$$

The left hand term provides all the thrust. It is identical to the purely rotational distribution (9.9). Its contribution is directly proportional to T' .

The right hand term is the viscous correction. It does not affect the thrust at all. It is proportional to ϵ , and it vanishes for $\epsilon = 0$. This term will always become zero at approximately the following radius :

$$x = \frac{2}{3} \cdot \frac{\cos^2\phi}{\kappa_\phi} \cdot X \cong \frac{2}{3} \cdot X \quad (15.16)$$

This is just over two thirds of the prop radius, say at $0.7R$. This location makes sense. The net thrust of the correction term is zero, and the cross-over location is almost exactly halfway the total disk area, since $(0.7)^2 = 0.49$.

The tip value of the distribution is :

$$b(X) = T' \cdot \frac{\cos^2\phi}{\kappa_\phi} + \epsilon \cdot \left(\frac{2}{3} \cdot X \cdot \frac{\cos^2\phi}{\kappa_\phi} - X \right) \quad (15.17)$$

For larger X the tip value of the distribution is well approximately by:

$$b(X) \cong T' - \frac{1}{3} \cdot \epsilon \cdot X \quad (15.18)$$

The tip chord vanishes for $T' < \frac{1}{3} \cdot \epsilon \cdot X$, and the blade will get a sharp point there. In general, $\epsilon \cdot X / T'$ is the shape factor for the viscous correction. The correction is most pronounced for lightly loaded propellers of high tip speed ratio, and for high values of ϵ . The change in shape of the thrust distribution is obvious. The notes in the appendix give the counterpart of these plots, for varying T' at a fixed value of ϵ .

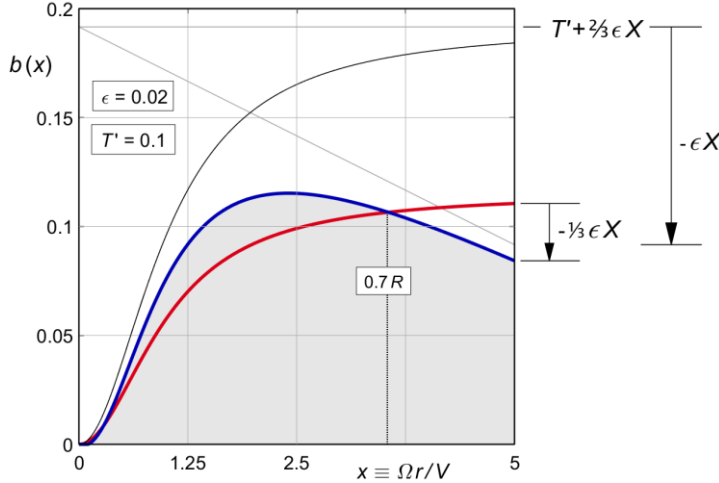


Figure 15.4: Basic diagram of the viscous optimization (15.12).

$T' = 0.1$. Red $\epsilon = 0$. Blue $\epsilon = 0.02$.

15.10 The net gain in the efficiency

15.10.1 A trade-off between viscous and momentum

Equation (14.7) gave the combined overall momentum and viscous loss of a propeller optimized only for rotation. We will calculate the improvement when adding viscous optimization.

We expect an increase in the momentum loss, because the distribution will no longer be fully optimized for momentum loss. In turn, we expect a stronger improvement in the viscous loss. The sum of the two losses should decrease. We will calculate the improvement in the viscous loss first.

15.10.2 The viscous loss integral

Equation (14.3) gave the viscous loss on a ring for an arbitrary (light loading) induction $b(x)$. Substituting the optimal $b(x)$ of (15.12), some of the $\cos^2\phi$ terms will cancel :

$$P'_v = \frac{1}{X^2} \int_0^X \left(\frac{T' \cdot \epsilon \cdot x}{\kappa_\phi} + \frac{\frac{2}{3} \cdot \epsilon^2 \cdot X \cdot x}{\kappa_\phi} - \frac{\epsilon^2 \cdot x^2}{\cos^2\phi} \right) \cdot 2x \cdot dx \quad (15.19)$$

The first term, together with the $2x$ outside the brackets, yields an integral of $2x^2$ with primitive $\frac{2}{3}x^3$. The result equals the non-optimized loss for a rotationally optimal propeller, (***). The second term has the same order of x , but with a different constant. In the third term, (2.7) changes the $x^2/\cos^2\phi$ term to $1+x^2$. Together with the $2x$ outside the brackets, the primitive of this term is $x^2 + \frac{1}{2}x^4$. The overall integral becomes :

$$P_v' = \frac{2/3 T' \cdot \epsilon \cdot X}{\kappa_\phi} + \frac{4/9 \cdot \epsilon^2 \cdot X^2}{\kappa_\phi} - \epsilon^2 \cdot (1 + 1/2 X^2) \quad (15.20)$$

Collecting terms, the non-dimensional loss ratio as in (3.9) is:

$$\frac{P_v'}{T'} = 2/3 \frac{\epsilon \cdot X}{\kappa_\phi} - \frac{\epsilon^2}{T'} \cdot \left(1 + X^2 \cdot \left(1/2 - \frac{4/9}{\kappa_\phi} \right) \right) \quad (15.21)$$

The left hand term is the viscous loss ratio of the rotationally optimal propeller of (14.6). The right hand term in is the decrease of the viscous loss ratio due to the viscous optimization. We will see below that exactly half of this gain is offset by an increase of the momentum loss ratio.

15.10.3 The momentum loss integral

Equation (7.11) gave the ring momentum loss for an arbitrary $b(x)$ for light loading. Substituting the newly optimized $b(x)$ from (15.12) yields :

$$P_m' = \frac{1}{X^2} \int_0^X 1/2 \cdot \frac{\left(\frac{T' + 2/3 \cdot \epsilon \cdot X}{\kappa_\phi} \cdot \cos^2 \phi - \epsilon \cdot x \right)^2}{\cos^2 \phi} \cdot 2x \cdot dx \quad (15.22)$$

Squaring the bracketed term gives three terms, as in $(a+b)^2 = a^2 + 2ab + b^2$. Dividing by $\cos^2 \phi$ and writing the three integrals separately yields :

$$\begin{aligned} P_m' &= \frac{1}{X^2} \int_0^X 1/2 \cdot \left(\frac{T' + 2/3 \cdot \epsilon \cdot X}{\kappa_\phi} \right)^2 \cdot \cos^2 \phi \cdot 2x \cdot dx \\ &+ \frac{1}{X^2} \int_0^X - \frac{T' + 2/3 \cdot \epsilon \cdot X}{\kappa_\phi} \cdot \epsilon \cdot x \cdot 2x \cdot dx \\ &+ \frac{1}{X^2} \int_0^X 1/2 \cdot \frac{\epsilon^2 \cdot x^2}{\cos^2 \phi} \cdot 2x \cdot dx \end{aligned} \quad (15.23)$$

The first integral is a constant times the integral of $\cos^2 \phi$ over the prop disk. By the definition of (9.3), that integral is κ_ϕ . The second integral is a constant times $2x^2$, which has the simple primitive $2/3 x^3$, which yields $2/3 X^3$. The third integral matches the last term in (15.19), which we solved in (15.20), only this time with a factor $+1/2$ instead of -1 . All in all this yields :

$$\begin{aligned} P_m' &= \frac{1/2 (T')^2 + 2/3 \cdot T' \cdot \epsilon \cdot X + 2/9 \cdot \epsilon^2 \cdot X^2}{\kappa_\phi} \\ &- \frac{2/3 T' \cdot \epsilon \cdot X + 4/9 \cdot \epsilon^2 \cdot X^2}{\kappa_\phi} \\ &+ 1/2 \epsilon^2 \cdot (1 + 1/2 X^2) \end{aligned} \quad (15.24)$$

Several terms cancel halfway or completely between the first and the second integral. One pair equals the non-optimized viscous loss. This pair cancels completely. Another term cancels halfway, leaving us with :

$$P_m' = \frac{(T')^2}{\kappa_\phi} - \frac{2/9 \cdot \epsilon^2 \cdot X^2}{\kappa_\phi} + 1/2 \epsilon^2 \cdot (1 + 1/2 X^2) \quad (15.25)$$

The non-dimensional ratio (3.9) for the overall momentum loss becomes:

$$\frac{P_m'}{T'} = 1/2 \cdot \frac{T'}{\kappa_\phi} + \frac{\epsilon^2}{T'} \cdot \left(1/2 + X^2 \cdot \left(1/4 - \frac{2/9}{\kappa_\phi} \right) \right) \quad (15.26)$$

The first term equals the "normal" rotationally optimal momentum loss ratio (***) . The other term, in ϵ^2/T' , is exactly half the decrease in the viscous loss. This result is too nice to be a coincidence, and in fact the same thing happens when GLAUERT's approximation is used. It may be related to the difference between figures 14.3 and 15.3, where figure 15.3 seems to split the deviation from a constant loss equally over the viscous and the momentum loss. Be that as it may, the net decrease in the combined loss ratio is :

$$\Delta \frac{P'_{m+v}}{T'} = -\frac{\epsilon^2}{T'} \cdot \left(1 + X^2 \cdot \left(\frac{1}{2} - \frac{2/9}{\kappa_\phi} \right) \right) \quad (15.27)$$

We will call the bracketed part of this term $h(X)$:

$$h(X) \equiv \frac{1}{2} + X^2 \cdot \left(\frac{1}{4} - \frac{2/9}{\kappa_\phi} \right) \quad (15.28)$$

The term depends only on X , since κ_ϕ also depends only on X .

Figure 15.5 shows the graph of $h(X)$ against X . The term is always positive. The dashed line shows that it can be approximated by a heuristic fit :

$$h(x) \cong \frac{(X - \frac{1}{2}\pi)^2}{32.25} \quad (15.29)$$

The factor of $\frac{1}{2}\pi$ is just a mnemonic. It has no physical meaning - the best fit just happens to be around 1.57. A "toy problem" ignoring all $\cos^2\phi$ terms gives $h(x) = 1/36 X^2$.

The gain in efficiency increases strongly with X . This makes sense, since for larger X the viscous loss is much more important.

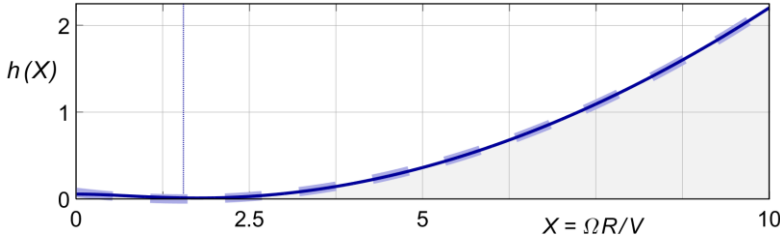


Figure 15.5 : The term $h(X)$ in the viscous optimization.

15.11 Numerical example

Our standard example has a tip speed ratio of $X = 5$. With (15.29), we find :

$$\Delta \frac{P'_{m+v}}{T \cdot V} = -0.0015 \quad \text{standard example} \quad (15.30)$$

The gain in efficiency is 0.15 % percentage point. This is small relative to the overall efficiency of 89 %, though not completely negligible.

For rotors of high tip speed ratio however, the viscous optimization is definitely worthwhile. At a tip speed ratio of $X = 10$, for the same $\epsilon = 0.02$ and $T' = 0.1$ as in the standard example, the overall loss ratio is reduced by 0.9 % percentage point. Such gains are not easily achieved in other ways.

For normal values of X , the main benefit of the viscous optimization is not so much the gain in efficiency, but rather the side effect on the blade shape, which visibly changes towards narrower and hence thinner tips. This gives mechanical and acoustical advantages.

16 Tip loss

16.1 The nature of the tip loss

Tip loss differs essentially from the losses we have encountered so far. Remember the thrust equation (6.1), repeated here :

$$dT = dm^* \cdot b(x) \cdot V \quad (16.1)$$

In this equation, the optimizations (9.9) and (15.12) for the rotational and viscous loss act only on the local induction $b(x)$. They are functions of the local pitch angle, and by the geometrical relations (2.3) to (2.7) they just depend on the non-dimensional radius x .

These optimizations are independent of the mass flow, at least under the light loading assumption. For equal efficiency, $b(x)$ is reduced locally by a certain factor, and for equal design thrust, $b(x)$ is uniformly inflated back again, both without touching the local mass flow dm^* .

Tip loss, by contrast works *only* on the local mass flow term dm^* , and *not* on the local induction $b(x)$, except by another uniform inflation factor on $b(x)$. This extra uniform inflation factor is needed to make up for the loss of thrust when lowering the local dm^* .

Changing dm^* in (16.1) does not change the equations for the optimal induction along the blades. It just affects the part of the mass passing through the disk which receives that induction.

Some of the air in the original momentum tube escapes around the blade tips and does not, or not fully, receive the induction. But even then, the efficiency of the blades still depends only on the local induction, and therefore the same optimal scheduling of $b(x)$ still applies. The only problem is that for a given efficiency, there will be less local, and hence less overall thrust.

Instead of speculating about the way the induction varies in the space between the blades, we can lump this effect by a reduction factor on the local mass flow. We define a function $K(x)$ for the reduction of the *effective* local mass flow. This changes the light loading momentum flow from (6.6) to :

$$dm'_K(x) \equiv K(x) \cdot \frac{2x \cdot dx}{X^2} \quad (16.2)$$

For heavy loading the contraction term in (6.6) can be retained as well.

We use the subscript K on dm' to indicate when it is reduced by $K(x)$.

Replacing the local mass flow in (16.1) by (16.2) changes the local thrust (6.12) into :

$$dT' = b(x) \cdot dm'_K = b(x) \cdot K(x) \cdot \frac{2x \cdot dx}{X^2} \quad (16.3)$$

The local efficiency of this propeller is fully determined by $b(x)$, but the local thrust at each radius will be reduced by the mass factor $K(x)$, relative to the dT' without tip loss of (10.1) .

16.2 Tip loss with optimal rotation

Equation (8.5) gave the local axial induction $b(x)$ for a lightly loaded, rotationally optimal propeller of constant marginal loss C . It is repeated here :

$$b(x) = C \cdot \cos^2 \phi \quad (16.4)$$

Substituting this into (16.3) yields :

$$dT' = C \cdot \cos^2 \phi \cdot K(x) \cdot \frac{2x \cdot dx}{X^2} \quad (16.5)$$

For a given C , this yields a propeller with the momentum loss ratio $\frac{1}{2}C$ of (8.9). But due to the presence of the term $K(x)$, at a given C this propeller will have less thrust than the same propeller before taking tip loss into account. With tip loss, the light loading, overall thrust integral changes from (9.2) to :

$$T' = C \cdot \int_0^X \cos^2 \phi \cdot K(x) \cdot \frac{2x \cdot dx}{X^2} \quad (16.6)$$

Like the κ_ϕ of (9.3), we define a reduction factor from C to T' :

$$\kappa_G \equiv \frac{1}{X^2} \int_0^X \cos^2 \phi \cdot K(x) \cdot 2x \cdot dx \quad (16.7)$$

The subscript G follows from a definition commonly used in the literature :

$$G(x) \equiv \cos^2 \phi \cdot K(x) \quad (16.8)$$

The factor $G(x)$ is the combination of the rotational reduction on the induction on the one hand, and the tip reduction of the mass term on the other hand.

We will find that in the exact solution, this function is in fact better behaved function than $K(x)$. Also, in the exact solution the combination $G(x)$ is easier to calculate than $K(x)$ itself.

16.3 The effect on the efficiency

The effect of the thrust reduction due to tip loss on the momentum efficiency is completely in parallel with the effect of the rotational reduction $\cos^2 \phi$ alone. Following through the steps from (9.7) to (9.12), once again using (8.9) to (8.11) but this time with the κ_G of (16.7) instead of the κ_ϕ of (9.3), we have :

$$T' = C \cdot \kappa_G \quad (16.9)$$

$$C = \frac{T'}{\kappa_G} \quad (16.10)$$

$$b(x) = T' \cdot \frac{\cos^2 \phi}{\kappa_G} \quad (16.11)$$

$$\frac{dP'_m}{dT'} = \frac{1}{2} \frac{T'}{\kappa_G} \quad (16.12)$$

$$\frac{P'_m}{T'} = \frac{1}{2} \frac{T'}{\kappa_G} \quad (16.13)$$

$$\eta_m = \frac{1}{1 + \frac{1}{2} \frac{T'}{\kappa_G}} \quad (16.14)$$

17 Two solutions

17.1 Two solutions for the shape of $K(x)$

Tip loss is by far the most complex part of propeller theory, and it comes down to determining the local mass reduction factor $K(x)$.

There are two classical ways of approaching this. One is an approximate method due to PRANDTL, and the other is an exact (vortex) method which is mainly associated with the name of GOLDSTEIN.

PRANDTL's method gives an intuitive insight into the problem, and with due care it will give a close approximation of the proper blade shape. But for propellers of larger pitch, it is too optimistic about the resulting momentum efficiency.

The vortex method is computationally intensive and far less intuitive, but it is the only way to get a good estimate of the true momentum efficiency. It also sheds a useful light on the inner workings of the propeller, from a different perspective than PRANDTL's method.

In line with all the literature, we will use PRANDTL's solution for our main argument. In a break with tradition, we will use the exact solution as a ground truth in the background, and even give an impression of the theory behind the vortex system, and a practical algorithm to solve it without the use of tables which made this method very unpopular in the past.

We will first very briefly outline the main ideas behind PRANDTL's solution and the exact one. Both are based on the shape of BETZ's ARCHIMEDES screw, and the next chapter will therefore go further into the flow around that wake.

17.2 Prandtl's solution

PRANDTL published his approximate solution for the tip loss in an appendix to BETZ's 1919 article on the rotational optimum, which also introduced the ARCHIMEDES screw wake. His derivation cleverly reverses cause and effect on the ARCHIMEDES screw. For the sake of his argument, PRANDTL considers the ARCHIMEDES screw as a physical object which drives the flow, instead of as a viscous wake sheet which drifts with the flow. He then approximates the shape of the screw by a straightened and simplified version of its outer edge, and from potential theory derives the mass transported.

Setting the induction flow in motion from rest (as if in switching on the thrust) requires an impulsive force from the spiral surface, and this force is proportional to the effective mass of the air set in motion. In this way, PRANDTL obtains an effective mass factor without having to consider the flow in between the spiral sheets. The limitation of his method is in the simplification of the screw shape.

Figure 17.1 gives a typical shape for PRANDTL's version of $K(x)$, which we will come to call $F(x)$ or $F'(x)$ for reasons explained later.

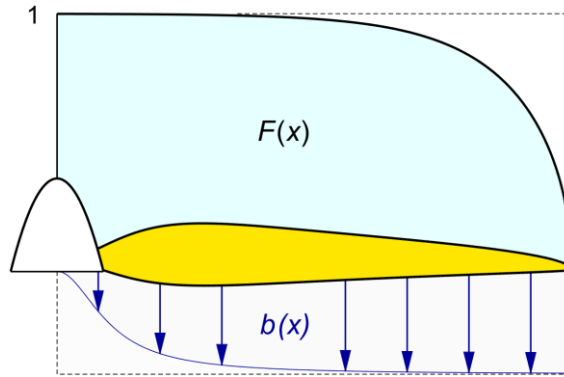


Figure 17.1 : Typical PRANDTL mass flow reduction factor.

17.3 The vortex solution

PRANDTL'S knew very well that his solution to the tip problem was an approximation, albeit a very good one. His method becomes less accurate for propellers of high pitch, and for low number of blades.

In the 1930's, the exact solution was calculated by GOLDSTEIN and others. Like PRANDTL'S, his solution is based on BETZ'S ARCHIMEDES screw shape. The method constrains the flow at right angles through the screw surface to be zero, and calculates the corresponding vortex pattern. Since this pattern respects the $\cos^2\phi$ function of the rotational optimization, it actually gives the combination $G(x)$ from **** for the rotational reduction and the tip reduction combined, in a single pass. Out of deference to GOLDSTEIN, the symbol $G(x)$ is often used for the combined reduction :

$$G(x) \equiv \cos^2\phi \cdot K(x) \quad (17.1)$$

The solution involved an expansion into BESSEL functions, which does not help the intuition much. Because the standard handbook on BESSEL functions did not contain all orders, the solution for three blades was not available. The two- and four-blade propellers were tabled for the outer 80% of the blade only. The hub region was considered less important because there is not much thrust there anyway. But it was also noted that $K(x)$ tends to go to infinity near the hub, which is both puzzling and inconvenient. The combination $G(x)$ does not suffer from this problem.

With the advent of digital computers in the 1980's better tables became available, but judging from the literature, they were rarely used in practice.

Recently, a relatively simple "brute force" algorithm has become available which solves the vortex system without resorting to tables for $G(x)$. The last chapters of this text give the algorithm, and an impression of the way it is derived.

Even in the 1930's it was clear that the function $K(x)$ is not as simple as PRANDTL'S approximation would indicate. The vortex solution of $K(x)$ in fact goes to infinity at the hub. Figure 17.2 gives an impression. For low number of blades, the combined reduction starts with a finite slope, instead of by $\cos^2\phi$ which gives the unpleasant result for $K(x)$. Unfortunately, the numerical solution does not give an intuitive reason for this.

The combined reduction factor $G(x)$ gives the regular ARCHIMEDES screw shape for optimal rotation, and the corresponding tip reduction.

Hier het bekende plaatje aanhalen van Goldstein vs. Prandtl, en dan de $K(x)$ afleiden die naar ∞ gaat bij $x=0$? En dan beloven dat je het gaat oplossen.

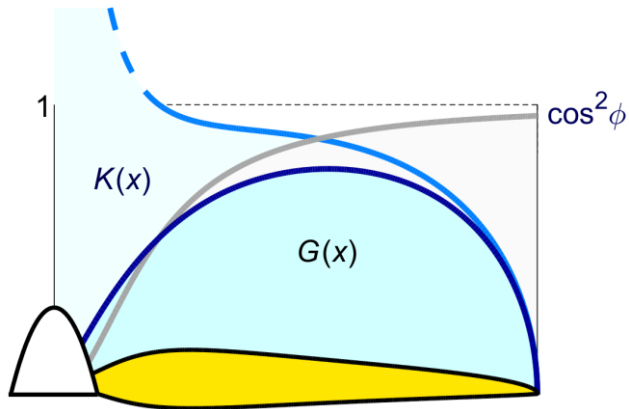
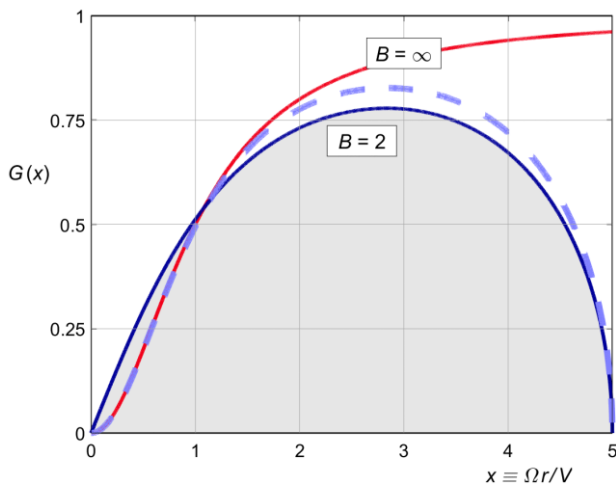


Figure 17.2 : A typical exact thrust reduction factor.

17.4 The difference between the two solutions

TBW.

Show the Glauert figure. Then show $K(x)$ going to infinity, without $G(x)$.



18 The spiral wake

18.1 Introduction

Both PRANDTL' solution and the exact one are based on BETZ's ARCHIMEDES screw wake. We will therefore look at the flow around this wake first.

18.2 The viscous wake sheet

Like a wing, a propeller leaves behind two distinct motions in the air :

- a large scale momentum wake, or propwash.
- a thin viscous wake sheet, sliding off the blade trailing edges.

For the propeller as for the wing, the velocities in the momentum wake and those in the viscous wake are orthogonal in both senses of the word. They are a right angles to each other, and they do not interact.

First we look at the viscous wake. Figure 18.1 shows the zero-thrust wake of a two-bladed propeller. Without blade lift or thrust there will be no induced velocity. The propeller "winds itself" effortlessly out of the picture, each blade leaving behind a thin viscous wake sheet in the air. This viscous "scar" separates the air which passed under the blade section from the air which passed over it. The friction between the passing air and the surface of the blade slows down a thin boundary layer of air, which trails off the trailing edges of the blades like treacle from a spoon. This thin viscous sheet momentarily follows the blade, then dissipates to zero velocity relative to the surrounding air after a few chord lengths. Figure 18.1 indicates the initial following velocity by a small arrow marked W . From the velocity diagram in figure 7.2 we know that for $X \gg 1$, the blade velocity W approximately equals Ωr .

The thickness of the viscous trailing sheet roughly equals the local viscous drag coefficient C_d of the blade section times its local chord. In propellers, this sheet thickness is on the order of a millimeter.

The viscous sheet has the shape of an ARCHIMEDES screw, like an old water pump, but screwing itself forward through the air instead of rotating in place. The air *between* the sheets remains completely undisturbed as the spiral slides through it. At zero thrust there is viscous drag, but there is no "tip loss".

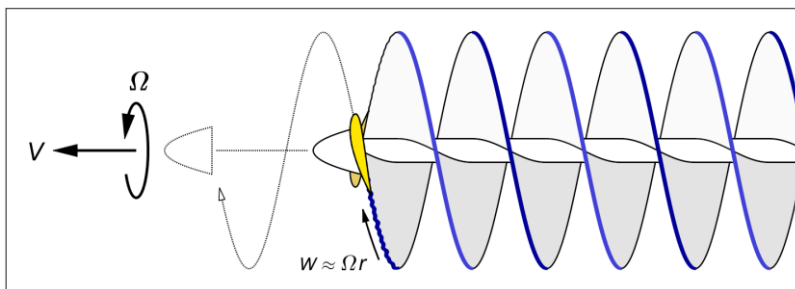


Figure 18.1 : ARCHIMEDES screw viscous wake of a two-bladed propeller ($B = 2$, $X = 5$).

18.3 The momentum wake

When the propeller starts delivering thrust, the wake at the propeller will drift backward with the local induced velocity v' from figure 7.2 at the propeller, and with the double velocity w' far behind it. Figure 18.2 shows this large scale velocity field.

The velocity v' is normally a small fraction of the flying velocity V . Because v' is tilted as in figure 7.2, the air is not just floating backward, it is also quietly rotating. The lift-induced velocity at the blades is by definition at right angles to the local blade velocity, and therefore orthogonal to the spiral surfaces. From section 0 we know that the wake sheets maintain the shape of a rigid ARCHIMEDES screw, even though they are stretched and rotated a bit by the blade lift.

The air does *not* stream backward behind the propeller with anything like the flying velocity V . This wake, drifting backward slowly, does not fit our usual intuitive image, where the propeller is stationary and the air is blowing backward fast. That image is based on an aircraft standing on the runway, or maybe from looking at a propeller in a wind tunnel. By contrast, in flight the propeller actually threads its way quietly like a corkscrew through an almost stationary atmosphere, adding very little rearward velocity.

Figure 18.2 shows the momentum wake viewed from a stationary reference on the ground. The small arrows represent the induced velocities v' from figure 7.2. They are orthogonal to the blade velocity and path, and spiral at an angle of ϕ' to the axis of the propeller.

Viewed from the airplane, the local velocity in the wake is not v' , but $V + v'$. Since V points straight to the rear and v' is relatively small, the swirl angle as seen from the aircraft is an order of magnitude less than the ϕ' tilt of v' alone. Stator vanes behind a propeller are sometimes used to recover some of the swirl. They typically point into the direction of flight without risk of stalling. The swirl from a wing mounted propeller is partially recovered by the wing itself, without a need to change the shape of the leading edge. Vertical fins at the tail are sometimes offset at an angle into the spiral slipstream to prevent a yawing moment from the prop spiral, but this offset is rarely more than one or two degrees.

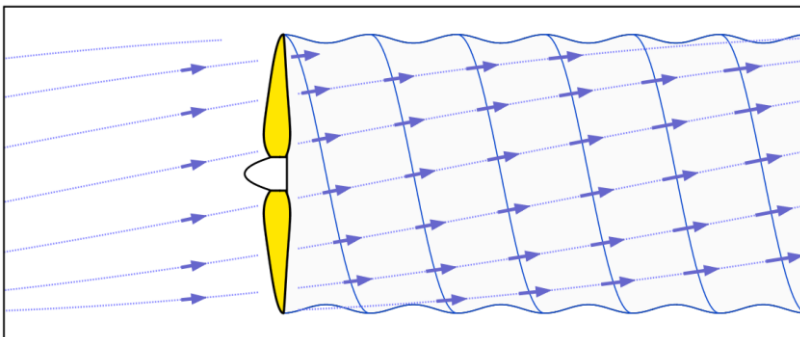


Figure 18.2 : Momentum wake of a two-bladed propeller when delivering thrust ($X = 5$). **Add V arrow.**

19 Prandtl's tip model

19.1 Introduction

In the same 1919 article where BETZ introduced his derivation of the optimal thrust distribution with respect to the rotational loss, PRANDTL added an approximate tip loss model. This model has stood the test of time remarkably well. Somewhat surprisingly, it does not use the concept of a tip vortex at all. Instead, it is based on the flow conditions around BETZ's optimal spiral wake.

Figure 18.2 showed BETZ's ARCHIMEDES screw wake from section 0, slowly translating and rotating. The spiral sheets are in actual fact drifting backward with the induced flow. It was PRANDTL's idea to invert cause and effect here. Since no air passes through the sheets, we are allowed to imagine that they are smooth, hard surfaces, themselves displacing the air to the rear instead of the other way around. In potential theory, if two flows meet the same boundary conditions, they are the same flow. If we can find a solution for this new flow, it must be identical to the flow we are looking for. There is also no sliding friction on the surfaces, so they can only push on the air at right angles to their surface.

This still does not make the problem trivial, for the ARCHIMEDES screw has a complex shape. At the center, the sheets are highly curved; but at the outer edges, they are almost flat. The left hand side of figure 19.1 replaces the ARCHIMEDES screw by a stack of flat disks, with the same spacing as the tip spirals in figure 18.1.

The disks move to the rear with the induction velocity v' , and they take the air with them, except around the edges. Some air is pushed to the side and circles back, creating a wavy outline to what was once a smooth flow tube.

PRANDTL further simplified this model by making it 2D. He made the edges into straight lines, like the straight fins of a heat sink. He then inverted cause and effect again, by considering the case where the fins are stationary, and the air flows by them at the velocity v' . The relative motion is still the same, and the flow pattern does not change. The right hand side of the figure shows this simplified model. The only sizing parameter it has is the spacing s between the fins.

From potential theory, PRANDTL was able to derive an equation for the force distribution on a fin when this system is set in motion. The outcome is an estimate for the **mass coefficient** $K(x)$ of (***), at least for the tip regions of this simple model.

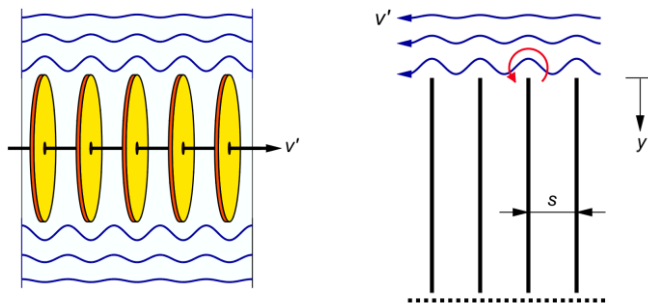


Figure 19.1 : Spiral sheets simplified to a stack of disks.

19.2 Prandtl's nearly elliptical tip shape

PRANDTL's approximate solution for $K(x)$ is traditionally denoted by $F(x)$. The Notes section discusses the shape, and PRANDTL's derivation of it in some more detail. With the y and s from figure 19.2, his equation reads:

$$F(y) = \frac{2}{\pi} \cdot \text{acos}\left(e^{-\pi \cdot \frac{y}{s}}\right) \quad (19.1)$$

As expected, $F(y)$ is zero at the edges. All of the air bypasses the fins there. Evaluating the equation at $y = 0$ in fact gives $\exp(0) = 1$ and $\text{acos}(1) = 0$.

$F(y)$ becomes 1 for $y \rightarrow \infty$. All of the air stays still deep between the fins. The equation gives $\exp(-\infty) = 0$ and $\text{acos}(0) = \frac{1}{2}\pi$ there.

In between these extremes, PRANDTL's tip shape is remarkably similar to a quarter circle with radius s . Figure 19.2 compares the two shapes. The figure confirms the strong family likeness between the elliptical mass coefficients of an optimal wing in figure 26.2, and the optimal propeller.

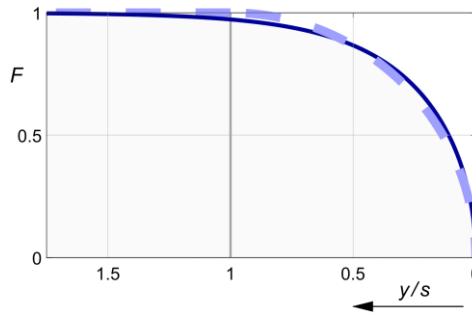


Figure 19.2 : PRANDTL's tip shape compared to a quarter circle.

19.3 The non-dimensional sheet spacing

We will fit PRANDTL's tip model (19.1) to the non-dimensional propeller. The distance between the "heat sink fins" is given by the spacing between the spiral sheets. PRANDTL applied his 2D model to the orthogonal distance between the sheet surfaces, because this is the direction in which the sheets displace the air, and it is the direction of the induction velocity v' . We will briefly follow PRANDTL's idea before converting back to the axial distance.

Figure 19.3 shows the "unrolled" outer surface of the original, smooth flow tube in non-dimensional units. All lengths are divided by the unit distance U of (2.9). The radius of the propeller is X , and the circumference of the flow tube is $2\pi X$. By figure 2.4, this makes the non-dimensional tip pitch 2π .

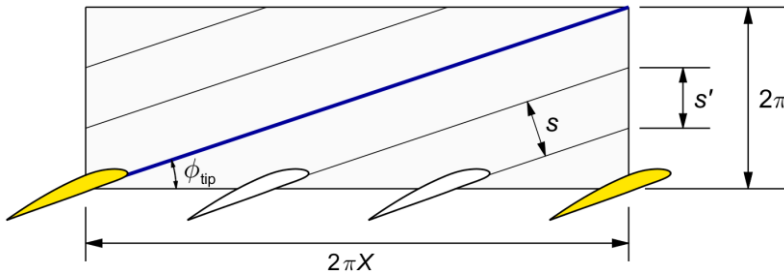


Figure 19.3 : The non-dimensional spacing between the tip spirals.

In a propeller of B blades, the tip spirals are spaced in the axial direction by the non-dimensional distance s' :

$$s' = \frac{2\pi}{B} \quad (19.2)$$

From the figure, the orthogonal distance between the sheets is shorter than the axial one by a factor of $\cos \Phi$. With (2.4) for the cosine, we have :

$$s = \frac{2\pi}{B} \cdot \cos \Phi = \frac{2\pi}{B} \cdot \frac{X}{\sqrt{1+X^2}} \quad (19.3)$$

PRANDTL based his model on this orthogonal sheet spacing, but in fact his function fits the exact vortex solution better if we use the axial spacing s' instead.

As a side benefit, this simplifies the equations. To distinguish between the two versions of s , we will use the notation $F'(x)$ (with a prime) for the version of F which uses the axial spacing of (19.2). Note $(X-x)$ is the distance from the tip at X , measured on the scale of $x \equiv \Omega r/V$.

$$F'(x) = \frac{2}{\pi} \cdot \text{acos} \left(e^{-\pi \cdot \frac{X-x}{s'}} \right) \quad (19.4)$$

This, then is PRANDTL's approximation for the mass flow factor $K(x)$ of (***).

19.4 The approximate clipped tip

Using (19.2), we can redraw figure 19.2 on the non-dimensional x scale. The tip shape will now start at a distance of $(X-x) = 2\pi/B$ from the tip.

Figure 19.4 shows the result. The plot runs from $(X-x) = 0$ at the right to $10/B$ on the left. For , i.e. from $x = 0$ to $X = 5$ for a two bladed propeller. This makes it a plot of $F'(x)$ for our standard running example of $B = 2$, $X = 5$. The tip shape starts from $(X-x) = 2\pi/2$, i.e. from $r/R = (1-\pi)/X$, or in other words from $r \cong 0.4R$. The "tip" therefore extends over more than half of the outer blade. It covers most of the disk area.

The figure also shows a simple approximation for the net area loss. PRANDTL noted that the blade tip can be replaced by a blade of full width, clipped off straight at a reduced diameter. The figure shows the cropping length ΔX which gives the same equivalent area loss as PRANDTL's tip shape. It is the following fraction of (19.2) :

$$\Delta X = \frac{\ln 2}{\pi} \cdot s \cong 0.22 \cdot s \quad (19.5)$$

The function value of $F'(x)$ at this reduced radius (before cropping) is $\frac{2}{3}$. The **math appendix** gives PRANDTL's analytical derivation.

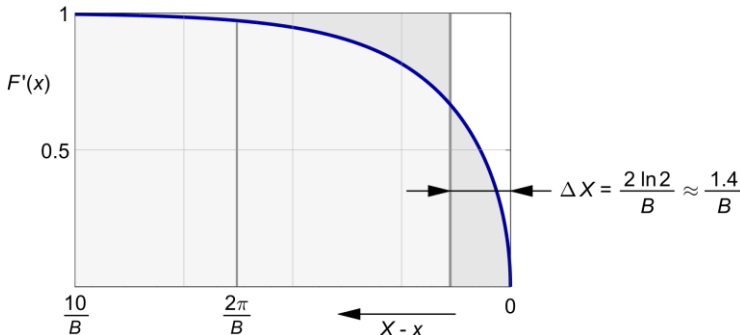


Figure 19.4 : Mass factor for $X = 5$, $B = 2$, showing cropping radius.

The quarter circle approximation gives almost the same area reduction, since the area loss of a quarter circle shape equals $1 - \frac{1}{4}\pi \cong 0.215$.

With $s' \cong 2\pi/B$, the cropping distance on the x scale becomes:

$$\Delta X = \frac{\ln 2}{\pi} \cdot \frac{2\pi}{B} = \frac{2 \ln 2}{B} \cong \frac{1.4}{B} \quad (19.6)$$

The "effective" radius of the propeller will be:

$$X' = X - \frac{2 \ln 2}{B} \cong X - \frac{1.4}{B} = X \cdot \left(1 - \frac{1.4}{B \cdot X}\right) \quad (19.7)$$

Figure 19.5 shows that the propeller is equivalent to a square tipped propeller inside a duct of this reduced radius, with no air escaping around the tips. With (19.7) for the radius, the effective disk area reduces to:

$$\begin{aligned} \pi X'^2 &= \pi X^2 \cdot \left(1 - \frac{2 \ln 2}{B \cdot X}\right)^2 \\ &= \pi X^2 \cdot \left(1 - \frac{4 \ln 2}{B \cdot X} + \left(\frac{2 \ln 2}{B \cdot X}\right)^2\right) \end{aligned} \quad (19.8)$$

If $B \cdot X \gg 1$, the quadratic term is small. PRANDTL therefore dropped it, keeping only the linear term. We will call his approximation of the overall tip mass reduction factor κ_F' :

$$\kappa_F' \cong 1 - \frac{2.77}{B \cdot X} \quad (19.9)$$

This κ_F' estimates the integral of the local mass factor $K(x)$ defined in (16.2). It is not the same as the integral κ_G in (16.7), because that contains the $\cos^2\phi$ term as well.

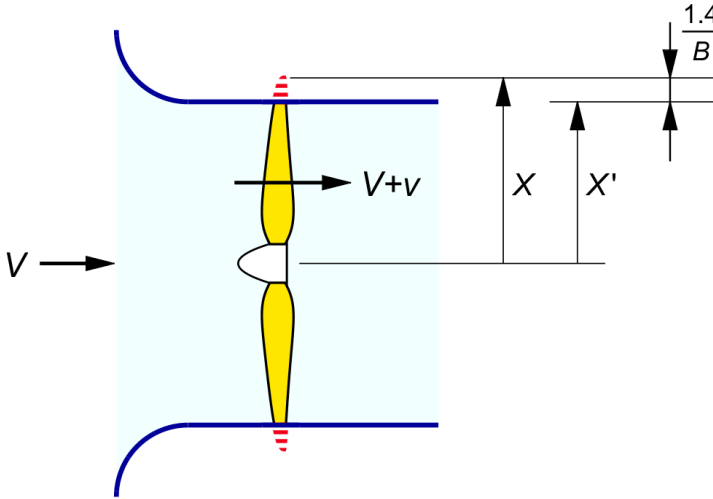


Figure 19.5 : Effective radius reduction for $B = 2, X = 5$.

19.5 The extent of Prandtl's tip

Figure 19.4 gave the factor $F'(x)$ scaled to our standard example of $X = 5$, for two blades. Figure 19.6 shows what happens when we vary these parameters. On a horizontal scale of $B.X$, the tip length is fixed at 2π , and the blade length shows up as $B.X$. This keeps the relative tip depth in the proper proportion. The tip has always the same shape, but with increasing $B.X$ it covers a shorter fraction of the blade.

Propellers with the same value of $B.X$ have the same tip length ratio, regardless whether this comes from B or from X . A propeller of $X = 10$ and $B = 2$ has the same tip ratio as a propeller of $X = 5$ and $B = 4$. Propellers with large $B.X$ have relatively small tip ratios and hence lower low tip loss.

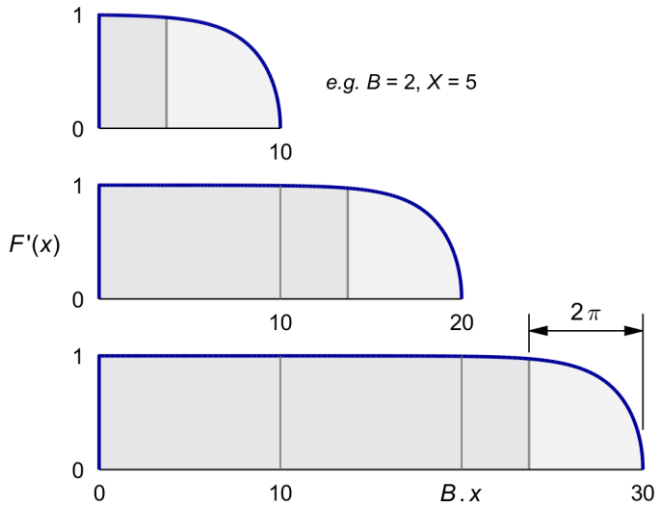


Figure 19.6: PRANDTL's tip extent as a function of $B.X$.

Figure 19.7 shows the same tip length reduction on a scale of r/R . This brings all the different blade lengths of figure 19.6 to the same length $R = 1$, compressing the tip sections $2\pi/(B.X)$ to ever blunter elliptical tips.

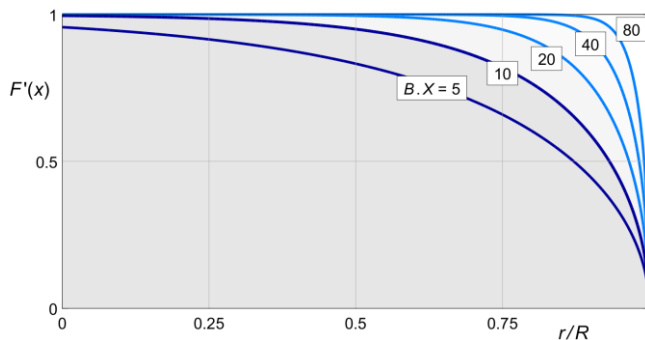


Figure 19.7: PRANDTL's tip shape for a fixed radius, at varying values of $B.X$.

20 Combining Prandtl's tip with rotation

20.1 Prandtl's tip on the x scale

So far we looked at Prandtl's approximation of the $K(x)$ as defined in (16.2). We will now look at the combined thrust reduction factor $G(x)$ of (16.8). With PRANDTL's approximation $F'(x)$ for $K(x)$, we define the obvious approximation to $G(x)$:

$$G'(x) \equiv \cos^2 \phi \cdot F'(x) \quad (20.1)$$

Figure 20.1 shows $F'(x)$ at yet another scale than figure 19.6 and figure 19.7. This figure shows $F'(x)$ on the natural scale for $\cos^2 \phi$, viz. the x scale. On this x scale, the blades have the length X . The figure shows the rotational factor $\cos^2 \phi$ factor as a red dotted line.

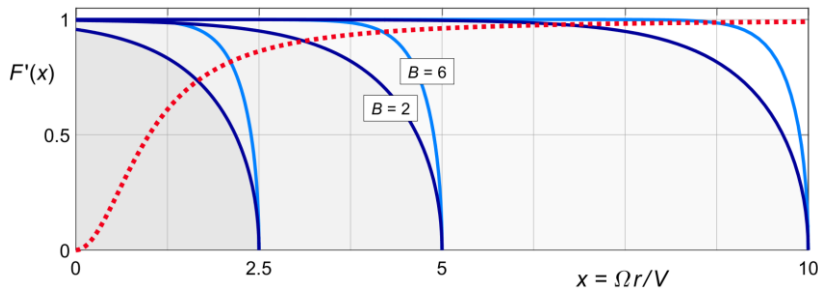


Figure 20.1 : PRANDTL's tip shape and the rotational reduction factor $\cos^2 \phi$ on the x scale.

20.2 The combined reduction

Figure 20.1 shows that the two reductions partly overlap. When they combine according to 20.1, they will both contribute to the reduction at a given radius.

The shape at the tip is mostly determined by $F'(x)$, not so much by $\cos^2 \phi$ since that factor approaches 1 for large values of x .

The shape near the root is mostly determined by the $\cos^2 \phi$ function, since the value of $F'(x)$ is nearly 1 there.

In the middle of the blade, the two reductions may overlap. When the combined value of $B \cdot X$ is small, both terms are well below 1, and the combined shape factor is strongly reduced along the whole blade.

Figure 20.2 shows the result. With large numbers of blades, the elliptical ends start looking like the straight-cropped tips of figure 8.1.

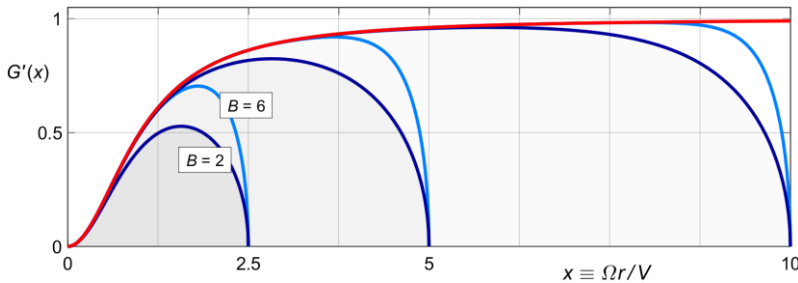


Figure 20.2 : The local thrust reduction (20.1) by $\cos^2 \phi$ and PRANDTL.

20.3 The overall reduction

In (16.7) we defined the overall thrust reduction factor κ_G , by analogy with the κ_ϕ of (9.3). We now define the equivalent for the $G'(x)$ of (20.1), calling this thrust reduction factor $\kappa_{G'}$:

$$\kappa_{G'} \equiv \frac{1}{X^2} \int_0^X \cos^2 \phi \cdot F'(x) \cdot 2x \cdot dx \quad (20.2)$$

There is no simple analytical solution to this, but we have relatively simple equations for both terms of the integrand (20.1) so this expression is easily integrated numerically by as many steps as we like (typically, 10,000), using the numerical integral formula (6.17) :

$$\kappa_{G'} \cong \frac{1}{X \cdot N} \sum_0^N \cos^2 \phi \cdot F'(x) \cdot 2x \quad (20.3)$$

...

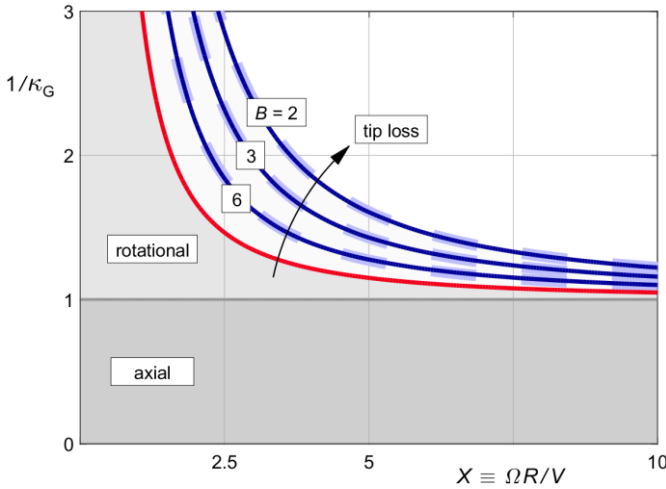


Figure 20.3 : Exact momentum loss ratio $1/\kappa_G$ in solid lines, approximation (22.2) shown dashed.

Make this for Prandtl alone here first.

21 Comparing Prandtl to the exact solution

21.1 Purpose

The previous chapters explained PRANDTL's approximate solution for the tip loss at some length. However, as we indicated in section ****, there is also an exact solution, and it has been known at least since the 1930's that there are appreciable differences between the two.

We will discuss the exact (vortex) solution in some detail in a later chapter. but we give enough of a preview here to justify the further use of PRANDTL's approximation to a far larger extent than would seem justified at first sight.

Figure 21.1 shows the exact solution in solid lines against the background of PRANDTL's approximation in thin outlines and shading. The two solutions differ markedly in the amplitude of the overall reduction, especially for lower values of X . Also, there is a real shape difference near the root of the blade.

The case for $B = 2$, $X = 5$ in the middle of the plot is the standard example in the literature since PRANDTL and GLAUERT. It shows a difference of about 10 % in the reduction between the exact $G(x)$ of ****, and PRANDTL's $G'(x)$ of ****. PRANDTL predicts visibly more thrust, and therefore better efficiency than the exact solution. A difference of 10% is really too much to be acceptable in aircraft design.

The inaccuracy is downplayed in the literature to a remarkable degree. The normal line of defense is that the theory is at least valid for larger values of X and B . But we can do much better than that.

We will now show that the difference in shape (though not in area), is in fact negligible if we apply a correction to restore the overall thrust. We will discuss the efficiency in the next chapter.

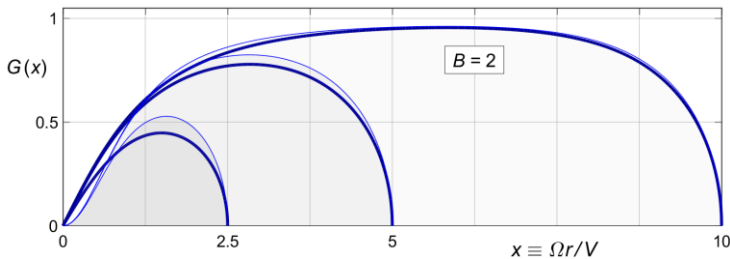


Figure 21.1 : The local thrust reduction exact (solid lines), compared to Prandtl (thin lines).

21.2 Inflating back to unit thrust

Redistribution, yes. Equal efficiency, no.

After finding $\kappa G'$, repeating the steps from (Error! Reference source not found.) to (Error! Reference source not found.) for this, yields :

$$dT' = \frac{T'}{\kappa_{G'}} \cdot \cos^2 \phi \cdot F'(x) \cdot \frac{2x \cdot dx}{X^2} \quad (21.1)$$

Do the steps, including C. Maybe use C_K .

Emphasize first the common ground and the difference with κ_ϕ .

Or decide on a place to tell the story about types of loss and refer to that.

Changing the reduction for equal loss into a redistribution for equal thrust, we have :

$$\frac{G'(x)}{\kappa_{G'}} = \frac{\cos^2 \phi \cdot F'(x)}{\kappa_{G'}} \quad (21.2)$$

We will now step ahead of the results of the exact vortex solution.

Figure 21.2 shows the result. It is the same as figure 10.1, **only this time not just for the induction**, but for the thrust. The fat lines show the results for the exact solution. The lighter lines and the shading give the results for PRANDTL's approximation (20.1). The figure normalizes both to unit thrust, each by their own correction factor $1/\kappa_G$ c.q. $1/\kappa_{G'}$.

When normalized like this, the exact solution and PRANDTL's approximation become very similar. In fact, they are virtually indistinguishable for $x = 2.5$ or larger. The only difference in shape is near the hub. For low blade counts ($B = 2$ and $B = 3$), the exact distribution starts with a straight slope from zero instead of the parabolic x^2 -like start of the $\cos^2 \phi$ function. It would be nice to have a physical explanation for this, but the literature does not provide one.

The difference near the hub is not a serious problem for several reasons : the area close to the hub is a very small part of the disk area in the first place (only a quarter of the disk area falls within half the radius), the blades are shaped for mechanical strength rather than aerodynamics near the hub, and the area closest to the hub is usually covered by a spinner.

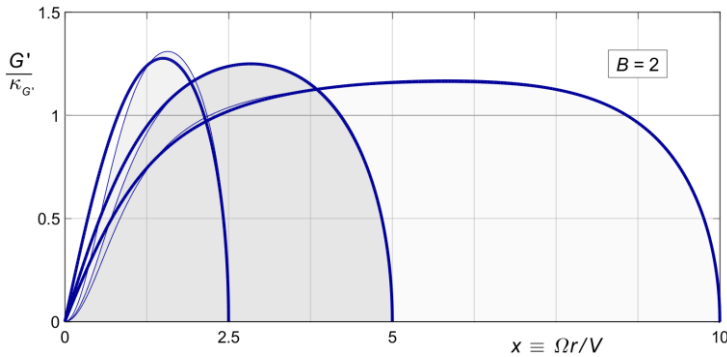


Figure 21.2 : Comparing exact solution (fat line) to Prandtl (thin), after inflating both to unit thrust.

22 Thrust reduction and efficiency

22.1 The inverse of the exact thrust reduction

Unlike κ_ϕ in (9.6), the exact κ_G of (**Error! Reference source not found.**) is not easily expressed analytically. Chapter **** gives an idea of the complexity of calculating $G(x)$. Even today, its calculation is somewhat slow. Once $G(x)$ is found however, the numerical integration into κ_G is easily accomplished by (6.17).

Like we found for κ_ϕ in section 9.4., the value of κ_G by itself is not very intuitive, It is the inverse $1/\kappa_G$ that determines the momentum loss ratio (**Error! Reference source not found.**), just as it was by (****) for rotation alone.

Figure 22.1 shows the graph of $1/\kappa_G$ versus X . for a few blade counts. The lowest line in the family is the one for $B = \infty$. This line is identical to $1/\kappa_\phi$ in figure 9.2. For infinitely many blades, there is no tip loss.

The visual impression of figure 22.1 is that the lines with tip loss look similar in shape to that of the rotational loss alone. The growth of the tip loss with decreasing number of blades looks roughly proportional to $1/B$. This is not completely unexpected, in view of the Prandtl's estimate of the effective area reduction in (19.9).

For a better intuition and as an aid to preliminary design, we explore a simple approximation for $1/\kappa_G$ from B and X . Equation (9.13) gave a simple approximation of the rotational momentum overhead on the axial momentum loss, which depends only on a power of $1/X$. We now find a similar but even simpler approximation for the extra overhead due to tip loss, by a linear multiplication factor in $1/B$:

$$\frac{1}{\kappa_G} \cong 1 + \left(\frac{\pi}{2 \cdot X} \right)^{1.7} \cdot \left(1 + \frac{6}{B} \right) \quad (22.1)$$

Approximation (22.1) does not have much physical meaning, but nevertheless it is very instructive.

For $B = \infty$, the overhead is just the rotational one. For $B = 6$, the factor $(1 + 6/B)$ equals 2. The extra tip loss equals the original rotational loss. For $B = 3$, the factor is 3, and for $B = 2$, the factor is 4. This simple rule of thumb is rarely noted in the literature.

Add relative merits of duct, number of blades, counterrotation, and swirl recovery (flow straightener) vanes.

At the cost of some simplicity, the fit can be made almost perfect by adding a term in B^2 and a little fudge factor:

$$\frac{1}{\kappa_G} \cong 1 + \left(\frac{\pi}{2 \cdot X} \right)^{1.7} \cdot \left(1 + \frac{4.5}{B} + \frac{3.15}{B^2} \right) + \frac{0.1}{B} \quad (22.2)$$

This formula can be used in (**Error! Reference source not found.** to get a near perfect estimate of the exact momentum efficiency of the rotationally optimal propeller with tip loss. There is no real need to calculate the shape of the distribution $G(x)$ for this.

Figure 22.1 shows (22.1) (well, actually (22.2)) in dashed lines under the exact solution. The fit is nearly perfect.

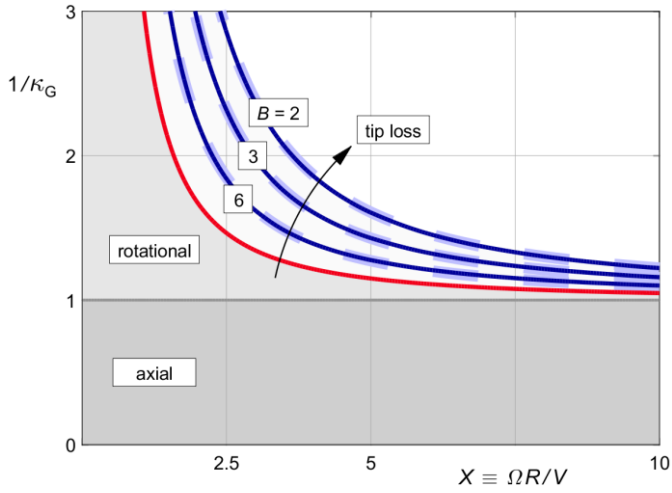


Figure 22.1 : Exact momentum loss ratio $1/\kappa_G$ in solid lines, approximation (22.2) shown dashed.

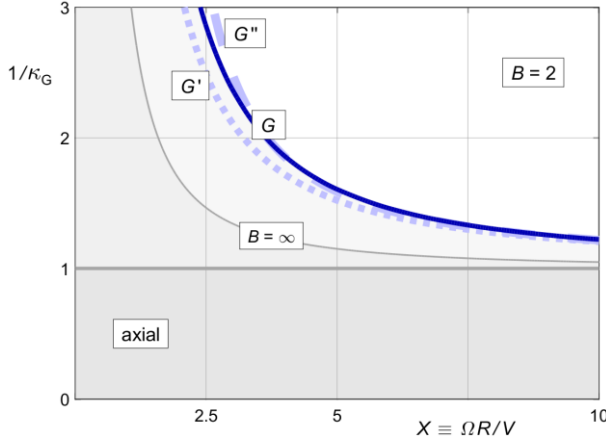


Figure 22.2 : Prandtl approximations to $1/\kappa_G$.

22.2 A simplistic Prandtl integration

It is worth mentioning another approximation of the exact $1/\kappa_G$, based on a simplistic view of PRANDTL's approximate thrust distribution.

We know that PRANDTL's approximation makes sense only if the rotational reduction area near the hub and the tip reduction area do not overlap. This requires roughly $X \geq 5$. The tip should also certainly not extend beyond the hub, which requires $X > 2\pi/B$. In other words, $B \cdot X \gg 2\pi$.

But we can lightheartedly ignore the fact, and pretend that the two effects are independent. We can then simply multiply the two reductions as if they were separate. We will call this combined reduction $\kappa_{G''}$:

$$\kappa_{G''} \equiv \kappa_\phi \cdot \kappa_{F'} \quad (22.3)$$

We know the analytical value of κ_ϕ from (9.6), and we found an approximation for the PRANDTL tip effect $\kappa_{F'}$ in (19.9). Taken together they give :

$$\kappa_{G''} \equiv \kappa_\phi \cdot \left(1 - \frac{2.77}{B \cdot X}\right) \quad (22.4)$$

Figure 22.2 shows $1/\kappa_{G''}$ as a dashed line. Somewhat surprisingly, it gives a decent match with the exact result, at least for $X \geq 4$, and certainly a better match throughout than the PRANDTL integral of (Error! Reference source not found.).

This very simplistic equation becomes quite exact for higher X and B . This is not surprising, since all of PRANDTL's simplifying assumptions apply there.

Approximations (22.2) and (22.4) each have their merits. The first gives better results for lower values of $B \cdot X$, and it provides useful information on the trade-off between rotational and tip loss. The second gives a better confirmation of the geometric relevance of the product $B \cdot X$ via (19.9).

23 Blade chords

23.1 The tip effect on the blade chord

Once κ_G is found, the chords that follow are :

$$B \cdot c' = \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_G} \cdot F'(x) \cdot \cos^2 \phi \cdot \sin \phi \quad (23.1)$$

This gives a very good approximation of the chords calculated by the exact method.

Equation (11.20) gave the blade chord for the rotationally optimized $b(x)$ of (9.9). The tip loss changes $b(x)$ to (21.2), which includes the term $K(x)$ and changes the integrated thrust reduction from κ_ϕ to κ_G . The new chord is:

$$B \cdot \frac{c}{R} = \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_G} \cdot K(x) \cdot \frac{1}{X} \cdot \cos^2 \phi \cdot \sin \phi \quad (23.2)$$

Figure 23.1 shows the result. It is the direct counterpart of figure 11.3 in terms of thrust and parameter variation, only this time including the tip loss.

Like in figure 21.2, the exact solution is drawn in solid lines, and the PRANDTL approximation in thin lines. The figure once again shows the difference in shape near the hub.

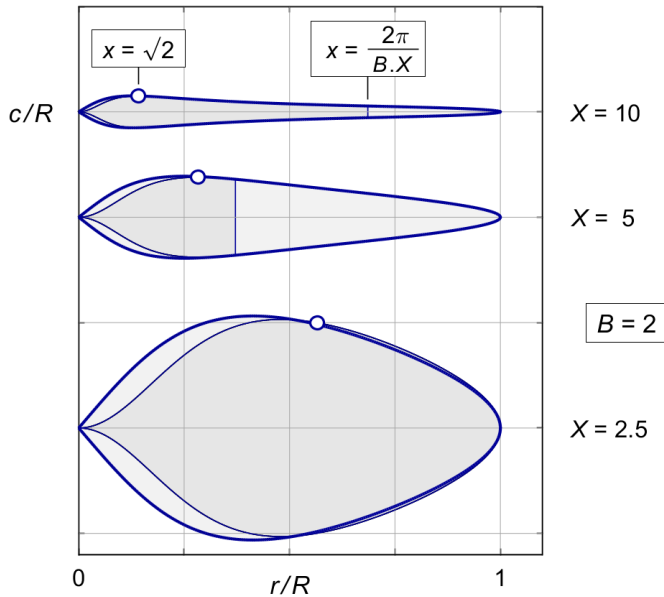


Figure 23.1 : Optimal chords with tip loss.
Thrust $T' = 0.1$, $C_L = 0.4$, $B = 2$.

23.2 Blade shape variation with X and B

Chapter 11 explored the basic parameters influencing the blade chord. It gave a simple approximation for the blade chord at $0.7R$ in (11.24). This approximation still holds if we replace κ_ϕ by κ_G , and take into account that the correction by $K(x)$ is usually small at $0.7R$.

The chord at $0.7R$ varies with $1/X^2$, but the largest chord width varies only by $1/X$. This means that there is a systematic variation with $1/X$ in the taper ratio between the widest point and the tip region. Figure 23.2 shows this variation. The tip speed ratio X decreases when moving from left to right between the pictures. The variation in taper ratio is most visible in the top row, for almost infinitely many blades. The location of the widest point at $r/R = \sqrt{2}/X$ is indicated by circle markers.

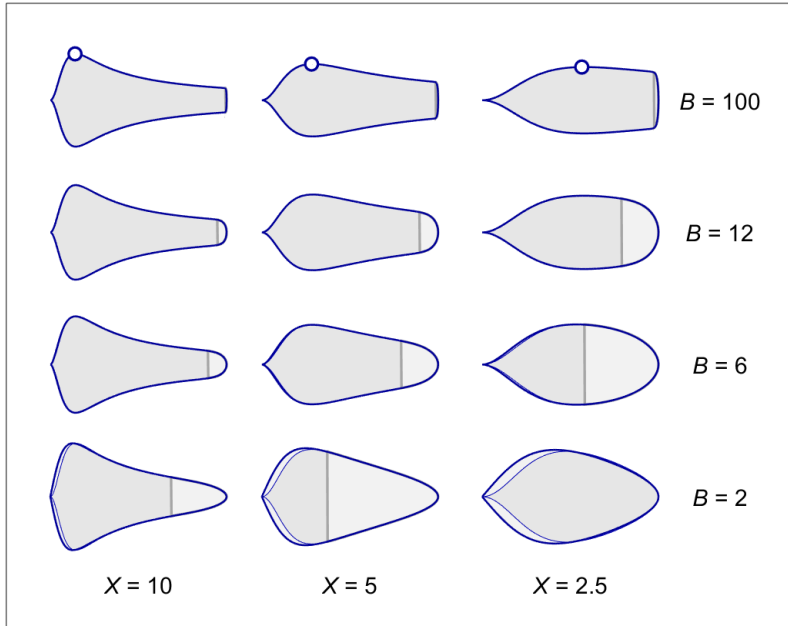


Figure 23.2 : Chord shapes for varying X and B .

Moving along the vertical axis, the plots show the effect of varying the number of blades. Figure 19.7 showed the relative extent $\Delta r/R = 2\pi/(B \cdot X)$ of PRANDTL's tip. For $X = 2.5$, $B = 2$ the whole blade is under the influence of the tip function. For our standard example of $X = 5$, $B = 2$ it extends over $2\pi/(2 \cdot 5) = 0.6$, which is more than half of the blade length.

Blades with high X are shaped like tapered, pointed saddles. Blades with low X and low B are shaped like a rounded paddle, or a spoon.

The blades in this figure 23.2 were all scaled to roughly the same blade width, to show the shape to good advantage. In reality, the variation in overall width with $1/X^2$ as shown in figure 20.2, completely dominates the shape effect.

23.3 Tip sweepback

The increase of the lift force on a wing section with angle of attack acts at the quarter chord point. The classical planform of most wings and many propeller blades therefore has a straight centerline at this percentage of chord.

More than wings though, propeller blade centerlines are subject to practical constraints such as bending moments, centrifugal forces, torsion, and

aero-elasticity, and from the beginning there has also been a school of thought, maybe for reasons of fabrication, to keep the trailing edge straight. This typically gives the leading edges sweepback near the tips.

The same straight trailing edges are seen traditionally in aircraft fins and stabilizers, and today in aircraft wings as well. There is some evidence that a straight trailing edge gives a flatter beginning of the wake sheet, and slightly less induced drag, than the "straight" wing.

In propellers, there may be a secondary advantage to the straight trailing edge. Even at moderate flying speeds, the blade tips can approach high Mach numbers with attendant noise and drag rise. Swept-back tips may help to alleviate these problems. In ships, they may help to shed cavitation bubbles.

A simple, though not necessarily optimal way to achieve some tip sweepback is by applying the tip reduction $K(x)$ one-sided from the trailing edge, instead of symmetrically around the centerline. This keeps the trailing edge as it was before the tip reduction, like in figure 11.3.

The exact solution does not give $K(x)$ as a separate factor, so we will use PRANDTL's approximation $F'(x)$ instead. With PRANDTL's approximation, the chord comes from (23.1). Leaving out $F'(x)$ gives :

$$B \cdot c' = \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_{G'}} \cdot \cos^2 \phi \cdot \sin \phi \quad (23.3)$$

This is a larger outline than (23.2), with a larger thrust, and with a straight cropped tip as in figure 11.3. Before sweepback, it will envelop the shape (23.3) as reduced by $K(x)$, c.q. $F'(x)$.

The PRANDTL tip will modify the chords by a factor of $F'(x)$. The difference with (23.3) will be :

$$\Delta(B \cdot c') = (F(x) - 1) \cdot \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_{G'}} \cdot \cos^2 \phi \cdot \sin \phi \quad (23.4)$$

Until now, we applied this difference symmetrically before and aft of the centerline. This will sweep the leading edge back by half, and the trailing edge forward by half. If we do not wish the trailing edge to sweep forward, then we will have to sweep back the centerline. We will call the sweepback $\Delta y'$, defined as positive to the rear :

$$\Delta y' = \frac{1}{2} \cdot (1 - F'(x)) \cdot \frac{1}{B} \cdot \frac{4\pi}{C_L} \cdot \frac{T'}{\kappa_{G'}} \cdot \cos^2 \phi \cdot \sin \phi \quad (23.5)$$

Figure 23.3 shows the result. The swept centerline is shown in grey dashes. It can be applied to PRANDTL's approximation, but also to the exact solution, since the two are very similar near the tip. The solid line shows the exact vortex solution after applying the sweepback. The dashed line shows the "envelope" (23.2) before the tip reduction.

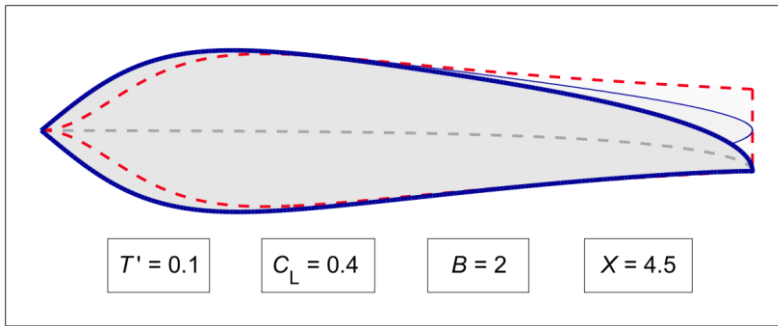


Figure 23.3 : Sweepback (23.5), applied to the exact solution.
Envelope (23.2) shown in red dashes.

24 The optimal tip speed ratio with tip loss

24.1 The viscous loss with tip loss

Viscous loss occurs mainly in the outboard sections of the blades. Tip loss moves thrust from outboard to inboard, so we can expect a beneficial side effect of the tip loss (for the same overall thrust) on the viscous loss. This turns out to be a modest effect, however.

From (14.2), the viscous loss is directly proportional to the local thrust. Equations (14.3) to (14.5) applied this to the rotationally optimal induction, using the effective local ring area dm' before tip loss. The integral P'_v turned out to be the very simple (14.6).

If we repeat the same steps with the $dm'_K(x)$ of (16.2) which includes the tip loss, the viscous loss integral for the rotationally optimal propeller changes from (14.5) to:

$$\frac{P'_v}{T'} = \frac{\epsilon}{\kappa_G} \cdot \frac{1}{X^2} \int_0^X K(x) \cdot 2x^2 \cdot dx \quad (24.1)$$

This differs from (14.5) in two ways: the thrust is reduced locally by $K(x)$ as well as by just the $\cos^2 \phi$ which led to the simple result (14.5) and it is scaled up uniformly by $1/\kappa_G$ instead of $1/\kappa_\phi$, to restore the original design thrust T' after both reductions, including $K(x)$.

Since $K(x)$ is not easily separated put of from the exact solution $G(x)$, and $F'(x)$ is a good alternative for larger values of x , we can replace this by:

$$\frac{P'_v}{T'} = \frac{\epsilon}{\kappa_{G'}} \cdot \frac{1}{X^2} \int_0^X F'(x) \cdot 2x^2 \cdot dx \quad (24.2)$$

Figure 24.1 shows the result. It repeats figure 14.2 for $B = \infty$, but this time including $B = 4$ and $B = 2$. The improvement is quite modest. For higher values of X , the effective X in the $\frac{2}{3}X$ of (14.6) is lowered by approximately $1.25/B$. This is reminiscent of PRANDTL's clipping distance $\Delta X = 1.4/B$ in (19.6). The improvement disappears for lower values of X .

Maybe plot ΔX after all.

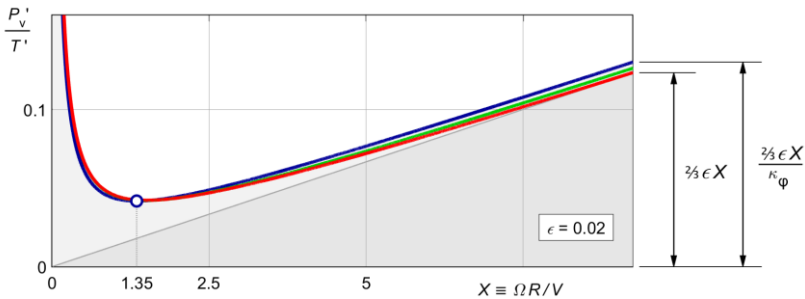


Figure 24.1 : The overall viscous loss ratio (24.2) as a function of X , for $B = 2$ (red), $B = 4$ (green) and $B = \infty$ (blue),

For the calculation of the optimum X , we only need the slope of (24.2) with X . For a first approximation we will assume that the slope does not change much with tip loss. For larger values of B , X at least, we will use a schematic version of (14.6) :

$$\frac{P'_v}{T'} \cong \frac{1}{\kappa_G} \cdot \frac{2}{3} \cdot \epsilon \cdot \left(X - \frac{1.25}{B} \right) \quad (24.3)$$

24.2 The optimum tip speed ratio with tip loss

Maybe open the chapter with this section.

Section 14.4 showed that there is an optimal balance between momentum loss and viscous, roughly near $X = 3$. In the presence of tip loss, the optimum shifts to higher values of X .

Figure 24.2 is similar to figure 14.3, only this time varying the number of blades around the standard example of $T' = 0.1$ and $\epsilon = 0.02$, instead of varying ϵ for $B = \infty$. The lowest line in the family of curves in figure 24.2 is the same as the solid line for $\epsilon = 0.02$ in figure 14.3.

The topmost line is the case for our standard example for $B = 2$. The optimum shifts from $X = 2.9$ for $B = \infty$ to $X = 4.8$ for the standard example.

It is probably no coincidence that our standard typical lightplane has a tip speed ratio of $X = 4.5$, and the examples used historically by both PRANDTL and GLAUERT have tip speed ratios of $X = 5$.

The optimum value for X shifts to the right with decreasing number of blades, and this is mainly due to the shift in the amplitude and slope of the momentum loss with tip loss as shown in figure 22.1. The parallel shift of the viscous loss in figure 24.1 by itself hardly affects the location of the optimum X . We can confirm this by repeating the derivation for the optimum X by the derivative of the combined loss, just like we did in section 14.4 before tip loss.

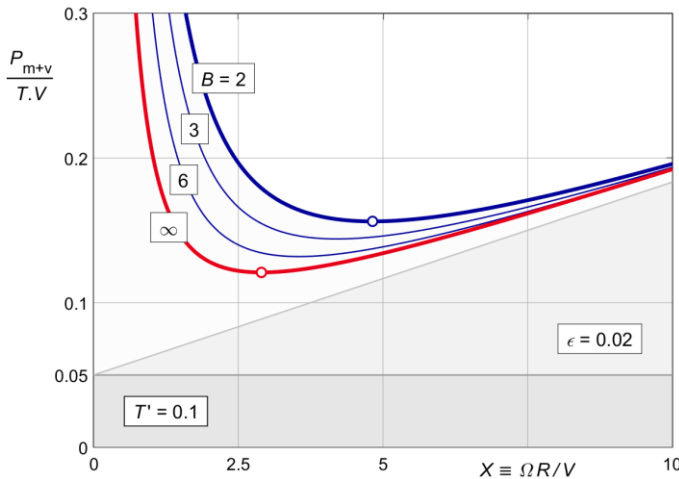


Figure 24.2 : Combined loss shape for $\epsilon/T' = 0.2$, for various blade counts.

The combined loss from (****) and (24.3) is :

$$\frac{P'_{m+v}}{T'} \equiv \frac{P'_m}{T'} + \frac{P'_v}{T'} = \frac{1/2 T'}{\kappa_G} + \frac{2/3 \cdot \epsilon \cdot \left(X - \frac{1.25}{B}\right)}{\kappa_G} \quad (24.4)$$

By the same steps leading to (14.9) we have from the derivative:

$$\frac{\kappa_G}{(\kappa_G)_X} = \frac{3}{4} \cdot \frac{T'}{\epsilon} + \left(X - \frac{1.25}{B}\right) \quad (24.5)$$

Figure 24.3 gives the result. The upper curve matches figure 14.4 for $B = \infty$. For decreasing B the curves are wider, like the curves for $1/\kappa_G$ in figure 22.1.

The optimum values for X shift to the right with decreasing number of blades. From the figure we see that this is almost entirely due to the change in the κ_G curves. The small shift in the viscous drag curves of figure 24.1 has far less influence.

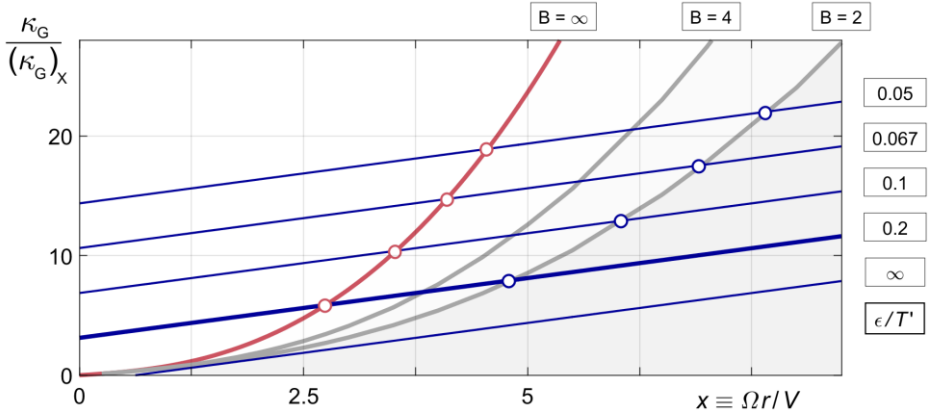


Figure 24.3 : Finding the optimum X with (24.5).
Generalization of figure 14.4.

24.3 A note on numerical differentiation

Equation (24.5) needed the derivative of κ_G with respect to X . The value of κ_G comes from the exact vortex solution and is only available at isolated points, usually gathered in a table. The derivative can therefore only be found by differencing these points. The points may not always be equidistant in X . The graph for the numerical derivative is noticeably smoother if more weight is given to the difference with the nearest point. If the points are ordered like X_0, X_1, X_2 etc., then the following formula will be useful :

$$\begin{aligned} \frac{d}{dX} \kappa_G(X_1) &\cong \frac{X_2 - X_1}{X_2 - X_0} \cdot \left(\frac{\kappa_G(X_1) - \kappa_G(X_0)}{X_1 - X_0} \right) \\ &+ \frac{X_1 - X_0}{X_2 - X_0} \cdot \left(\frac{\kappa_G(X_2) - \kappa_G(X_1)}{X_2 - X_1} \right) \end{aligned} \quad (24.6)$$

- use $G(x) / \cos^2 \phi / \kappa_G$ (???)
- make a plot of X_{opt} versus ϵ / T' .

25 Viscous optimization with tip loss

25.1 Introduction

Chapter 15 gave the combined optimization for rotational and viscous loss by the optimal distribution (15.7). As we have seen several times before, the local reduction of the induction will not change with tip loss, but the overall thrust reduction and inflation factor will. (15.9)

The local loss ratio depends on the local radius x (via the local pitch angle ϕ), and on the design value of the marginal loss ratio A . This controls the overall thrust via the solution (15.11) of the thrust integral (15.9).

With tip loss, the local induction scheduling (15.7) does not change, but the thrust integral (15.9) does, because it will now contain the mass flow term $K(x)$. Combining the $b(x)$ from (15.7) with the light loading dM'_K including tip loss of (26.8), the thrust integral (6.13) becomes:

$$\begin{aligned} dT' &= (A \cdot \cos^2 \phi - \epsilon \cdot x) \cdot dm'_K \\ &= (A \cdot \cos^2 \phi - \epsilon \cdot x) \cdot K(x) \cdot \frac{2x \cdot dx}{X^2} \end{aligned} \quad (25.1)$$

Strictly speaking, the function $K(x)$ is normally derived for BETZ's regular ARCHIMEDES screw, which is itself derived from the unmodified $\cos^2 \phi$ distribution. But the exact shape of $K(x)$ is probably not too critical to the viscous loss improvement. We will assume that something close to $K(x)$ also holds for the distribution (15.7) as modified by the viscous optimization.

In fact, we will use PRANDTL's approximation $F'(x)$ for a first assessment. The integration of dT' from (25.1) then changes to :

$$T' = \frac{1}{X^2} \int_0^X (A \cdot \cos^2 \phi - \epsilon \cdot x) \cdot F'(x) \cdot 2x \cdot dx \quad (25.2)$$

We met both terms under the integral before. For the left hand term, in (20.2) we defined :

$$A \cdot \frac{1}{X^2} \int_0^X \cos^2 \phi \cdot F'(x) \cdot 2x \cdot dx \equiv A \cdot \kappa_{G'} \quad (25.3)$$

For the right hand term, (24.3) gave an approximation for (24.2):

$$\frac{1}{X^2} \int_0^X -\epsilon \cdot x \cdot F'(x) \cdot 2x \cdot dx \equiv -\epsilon \cdot X_\mu \quad (25.4)$$

Combining the two gives an overall thrust of :

$$T' = A \cdot \kappa_{G'} - \epsilon \cdot X_\mu \quad (25.5)$$

The marginal loss ratio A then becomes :

$$A = \frac{T'}{\kappa_{G'}} + \frac{\epsilon \cdot X_\mu}{\kappa_{G'}} \quad (25.6)$$

This is a combination of the C with tip loss from (9.8) with the viscous optimization.

The value of A is higher the (15.11) before tip loss, because the reduced area at the tips gives higher local loading on the active disk area. As a small compensation, X_μ replaces the $\frac{2}{3} X$ of (15.11) because on average, the thrust is delivered slightly more inboard.

Substituting the A of (25.6) into (****) changes the distribution (15.12) into:

$$b(x) = (b_o + \epsilon \cdot X_\mu) \cdot \frac{\cos^2 \phi \cdot F'(x)}{\kappa_{G'}} - \epsilon \cdot x \cdot F'(x) \quad (25.7)$$

We can make this solution more exact by replacing the $G'(x)/\kappa_{G'} \equiv \cos^2 \phi \cdot F'(x)/\kappa_{G'}$ from (****) by the exact solution $G(x)/\kappa_G$ from (****):

$$b(x) = (b_o + \epsilon \cdot X_\mu) \cdot \frac{G(x)}{\kappa_G} - \epsilon \cdot x \cdot F'(x) \quad (25.8)$$

The right hand term $\epsilon \cdot x \cdot F'(x)$ is not so easily replaced by the exact version, but this is less critical. The correction term is only important near the tips, where PRANDTL's approximation is already reasonably accurate.

Figure 25.1 gives a sketch of the principle, similar to 15.4 for the case without tip loss.

Figure 25.2 gives the result for our standard example of $X = 5$, $T' = 0.1$ and $\epsilon = 0.02$.

old remark : Find integral of F'/κ_F , see [24089 PLdocx](#).

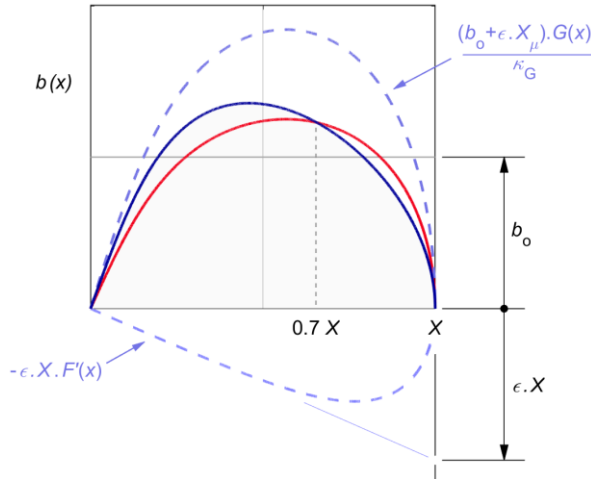


Figure 25.1: Sketch of equation (25.8).
Red line without viscous optimization.
Leave that for next figure.

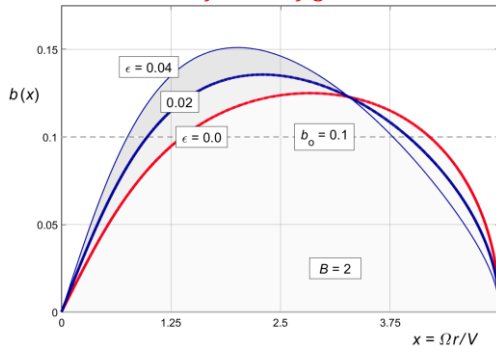


Figure 25.2: Optimal thrust distributions for $B = 2$,

varying profile drag.

Design friction $\epsilon = 0$ equals the non-optimized case.

26 The vortex solution for the tip loss

26.1 Introduction

TBW.

The exact vortex solution (for the optimal case) was given in the 1930's by GOLDSTEIN and others, but it was not available in an easy to use form until fairly recently.

Before leaving the subject of tip loss, we will give a quick impression of the "exact" vortex solution for the wake of the optimal propeller.

This chapter is for interested readers only, others can skip it.. It is not needed to follow the rest of the text.

- Biot-Savart. line vortex.
- spiral vortex / vortex rings / solenoid.
- solution for the Archimedes screw wake (à la the elliptical wing).
 - historical note :
 - analytical method, Bessel functions that were not all tabulated.
 - Tibery & Wrench, Wald.
 - mutual induction. Solve an $n \times n$ matrix.

The essence of the tip problem is finding the function $K(x)$ defined in (****). Like in the wing, the only true solution is to find the vortex distribution which matches the desired induction pattern.

26.2 The parallel with the wing

Propellers are rotating wings. For a better understanding of the tip loss in propellers, a quick impression of the tip loss of a wing is in order.

A wing deflects air downward. It leaves a downward velocity or momentum in its wake, just like the propeller leaves behind a horizontal velocity and momentum. NEWTON's law for the lift of a wing is exactly the same as (4.7) for the thrust of a propeller :

$$L = m \cdot w \quad (26.1)$$

This time w is the vertical downwash velocity in the far wake of the wing. For reasons entirely similar to those in the propeller, the downwash velocity v at the wing is exactly half that in the far wake. Adding the vertical downwash velocity v to the horizontal flying velocity V , the wing deflects the air passing over it by the so-called "induced" angle α_i : *at the lifting line*

$$\alpha_i = \frac{v}{V} = \frac{1}{2} \frac{w}{V} \quad (26.2)$$

At every wing cross-section, the lift vector is tilted backward by this local downwash angle α_i . The momentum loss ratio (the induced power loss per lift) is equal to this tilt angle. By the same reasoning as for the marginal loss of a propeller blade element, it pays to move wing lift from a region with larger (marginal) loss to a region with lower loss. This process ends when the downwash angles are the same all along the span, and so this is the optimum downwash distribution. For a uniform downwash velocity, the entire wing "sees" a flow which is inclined by the same angle. Since the wing is effectively "climbing" in its own local downwash, or alternatively: we can imag-

ine the lift in horizontal flight inclined backward by the local downwash angle, we obtain for the horizontal force associated with the lift, the so-called "induced drag":

$$\frac{D}{L} = \alpha_i \equiv \frac{1}{2} \frac{w}{V} \quad (26.3)$$

Though not strictly an efficiency, this is the direct equivalent of the momentum loss ratio of the ideal actuator disk in (**Error! Reference source not found.**).

This equation gives us the momentum loss ratio of the wing as a function of the downwash velocity. But it does not give us the lift. According to (26.1), for this we also need the deflected mass flow m^* . By the reasoning so far, we have no way of knowing the size of this mass flow.

PRANDTL established a model for the flow around the wing based on a system of line vortices. New vorticity along the span of the wing is continuously created in the boundary layer on the wing surface. This vorticity gathers in a line vortex across the span which is called the "lifting line". This vortex bends down air, and this creates lift. Details on this mechanism are found in Appendix C on "Vortex Theory".

The constantly created vorticity does not accumulate indefinitely on the wing. Its growth is continuously left behind in the atmosphere as "tip" vortices trailing straight back. Figure 26.1 gives an impression of the vortex system, and the flow it induces in the wake. In reality, "tip" vortices stream backward from every location where the lift changes, not just from the tips.

The system with only a single tip vortex gives more downflow near the tips than in the center, so it is definitely not optimal.

"After various attempts" to quote PRANDTL, his students found a spanwise lift distribution which gives the desired uniform downwash at the lifting line. The shape turned out to be an ellipse.

For mathematical reasons, vortices cannot change strength in mid air. The only vortices possible in the theory are continuous vortex lines, either in a closed loop or extending to infinity.

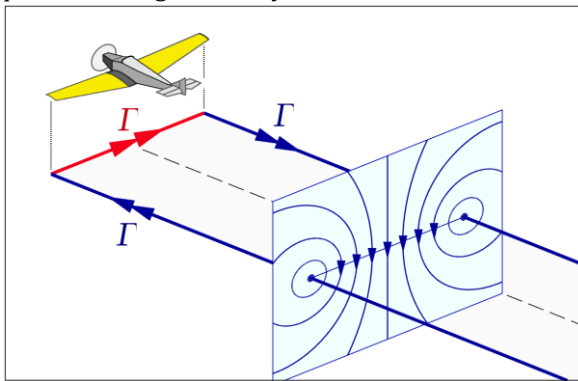


Figure 26.1 : The downwash of a single tip vortex.

The bound vortex and the tip vortices form a continuous vortex line. The tip vortices, and only the tip vortices, explain the downwash w in the far wake. The wing and its lifting line have receded far away into the distance and can no longer influence the nearby air.

On the other hand, the bound vortex, and only the bound vortex, explains the lift L at the wing. It is the bound vortex which bends down the air at the lifting line. Since the bound vortex and the tip vortices are a single closed loop, they are identical in strength. This gives a direct relation between the lift L and the downwash w . Using these in (26.1) gives the apparent value of m^* . The value for the elliptical distribution turns out to be :

$$m^* = \frac{L}{w} = \rho \cdot V \cdot \frac{1}{4} \pi b^2 \quad (26.4)$$

Here b is the span of the wing. The mental model (but it is no more than that) is that the wing bends down a circular tube of air, with the wing as its diameter, by a constant local angle of α_i . *Perhaps define an m' !!*

A similar mass ratio between downflow and lift must also hold for every slice of the wing :

$$dm^* = \frac{dL}{w} \quad (26.5)$$

The solution for uniform downwash w gives an elliptical distribution of the lift dL . This means that the same holds for the mass flow dm^* . This neatly fits the idea of deflecting the mass in vertical slices of a circular tube of air.

Figure 26.2 shows the idea schematically. The nice match between the circle in (26.4) and elliptical lift and mass distribution is not much more than a coincidence, as we can see from figure 26.1. The actual flow pattern extends for several span lengths in all directions, and the only place where it really matches the simplified model of figure 26.2 is at the lifting line.

Applying a constant downwash w at the lifting line allows us to calculate the optimum wing efficiency. Using (26.4) in (26.3) we have :

$$\frac{D}{L} = \frac{1}{2} \frac{w}{V} = \frac{1}{4} \cdot \frac{L}{\frac{1}{2} \rho V^2 \cdot \frac{1}{4} \pi b^2} \quad (26.6)$$

This is the well known equation for the induced drag of a wing. It is the direct equivalent of (**Error! Reference source not found.**) and (5.12) for the momentum loss ratio of an ideal actuator disk. In aerodynamics textbooks, this so-called "induced drag" is often rewritten in terms of the wing aspect ratio. This is unfortunate, because it confuses the issue. The induced drag depends on the square of the span, and this is independent of the wing aspect ratio.

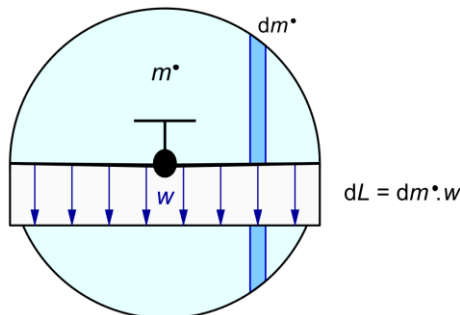


Figure 26.2 : The lift on an elliptical wing.

26.3 The wing analogy applied to the propeller

There is a close analogy between the tip loss of a wing and that of a propeller. They both reduce the local mass flow dm^* , and therefore they both reduce the thrust, c.q. the lift, towards the tip.

The optimal wing has uniform downflow. For the propeller, $b(x)$ is scheduled with $\cos^2\phi$ for minimal rotational loss.

Like in the wing, we can only find the thrust distribution by finding the appropriate vortex pattern. Once the thrust distribution is found from this, the local ratio between thrust and induction is by definition the effective value of the local mass flow dm^* .

The vortex field around an optimal propeller is more complicated than that around a wing. Figure 26.3 shows a single vortex line for a single propeller blade, similar to figure 26.1 for a wing. The path of the blade tip is a spiral, and the "outer" tip vortex of the blade trails back along this spiral path. The frontispiece photograph of a Hawker Hurricane ticking over on a moist day shows that these spiral tip vortices are more than just a theoretical model.

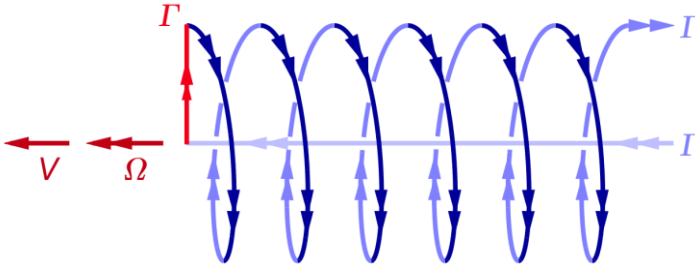


Figure 26.3 : The vortex system of a single propeller blade.

The spiral lines makes the math more complex than for the straight vortices trailing from the wing tips. But the optimal pattern has been solved, and in some ways the result is very similar to the elliptical lift distribution on a wing.

Like in the finite wing, some of the air will escape around the tips. In the optimal wing, the solution is an elliptical spanwise reduction of the deflected mass flow. For the propeller, we define a similar function $K(x)$. It can be seen as the local reduction of the ring mass flow, of the incoming momentum flow, or of the effective ring frontal area; these are all valid interpretations. The effective (non-dimensional) frontal area (6.6) is reduced by the "mass factor" $K(x)$:

$$dm'_K(x) = K(x) \cdot (1 + a(x)) \cdot \frac{2x \cdot dx}{X^2} \quad (26.7)$$

We use the subscript K to indicate that m'_K has been reduced by this factor. The local ring area (6.8) for light loading likewise changes to :

$$dm'_K(x) = K(x) \cdot \frac{2x \cdot dx}{X^2} \quad (26.8)$$

Figure 26.4 shows a typical shape for $K(x)$. Unlike in the optimal wing, $K(x)$ is *not* identical for every optimal propeller, but there is a clear family likeness. Near the tip, the distribution is almost elliptical. In the center of the prop disk however it becomes flat, tending towards a constant value of $K(x) = 1$.

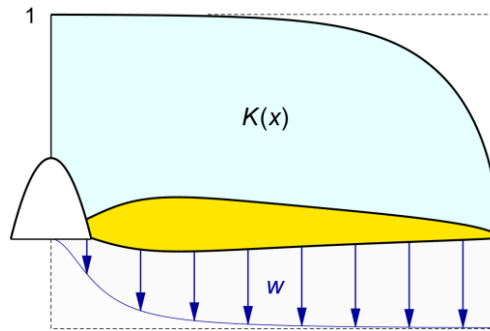


Figure 26.4 : Typical mass flow reduction factor $K(x)$. $b(x)$?

26.4 Biot-Savart on a spiral

TBW. Get it from old versions, or from earlier attempts at an Appendix.

27 Types of loss

Either here, or in 0.

Somewhere, describe the viscous (linear),
versus momentum (quadratic) thing.

Also, thrust reduction for equal loss ratio,
and thrust redistribution for equal thrust.

And : induction reduction
versus effective area reduction.

28 Overall design procedure

A practical way to handle ~~tip loss in~~ preliminary design is as follows :

- use PRANDTL's approximate thrust distribution $G'(x)$ of (20.1) for the shape.
This shape has a reduced thrust relative to unit thrust.
- find the exact κ_G' for this shape by numerical integration of (20.1).
This is simple enough, and it gives the exact thrust reduction for normalization.
- scale $G'(x)$ to the design thrust C_T by (****) and (****).
Do not use the "exact" κ_G equations for this step.
Using the numerical integral of κ_G' will give the propeller the exact design thrust.
The shape will be close to the exact solution (but not exact),
*certainly for propellers of $X \geq ****$.*
- find the estimated momentum efficiency of the propeller with (22.2).
Do not use the κ_G' used in the normalization for this step.
The κ_G value of (22.2) will give a better (slightly lower) estimate
of the actual efficiency for the shape, since it is close to the exact one.

There will be a subtle difference in blade shape between this solution and the exact one. If the exact solution is available then of course it should be used. However, the next chapters will show that viscous optimization affects the blade shape much more than any difference between PRANDTL'S tip solution and the exact one ever would.

Table 28.1 gives the exact numbers for our standard example of $X = 5$. The ratio at $B = \infty$ is the same as the purely rotational increase $1/\kappa_\phi$. At this tip speed ratio, the power increase is a factor of 0.15 on the axial loss. Going from infinity to six blades doubles the increase from 0.15 to almost 0.3. Going to three blades triples the increase to almost 0.45. Going to two blades bring the increase factor to just over 0.60, which is four times the factor due to rotational increase alone. Clearly, tip loss is a larger influence on the momentum loss than rotational loss. Fortunately, it is much easier to remedy. While it is possible to reduce rotational loss by recovering swirl with stator vanes or by applying counter-rotating propellers, the same gain in efficiency can be obtained by increasing the number of blades from 2 to 3. This gain can be doubled by going from 3 to 6 blades. Beyond 6 blades, we are in the realm of diminishing returns.

Since tip loss is just a loss of effective prop disk area, the easiest way to compensate it is by increasing the diameter of the propeller by the square root of $1/\kappa_G$. But since increasing the diameter is the best way to reduce momentum loss in the first place, the diameter should always be maximized even before looking at the tip loss.

Table 28.1 : Momentum loss ratio relative to axial.

<i>Tip speed ratio $X = 5$.</i>					
B	∞	6	4	3	2
$1/\kappa_G$	1.15	1.28	1.35	1.46	1.62

29 Installation effects

29.1 Introduction

ground clearance and diameter.

- diameter as large as possible.
- high prop axis can fight a bit with spinners.

spinners.

ducts.

counterrotation.

swirl recovery.

- wing LE.
- fins, swerve at take-off.
- same rotation fan in ships.

overlapping propellers (counter-rotating ?)

- Kaman helicopters, roomklopper / hand mixer.

pushers.

- noise, difficult installation, cooling, ground clearance, stability (tailless, tail booms)

noise.

- point to Ffowcs Williams, but refer to $Mach < 0.5$ mostly.
- more noise when inclined flow.

fin effect (side force)

- side projection, Dutch roll, front fin, stabilizer action, dead stick landings.

powered lift.

helicopters and lift fans.

30 Hubs and ducts

30.1 Introduction

Displacement and velocity disturbances. Rankine body, hubs. Overwing propellers.

WALD stresses that the wake closes behind the nacelle or pusher propeller, and that is where you want the ARCHIMEDES screw wake.

Displacement by the hub is not important except for pitch (and local velocity). The wake closes again. Best seen as an effective diameter.

First discuss the natural shape of the wake, before modifying it with a duct.

- *what about the core vortex of a hub ? Is the spinner a mirror plane like the duct ?*
- *does it turn wildly after contraction in the far wake ?*
- *counterrotation has no core vortex because they cancel / it closes the bound vortices. Stator vanes also have the reverse circulation.*

Figure **** for "infinitely many blades" is also valid for propellers inside a duct or a channel. Ducted propellers are often called fans. They do not suffer from tip effect except for practical problems around wall gaps and boundary layers. The bound vortex for each blade can be thought of as extending straight through the duct wall to infinity, like the bound vortex of an "infinite" wing mirrored by the wind tunnel wall, as discussed in a later chapter on vortex theory. The load distribution simply continues "through" the wall, resulting in square cropped tips like in figure 10.1.

Ducted fans may also have stator blades behind them, if only to support the nacelle. This, and the blocking of the center of the prop disk by the nacelle carrying the fan would tend to move the distribution toward a uniform axial thrust instead of the rotationally optimal scheduling with $\cos^2 \phi$. The distribution would move towards the viscous optimum around $x = 1$, i.e. the optimum X would move to lower values.

30.2 The natural wake contraction

Before going into the shape of a duct, we look at the shape of the flow tube without one. From continuity, we have the exact version of the mass flow equation (****):

$$m^{\bullet} = \rho \cdot A_{\text{wake}} \cdot (V + w) = \rho \cdot A_{\text{prop}} \cdot (V + v) \quad (30.1)$$

From (****) we know that $v = \frac{1}{2} w$. Since air is "incompressible", the density ρ is a constant. The natural contraction ratio of the slipstream will be:

$$\begin{aligned} \alpha &= \frac{A_{\text{wake}}}{A_{\text{prop}}} && \text{wake (area) contraction ratio} && (30.2) \\ &= \frac{V + \frac{1}{2} w}{V + w} \end{aligned}$$

In lightly loaded propellers, $w \ll V$, and the wake contraction will be negligible:

$$\alpha \cong 1 \quad \text{light loading} \quad (30.3)$$

In the case of static thrust, $V = 0$, and we have :

$$\alpha = 0.5 \text{ static thrust} \quad (30.4)$$

All other thrust conditions will have wake contraction values between these extremes.

helicopter ?

30.3 Ducts

Propellers can be mounted in ducts, in which case they are often called fans. The most obvious case is when the duct is closed, like in a wind tunnel or a mechanical ventilation system. Ships and aircraft can have short, open ducts which can have two purposes :

- mitigate or eliminate tip and rotational loss.
- draw more air into the fan than in an open installation.

The first will work. Tip cut off straight like in figure 10.1, the extreme case of **Error! Reference source not found.**

Make a drawing of the tube as a mirror plane, à la figure ****.

Things like κ_ϕ apply, no need for κ_G . With stator vanes, much of that gets eliminated too.

The duct exit fixes the size of the wake, almost regardless of the propeller thrust. From a straight duct exit the flow continues backward in a straight line without any further contraction. Even if the exit is flared open, the air almost immediately stops expanding. Air will not follow exit angles over 4° , and the wake area will never exceed 120% of the duct exit area. Inside the duct channel however, larger expansions are possible. Such ducts overrule the natural contraction ratio α of (30.2). The forced will fix the fan-to-wake contraction ratio to some arbitrary value α_{duct} .

A ducted fan is functionally equivalent to a propeller of the same wake size and wake velocity. According to (4.7), (****) and (30.1), it will have the exact same thrust, efficiency and power, both in forward flight and in hover, as the equivalent propeller. The difference is that the flow at the fan may be faster (or slower) than that in the open propeller plane, depending on the relative values of α in the propeller and in the duct.

If the flow at the fan is faster, then the fan thrust will have be lower in the same proportion, for the same power. To confirm this, we will calculate the fan thrust independently, by applying Bernoulli's law on a streamline through the ducted fan. Far ahead and far behind the system, the static pressure is atmospheric, so the difference in total head between them follows from the velocities alone. The fan in the middle creates a jump in the total head $p_{\text{total}} \equiv (p_{\text{local}} + \frac{1}{2} \rho V_{\text{local}}^2)$:

$$\Delta p_{\text{fan}} = \frac{1}{2} \rho (V+w)^2 - \frac{1}{2} \rho (V+w)^2 = \rho \cdot (V + \frac{1}{2} w) \cdot w \quad (30.5)$$

This pressure step at the fan is independent of the contraction ratio.

The thrust of the fan equals pressure step times fan area :

$$T_{\text{fan}} = \rho (V + \frac{1}{2} w) \cdot w \cdot A_{\text{fan}} \quad (30.6)$$

At the same time, the total thrust will always be, with (4.7) and (30.1) :

$$T = \rho \cdot (V + w) \cdot w \cdot A_{\text{wake}} \quad (30.7)$$

The ratio between fan thrust and overall thrust is :

$$\frac{T_{\text{fan}}}{T} = \frac{1}{\alpha_{\text{duct}}} \cdot \frac{V + \frac{1}{2} w}{V + w} = \frac{\alpha_{\text{open}}}{\alpha_{\text{duct}}} \quad (30.8)$$

In an accelerating duct (where the fan is smaller than the equivalent propeller), only part of the thrust is delivered on the fan, and the rest on the

duct. The fan power remains the same, because as the fan thrust is reduced, the velocity at the fan is increased by the same ratio. The thrust on the duct is definitely not "free", as is sometimes suggested !

The chapter on special application will discuss in more detail the well known application of ducted fans in vertical lifting platforms. **These fans have a bell mouth at the opening for taking in air from all sides, and a straight section after the fan, setting the contraction ratio α_{duct} to 1. The velocity at the fan is equal to the velocity w in the wake, instead of $\frac{1}{2}w$ at hover. For the same power and lift, the thrust on the fan is halved. The other half of the lift is delivered on the duct.** The advantage is that the duct is $1/\sqrt{2}$ smaller in diameter than an open rotor. The fan is better protected, and maybe shielded for noise a little.

In all other respects, ducts have serious disadvantages. The bell mouth inlet needed for low speeds does not fit the high speed condition, and the duct itself has considerable drag. Ducts are only suited to very special requirements, but their study gives some useful insight into the effects of local accelerated and decelerated flow.

fenestron, hovercraft, swivelling tug boats, shallow water

30.4 Hubs

TODO They cause displacement and velocity. And some spanwise flow.

Physical propellers have hubs of a finite size. The axle and blade roots are usually covered by a rotating spinner, which ideally blends into the rest of the streamline shape of a fuselage or nacelle. In a sense, the core of the propeller is "not there".

First look at a schematic fuselage or nacelle. Displacement and velocity (and direction).

TBW Fuselage turns inner disk into a ring, and pushes out the next rings. If just displacement, 15% spinner pushes out the outer diameter by 1.2%.

But in practice much less, displacement is mostly near the center of the object.

TBW If higher velocity, the local blade will have to add the same momentum, and the same rotation, while working locally at a higher axial and rotational velocity.

note : The same balance between axial and rotational will be impossible because ϕ is different. First discuss the purely axial case. Leave open the rotational details as long as I **don't know the answers**. But do mention that any rotation at the core is impossible, or at least it will leave an infinite core vortex in the far wake. We will always have zero thrust at the root of the propeller blade ?

Propellers can be enclosed inside a shroud (a short ring), or a duct (a longer tube).

The hub or core is suspended inside the duct by struts, which invariably double as stator vanes, taking out most of the rotation from the slipstream.

Swirl recovery - stators and counterrotation

Swirl spiral angle is much, much smaller than in figure 18.2. See wind tunnels. Different Goldstein function. Not "only" tip loss.

See Theodorsen Fig. 5 for $\lambda = 0.602$. No good approximation found in Tiberius*.m.

But Branlard can no doubt be expanded with counter-rotation.

Ducts can be of the accelerating type (like a venturi tube) for increased momentum efficiency, or of the decelerating type, for reduced Mach number at the expense of some efficiency. We will briefly look at the principle.

The flow behind an open propeller shows some contraction, depending on the thrust coefficient. Let us call the relative reduction in wake area $\alpha = A_{\text{wake}}/A_{\text{disk}}$. the factor α is close to 1 in a lightly loaded propeller, and it is 0.5 in static thrust,

- tip effect.

Maybe not as great as it seems if the duct is short. The duct itself has considerable drag.

And the intake lip gets all sorts of inflow angles at different flight regimes.

- local velocity increase (open Venturi duct).

Example of a duct which gives an N times higher local velocity.

Same power gives N times lower thrust on the propeller, $T = T_o/N$.

Same overall thrust T_o means that the duct gets a thrust of $T_o \cdot (1 - 1/N)$.

If $n = 2$, then the fan has only half of the total thrust, and the pressure field on the duct produces the other half. The duct thrust is not "free" or extra". The pressure field on the duct is simply induced by the fan acting in a faster local flow.

Off-design performance

31 Design considerations / constraints

31.1 Limitations on the diameter

Move this whole story to installation effects ? And/or off-design, and/or design constraints ?

We will find that the efficiency of a propeller improves with the diameter, or rather, with lower thrust loading. There are however several practical limitations to the diameter of the propeller, and therefore to the thrust loading.

Airplane propellers are limited in diameter primarily for reasons of ground clearance, but also because the allowable tip Mach number limits the prop diameter. Tip speeds above $M = 0.8$ are prohibitively noisy and inefficient, and $M = 0.4$ or lower would be ideal.

Traditional aircraft engines have more or less fixed rotation rates around 2200 RPM, and are direct drive. Smaller engines may have higher RPM and gearing to make the prop turn slower than the engine, but the propeller RPM is often in the same region. Tip speeds are proportional to the engine RPM and more or less proportional to the diameter, and this tends to limit the prop diameter, and the possibility of light loading at low speeds.

Ship propellers are even more severely limited in diameter. The tip path at the top has to be at some minimum depth to avoid cavitation, and at the bottom it cannot be deeper than the keel, otherwise the propeller would run aground before the ship does, causing extensive damage. The blade lift coefficients are also limited due to cavitation. *Wageningen B series, KT to CT, see ship section.*

move wind turbines to special section

Wind turbines have negative thrust, and they operate at a very specific optimum loading, called the BETZ optimum. At this loading, the outflow velocity is $\frac{1}{3}$ of the free wind speed, and the speed at the disk is $\frac{2}{3}$ of the free wind speed. This is in a way equivalent to a normal propeller which speeds up the air by a factor of $1\frac{1}{2}$ at the disk running in reverse, or even a factor of 3 from free air to far wake. Anyway, the C_T of a wind turbine is equivalent to 1 or 1.5 on a normal propeller. Relative to aircraft, wind turbines are very heavily loaded.

31.2 Typical values for the thrust coefficient

The C_T of a propeller in cruising flight is likely to be on the same order as the C_L of the wing, since the ratio between the diameter of the prop disk and the span of the wing is usually on the same order as the drag to lift ratio of the aircraft, somewhere in the range of 1:10. *But this interesting comparison belongs ... vortex ...*

31.3 Typical tip speed ratios

A small aerodynamic pitch angle ϕ gives high tip speeds for a given forward velocity V . But high tip speeds can lead to trouble. The highest tip speeds in aircraft occur in so-called propfans, and in high Mach propellers. These suffer from acoustic noise and reduced blade section efficiency due to shock waves. Their blades typically have swept back tips.

Conventional aircraft have tip speed ratios around 5 in cruising flight, rising to 10 in the climb. Historically, these seem to be the optimum values.

Ship propellers also have fairly low tip speed ratios, in this case to avoid cavitation.

Modern wind turbines have very flat pitch, for reasons explained later. Table 31.1 gives some typical values for the tip speed ratios in various fields of application. The literature often uses the definition $J = V/nD$ for the advance ratio of the propeller, where n is the rate of rotation in revolutions per second. This value differs from $V/\Omega R$ by a factor of π , since Ω is in radians per second, and one full revolution equals 2π radians of angle.

Table 31.1 Some typical tip speed ratios				
	<i>tip speed ratio</i>	<i>tip advance ratio</i>		ϕ_{tip}
	$X \equiv \Omega R/V$	$\lambda \equiv 1/X$	$J \equiv V/nD$	
<i>propfans</i>	1	1	3.0	45°
<i>ships / high Mach</i>	2.5	0.4	1.2	22°
<i>aircraft in cruise</i>	5	0.2	0.6	12°
<i>wind turbines</i>	10	0.1	0.3	6°

31.4 From cruise to climb

Wing C_L from 0.4 to 1.0 ? Check Cessna 172 cruise speed vs. best climb speed. Same difference in C_T already because of $\frac{1}{2} \rho V^2$. Then once more with V because of constant power of engines. Check best climb angle and climb rate, for thrust.

Find b_0 in climb. See what this does to effective pitch.

Blade angle of attack for constant pitch, draw parallel with PRANDTL's lifting line theory.

Skewed blades ? Relation to variable pitch propeller, blade C_L (as a function of C_T and X ?).

Look at blade chord chapter for C_L .

more to the supplementary chapters on vortex theory.

As a result, the thrust coefficient in cruising flight is normally well below 1. In climb, the situation may be different. The climb speed is maybe half the cruising speed. Aero engines have more or less fixed maximum power, and since power equals thrust times velocity, the available thrust will double. With the square of the flying speed going down by a factor of four and the

thrust doubled, the C_T of the propeller is eight times higher in climb than it is in cruise.

The thrust loading in the take-off run is even higher, and will affect take-off performance to the point that for fixed pitch propellers, a choice has to be made between good take-off and climb performance, and good cruising efficiency.

Table 31.2 Typical thrust coefficients	
	C_T
<i>ship</i>	0.3
<i>aircraft, climb</i>	0.5
<i>aircraft, cruise</i>	0.25

This fits a design process where the starting point is the cruise velocity and drag of an airplane or ship, and a maximum allowable propeller diameter. We develop the optimal propeller and RPM for that design condition. This gives the cleanest approach to optimal propeller design.

In practice, the outcome may not always fit the available power and RPM of the selected engine, and iterations may be needed. The present text covers only the ideal solution. It does not provide methods for an iteration process taking into account other constraints.

A practical lightplane example

Vooruitwijzend, iets over gebruikelijke waarden voor C_T en X ?

Cessna example to a table at the end?

In what follows, we will use as a running numerical example the propeller of a typical lightplane. Table 31.3 summarizes the data for a lightplane of the Cessna 172 / Piper Arrow class, collected from various sources, and averaged to some typical values.

PRANDTL and GLAUERT both use a **tip speed ratio of $X = 5$** in their examples.

Table 31.3 Numerical values for a typical lightplane		
<i>airplane</i>		
<i>take-off mass</i>	1,100	[kg]
<i>wing area</i>	16.2	[m ²]
<i>wing span</i>	11.0	[m]
<i>wing section (typical)</i>	NACA 2412	-
<i>wing C_L cruise</i>	0.4	
<i>cruise power (75 % max)</i>	90 000 (120 hp)	[W]
<i>cruise speed (8,000 ft)</i>	63 (120 kts)	[m/s]
<i>propeller</i>		
<i>diameter</i>	1.90	[m]
<i>geometrical pitch</i>	1.25	[m]
<i>$X = \Omega R / V$ cruise</i>	4.5	-
<i>typical blade section</i>	NACA 4412	-

blade C_l cruise	0.6	-
blade C_d cruise	0.006	-

32 Arbitrary propeller shapes

Thrust, torque and RPM. Separate rings. Iterate from α to thrust ??

Off-design is an "arbitrary shape". TBW.

Blade element method (BEM).

Rings are independent. Show ring picture again ?

Variation of pitch angle.

Off-design is a special case of an arbitrary propeller shape.

33 Variable pitch propellers

TBW.

34 Helicopters and lifting ducts

34.1 Helicopters and the hover equation

The equations for a propeller in static thrust can also be used for a helicopter in hover. The propulsive "efficiency" of (4.14) is not a meaningful figure here. It is zero by definition, since $V = 0$. A more useful figure of merit is : "power per thrust".

The (momentum) power required for a helicopter to hover can be related to the rotor disk loading by the so-called "hover equation" :

$$P_{\text{hover}} = \sqrt{\frac{T^3}{2\rho A_{\text{rotor}}}} \quad (34.1)$$

We derive a slightly different form here, based on the size of the wake instead of the size of the rotor, because this is needed for ducted (lift) fans. The total power equals the kinetic energy left behind in the wake, as in (***). With $m^\bullet$ as in (30.1) , we have :

$$P = \frac{1}{2} m^\bullet w^2 = \frac{1}{2} \cdot \rho \cdot A_{\text{wake}} \cdot w^3 \quad (34.2)$$

Now we need the value of w . From (4.7) and (30.1) we have :

$$T = m^\bullet \cdot w = \rho \cdot A_{\text{wake}} \cdot w^2$$

hence

$$w = \sqrt{\frac{T}{\rho A_{\text{wake}}}} \quad (34.4)$$

Substituting this into (***) yields :

$$P = \sqrt{(T^3 / (4 \cdot \rho \cdot A_{\text{wake}}))} \quad (34.5)$$

This is the hover equation in terms of the wake size. For an open rotor $A_{\text{wake}} = \frac{1}{2} A_{\text{rotor}}$, so (34.5) is consistent with the classical hover equation (34.1).

However, this new version will also hold for a ducted rotor, where α may be different from $\frac{1}{2}$.

Helicopters can have other rotor states very different from hover, like forward flight and even, descending into their own wake; but these states will not be discussed here.

34.2 Lifting ducts

TBW. Refer to helicopter chapter and duct chapter.

35 High Mach propellers

TBW. Decelerating ducts.

36 Ship propellers

TBW. High loading. Cavitation, high pitch. Ducts, stators. Wake adaptation. Wageningen, $K_T \rightarrow C_T$. Tell us why the blades are so wide, is it because of the high P/D ?

Wageningen B series, K_T to C_T .

P/D	$\Omega R/V$	K_T	C_T
1	3	0.08	
1.2	2.6	0.12	
1.4	2.25	0.15	
$C_T = \frac{T}{\frac{1}{2}\rho V^2 \cdot \pi R^2} = \frac{K_T}{j^2} \cdot \frac{8}{\pi}$			

37 Wind turbines

Fixed load. Wind mills are extremely heavily loaded. It's not only about efficiency, it's about power.

Maybe that's why they have flat pitch : momentum loss is more important than in most airplanes. *Is this true for $\cos^2$?*

But think about the viscous drag. The self locking band is near the tip, near $\phi = 0^\circ$ instead of near the hub at 90° . Interesting to look at this for a moment ! Left as an exercise to the reader, but give the equation for a driven leadscrew, with the other sign of ϵ .

Loss w.r.t. small inflow tube, w.r.t. to disk, and inverted b for prop, C_T w.r.t. disk.

Contraction $\sqrt{1} \quad \sqrt{2} \quad \sqrt{3}$

Torque rotates more slowly due to induced rotation in a windmill (faster in propeller), hence less power delivered. In a way, it's a propeller running backwards in time. But blade section LE and TE are reversed, and the spirals are in the expanding tube now.

Ik heb zorgen. Ik zie steeds maar niet of je $\cos^2$ wel moet toepassen terwijl in iedere ring misschien het optimum van Betz geldt, en dat werkt alleen op de axiale inductie.

Maar moet je toch niet iets met het rotatie-verlies ?? Vraag dat eens aan iemand. Branlard ? Okulov ?

References

- [1] Glauert (in Durand). Wald. Adkins. Larrabee.
- [2] Weinig.
- [3] Abbott & von Doenhoff.

37.1 Comparing Prandtl to the exact solution

... the "exact" vortex solution is available. It is associated with the name of GOLDSTEIN. It has always been known that the outcomes differ visibly, in particular for propellers of high pitch and low blade count. Before the age of computers however, the exact method was too complex for general use. In recent years, computer methods have become available to calculate this solution in real time, although it is still relatively time consuming.

PRANDTL's reduction factor is a bit too optimistic about the amount of air trapped between the sheets for propellers of high pitch and low blade count. The value in the exact solution is lower; maybe the "floor" between the fins in figure **** pushes the air out a bit. Be that as it may, PRANDTL's tip formula gives more accurate results if in the function (19.1) for $F(x)$ we use the axial distance s' , instead of the theoretically more accurate orthogonal distance s . As a side benefit, the equations are simpler.

The vortex solution is not just a "tip loss". Unlike PRANDTL's approximation, affects the root region as well. This fact already surprised GLAUERT, and we still have no quick intuition on this.

Figure **Error! Reference source not found.** showed how little of the overall thrust is actually produced in the root area. The blade roots cannot have a good blade section shape for structural reasons anyway, and they are usually covered by a streamlined hub or spinner. For most propellers, the difference in blade shape between the two solutions could well be ignored. However, the same cannot be said for the average thrust loss, and hence for the predicted efficiency.

We will discuss the best way to mix the exact and the approximate strategies in a later chapter.

From reduction to redistribution, blow up $b(x)$ to the design thrust. But not the mass flow. This all acts on the $b(x)$ term of the thrust (6.1), but it does not affect the local mass flow, at least not in the light loading assumption where $(1 + \frac{1}{2} b(x)) \cong 1$.